

KUDLA-RAPOPORT CONJECTURE FOR UNRAMIFIED MAXIMAL PARAHORIC LEVEL

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ABSTRACT. We prove the Kudla-Rapoport conjecture for unramified unitary groups with maximal parahoric level structure. Our approach differs from the local proof given in [LZ22a]. We reduce the conjecture to a global intersection problem using local-global compatibility. Then we apply an inductive procedure based on the modularity of generating series of global special divisors. This strategy follows the framework developed in the proof of the arithmetic fundamental lemma from [Zha21, MZ24] and arithmetic transfer identities from [Zha25, LMZ25].

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1. INTRODUCTION

1.1. Background. The classical *Siegel–Weil formula* ([Sie35, Sie51, Wei64, Wei65]) establishes a fundamental connection between central values of Siegel Eisenstein series and theta functions – generating series that encode the arithmetic of quadratic forms. In a groundbreaking program initiated by Kudla ([Kud97, Kud04]), the *arithmetic Siegel–Weil formula* extends this relationship into arithmetic geometry. It aims to express the central derivative of Siegel Eisenstein series as arithmetic theta functions – generating series of arithmetic intersection numbers of $\clubsuit$ special divisors on Shimura varieties associated to $\clubsuit(\clubsuit - \heartsuit, \spadesuit)$ or $\heartsuit(\clubsuit - \heartsuit, \heartsuit)$. These special divisors include Heegner points on modular or Shimura curves appearing in the Gross–Zagier formula ([GZ86, YZZ13]) ($\clubsuit = \spadesuit$), modular correspondences on the product of two modular curves in the Gross–Keating formula ([GK93]) and Hirzebruch–Zagier cycles on Hilbert modular surfaces ([HZ76]) ($\clubsuit = \heartsuit$).

The arithmetic Siegel–Weil formula has been established in great generality for $\clubsuit = \heartsuit, \spadesuit$ (orthogonal case) through the seminal work of Kudla, Rapoport, and Yang ([Kud97, KR99, KRY99, KR00, KRY04, KRY06]). The *archimedean* component of the formula is also known, due to Liu

[Liu11a] (unitary case), and Garcia–Sankaran [GS19] in full generality (cf. Bruinier–Yang [BY21] for an alternative proof in the orthogonal case).

In their works [KR11, KR14], Kudla and Rapoport refined the non-archimedean component of the conjectural formula by defining arithmetic models of special cycles in the unitary case for arbitrary $\mathfrak{p}$. These models, now known as *Kudla–Rapoport special cycles*, led to their formulation of the *global Kudla–Rapoport conjecture*, addressing the nonsingular part of the arithmetic Siegel–Weil formula. They prove that this global conjecture would follow from the *local Kudla–Rapoport conjecture* — a deep relationship between derivatives of local representation densities of hermitian forms and arithmetic intersection numbers of Kudla–Rapoport cycles on unitary Rapoport–Zink spaces.

This conjecture was proved in the celebrated paper of Li and W. Zhang [LZ22a] for unramified unitary groups with hyperspecial level and nonsingular coefficients. Their breakthrough has inspired numerous extensions to new cases: Li and Liu [LL22] and Yao [Yao24] addressed ramified unitary groups with exotic good reduction, He, Li, Shi, and Yang [HLSY23] explored the Krämer model, and Li and W. Zhang [LZ22b] extended the results to orthogonal groups. For a comprehensive introduction to these developments, we refer the reader to [Li23]. In parallel developments, Bruinier and Howard [BH24] and Chen [Che24a, Che24b, Che24c, Che24d] have made substantial progress in understanding singular coefficients.

The present paper focuses on the Kudla–Rapoport conjectures with maximal parahoric level structures for unramified unitary groups, building on the work of Cho [Cho22] and Cho, He, and Z. Zhang [CHZ23]. In this setting, special divisors appear in two distinct classes: $\mathfrak{p}$ -divisors and $\mathfrak{p}^2$ -divisors. Cho [Cho22] formulated an extended version of the local Kudla–Rapoport conjecture for intersection products of arbitrary combinations of these $\mathfrak{p}$ - and $\mathfrak{p}^2$ -divisors. Cho, He, and Z. Zhang [CHZ23] proved the conjectures for $\mathfrak{p}$ -cycles, conditionally on a strong version of the Tate conjecture, while obtaining unconditional results for the cases $\mathfrak{p}^2$ and $\mathfrak{p}$. Their method follows [LZ22a], where the Tate conjecture is used to prove local modularity, which is essential for their induction argument.

In this paper, we reformulate and prove the Kudla–Rapoport conjectures with maximal parahoric level structures for unramified unitary groups and arbitrary mixed cycles. Our approach differs from the local inductive method of [LZ22a], which relies on local modularity, see [LZ22a, §1.4]. Instead, we employ an induction argument based on local-global compatibility and global modularity. Our method builds on W. Zhang’s proof of the arithmetic fundamental lemma [Zha21], which has been subsequently generalized by Mihatsch and W. Zhang [MZ24] to arbitrary $\mathfrak{p}$ -adic fields for $\mathfrak{p}$ large enough, by Z. Zhang [Zha25] to maximal parahoric level over finite unramified extensions of $\mathfrak{p}$, and by the author, Mihatsch, and Z. Zhang [LMZ25] to maximal parahoric level over arbitrary $\mathfrak{p}$ -adic fields.

1.2. The Kudla–Rapoport conjecture. Let $\mathfrak{p}$ be an odd prime. Let $\mathfrak{K}$ be a finite extension of $\mathfrak{F}$ with residue field $\mathfrak{k}$ and a uniformizer ϖ . Let $\mathfrak{L}$ be the unramified quadratic extension of $\mathfrak{K}$, and let $\mathfrak{L}^\flat$ be the completion of the maximal unramified extension of $\mathfrak{L}$. For any integers $\mathfrak{r} \geq \mathfrak{s}$ and $\mathfrak{t} \leq \mathfrak{u} \leq \mathfrak{r}$, the *unitary Rapoport–Zink space* $\mathfrak{RZ}^{[\mathfrak{t}]}(\mathfrak{r}, \mathfrak{s})$ (§2.2) is the formal scheme over $\mathfrak{F} = \mathfrak{K}[\varpi]$ parameterizing hermitian $\mathfrak{H}^\flat$ -modules of signature $(\mathfrak{r} - \mathfrak{s}, \mathfrak{s})$ and type $\mathfrak{t}$, within the supersingular quasi-isogeny class (§2.1).

Let $\mathfrak{H}$ (resp. $\mathfrak{H}^{[\mathfrak{t}]}$) be the framing hermitian $\mathfrak{H}^\flat$ -module of signature $(\mathfrak{s}, \mathfrak{s})$ and type $\mathfrak{s}$ (resp. $(\mathfrak{r} - \mathfrak{s}, \mathfrak{s})$ and type $\mathfrak{t}$) over $\mathfrak{K}$. The space of *quasi-homomorphisms* $\mathfrak{H} := \mathfrak{H}^\flat \otimes_{\mathfrak{K}} \mathfrak{H}$ carries a natural $\mathfrak{H}^\flat/\mathfrak{H}$ -hermitian form, which makes $\mathfrak{H}$ the unique (up to isomorphism) non-degenerate hermitian space that does not contain any type $\mathfrak{t}$ vertex lattice, i.e., a hermitian lattice $\mathfrak{M} \subset \mathfrak{H}$ such that $\mathfrak{M} \subset \mathfrak{M}^\sharp \subset \mathfrak{H}^\sharp$, where $\mathfrak{M}^\sharp$ is the hermitian dual of $\mathfrak{M}$. Let $\mathfrak{H}^\flat$ be its nearby hermitian space which contains a type $\mathfrak{t}$ vertex lattice $\mathfrak{M}^{[\mathfrak{t}]}$.

When $\mathfrak{t} \neq \mathfrak{s}$, for any $\mathfrak{M} \in \mathfrak{H}^{[\mathfrak{t}]}$, there are two classes of special divisors on $\mathfrak{RZ}^{[\mathfrak{t}]}(\mathfrak{r}, \mathfrak{s})$: $\mathfrak{p}$ -divisors $\mathfrak{p}(\mathfrak{M})$ and $\mathfrak{p}^2$ -divisors $\mathfrak{p}^2(\mathfrak{M})$ (§2.3.1), arising due to the non-principal polarization. Let $\mathfrak{M}^{[\mathfrak{t}]}$ be a vertex lattice of type $\mathfrak{t}$ and let

$$\mathfrak{p}(\mathfrak{M}, \mathfrak{M}^{[\mathfrak{t}]}) := \mathfrak{p}(\mathfrak{M}), \quad \text{and} \quad \mathfrak{p}^2(\mathfrak{M}, \mathfrak{M}^{[\mathfrak{t}], \sharp}) := \mathfrak{p}^2(\mathfrak{M}).$$

Let $\bullet \leq \ast \leq \spadesuit$ be an integer. A *weight vector* is an element $\spadesuit \in \{\spadesuit, \clubsuit\}^\ast$. An¹ *admissible Schwartz function* (of weight $\spadesuit$ and type $\clubsuit$) is a function $\spadesuit := \spadesuit^\clubsuit \in \mathcal{S}(\mathbb{A}^\ast)$ such that

$$\spadesuit = \bigotimes_{\mathfrak{p} \in \mathbb{A}} \spadesuit_{\mathfrak{p}}, \quad \text{where} \quad \spadesuit_{\mathfrak{p}} = \begin{cases} \spadesuit_{\mathfrak{p}}^\clubsuit & \text{if } \spadesuit_{\mathfrak{p}} = \spadesuit; \\ \spadesuit_{\mathfrak{p}}^\clubsuit \cdot \spadesuit_{\mathfrak{p}}^\spadesuit & \text{if } \spadesuit_{\mathfrak{p}} = \clubsuit. \end{cases}$$

For any $\spadesuit \in \mathbb{A}^\ast$ and any admissible Schwartz function $\spadesuit^\clubsuit = \spadesuit^\clubsuit$, we define the *arithmetic intersection number* for mixed cycles (Definition 2.3.3)

$$\mathbb{I}(\spadesuit, \spadesuit^\clubsuit) := \mathbb{I}(\spadesuit^\clubsuit, \spadesuit^\clubsuit) \otimes \cdots \otimes \spadesuit^\clubsuit(\spadesuit, \spadesuit^\clubsuit) \in \mathbb{I}, \quad (1.2.1)$$

where $\spadesuit^\clubsuit(\spadesuit, \spadesuit^\clubsuit)$ is the structure sheaf of the special divisor $\spadesuit(\spadesuit, \spadesuit^\clubsuit)$ and $\otimes^\ast$ is the derived tensor product of coherent sheaves on $\mathbb{A}^\ast$ and $\mathbb{I}$ is the Euler-Poincaré characteristic.

When $\spadesuit = \clubsuit$, the hermitian lattice $\mathbb{I} = \mathbb{I}^\clubsuit$ is *unimodular*, meaning that $\mathbb{I} = \mathbb{I}^\clubsuit$. Let $\spadesuit^\clubsuit = \spadesuit^\clubsuit \in \mathcal{S}(\mathbb{A}^\ast)$, the work of Li and W. Zhang [LZ22a] shows the identity

$$\mathbb{I}(\spadesuit, \spadesuit^\clubsuit) = \frac{\mathcal{D}'(\spadesuit, \spadesuit^\clubsuit)}{\mathcal{D}(\spadesuit, \spadesuit^\clubsuit)} \quad (1.2.2)$$

where $\spadesuit$ is the identity matrix of rank $\spadesuit$, and $\spadesuit^\clubsuit = (\spadesuit, \spadesuit^\clubsuit)$ is the moment matrix associated with a collection of linearly independent vectors $\spadesuit \in \mathbb{A}^\ast$. The function $\mathcal{D}(\spadesuit, \spadesuit^\clubsuit)$ is the *representation density function* and $\mathcal{D}'(\spadesuit, \spadesuit^\clubsuit)$ is its central derivative, see [KR14, §10] or §3.5 for precise definitions.

However, in the general maximal parahoric case, the intersections between $\spadesuit$ -divisors and $\clubsuit$ -divisors fail to satisfy the *linear invariance property*, in contrast to the representation density function, which always does; see [LZ22a, §3.3]. To address this, we introduce the *weighted local density function*, as defined by Kudla [Kud97, Appendix], which generalizes the representation density functions appearing on the right-hand side of (1.2.2).

Definition 1.2.1. Let $\ast$ be any non-negative integer and let $\mathbb{A}^\ast$ be the split hermitian space of dimension $\ast$. Given a Schwartz function $\spadesuit \in \mathcal{S}(\mathbb{A}^\ast)$ in a hermitian space $\mathbb{A}$ of dimension $\spadesuit$ and a hermitian matrix $\spadesuit \in \mathbb{O}(\spadesuit)$, we define the *weighted local density function* as

$$\mathcal{D}_\ast(\spadesuit, \spadesuit) := \int_{\mathbb{O}(\spadesuit)} \int_{\mathbb{A}^\ast} \mathcal{D}(\spadesuit((\spadesuit, \spadesuit) - \spadesuit)) \spadesuit^\ast(\spadesuit) \spadesuit \spadesuit.$$

where $\mathbb{A}^\ast := \mathbb{A} \oplus \mathbb{A}^\ast$ and $\spadesuit^\ast(\spadesuit) := \spadesuit(\spadesuit) \otimes \spadesuit_{\mathbb{A}^\ast} \in \mathcal{S}((\mathbb{A}^\ast)^\ast)$, with $\mathbb{I}_{\mathbb{A}^\ast} \subset \mathbb{A}^\ast$ being a self-dual lattice. Here $\spadesuit$ and $\spadesuit$ are self-dual Haar measures on $\mathbb{A}^\ast$ and $\mathbb{O}(\spadesuit)$, respectively.

This provides a natural generalization of the representation density function: when $\spadesuit = \spadesuit$ is the characteristic function of a lattice $\mathbb{I} \subset \mathbb{A}$, for any $\ast \geq \spadesuit$, let $\spadesuit^\ast$ be a basis of the lattice $\mathbb{I} \oplus \mathbb{I}_{\mathbb{A}^\ast}$ and let $\mathcal{D}_\ast(\spadesuit, \spadesuit) = (\spadesuit^\ast, \spadesuit^\ast)$ be its moment matrix. By [Kud97, Proposition A.6], for any hermitian matrix $\spadesuit \in \mathbb{O}(\spadesuit)$, we have

$$\mathcal{D}_\ast(\spadesuit, \spadesuit) = |\spadesuit|_{\mathbb{A}^\ast} \cdot \mathcal{D}(\spadesuit, \spadesuit). \quad (1.2.3)$$

where $|\spadesuit|_{\mathbb{A}^\ast}$ is the metric in $\mathbb{A}^\ast$. See also [KR14, §10] for further details of this relation.

Let $\spadesuit^\clubsuit \in \mathcal{S}(\mathbb{A}^\ast)$ be an admissible Schwartz function of type $\clubsuit$. It is known that there exists a polynomial $\spadesuit^\clubsuit(\spadesuit, \spadesuit^\clubsuit) \in \mathbb{A}[\spadesuit]$ such that for each non-negative integer $\ast$, we have $\mathcal{D}_\ast(\spadesuit, \spadesuit^\clubsuit) = \spadesuit^\clubsuit(\spadesuit^\ast, \spadesuit^\clubsuit)$. Let $\spadesuit \in \mathbb{O}(\spadesuit)$ be any hermitian matrix that does not represent the space $\mathbb{A}$. By definition $\mathcal{D}_\ast(\spadesuit, \spadesuit^\clubsuit) = \spadesuit$ vanishes at zero. In this case, we define the *derivative of the weighted local density function*

$$\mathcal{D}'_\ast(\spadesuit, \spadesuit^\clubsuit) := - \frac{\spadesuit^\clubsuit}{\spadesuit} \Big|_{\spadesuit=0} \spadesuit^\clubsuit(\spadesuit, \spadesuit^\clubsuit).$$

As already observed in [KR00, KRY06, HLSY23, Cho22, CHZ23], the naive identity analogous to (1.2.2) does not hold in the general maximal parahoric setting, and correction terms must be added to the analytic side. We will explain it after stating the main result.

¹A priori, the function also depends on the choice of the vertex lattice $\mathbb{I}^\clubsuit$, but all statements are in fact independent of this choice.

Theorem 1.2.2 (Theorem 3.4.4, local Kudla-Rapoport conjecture). *For any $\mathfrak{o} \leq \mathfrak{b} \leq \mathfrak{r}$ and for any weight vector $\mathfrak{g} \in \{\mathfrak{p}, \mathfrak{q}\}^*$, let $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}$ be the explicit constants for $\mathfrak{o} \leq \mathfrak{g} < \mathfrak{b}$, $\mathfrak{g} \equiv \mathfrak{b} - \mathfrak{a}$ ($\mathfrak{a} \in \mathfrak{p}, \mathfrak{q}$), given by Definition 3.4.2. Then for any collection of linearly independent vectors $\mathfrak{v} \in \mathfrak{V}^*$, we have*

$$\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{o}, \mathfrak{v}^{[\mathfrak{b}]}) = \frac{\mathfrak{a}}{\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]})} \left(\mathfrak{a}'(\mathfrak{o}, \mathfrak{v}^{[\mathfrak{b}]}) - \sum_{\substack{\mathfrak{o} \leq \mathfrak{g} < \mathfrak{b} \text{ and} \\ \mathfrak{g} \equiv \mathfrak{b} - \mathfrak{a} \text{ (}\mathfrak{a} \in \mathfrak{p}, \mathfrak{q}\text{)}}} \mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]}) \right), \quad (1.2.4)$$

where $\mathfrak{g} = (\mathfrak{p}, \mathfrak{q})$ and $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]})$ is the volume of the stabilizer $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) = \mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]})$ with respect to the Haar measure $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}$ constructed from self-dual measures, see (H5).

Remark 1.2.3. The coefficients of the error terms are determined in Proposition 3.4.1. We illustrate the construction in the case where $\mathfrak{g} = (\mathfrak{p}, \mathfrak{q})$ and $\mathfrak{v}^{[\mathfrak{b}]} = \mathfrak{v}^{[\mathfrak{b}], \mathfrak{r}}$ following [CHZ23, Definition 1.2].

For each $\mathfrak{o} \leq \mathfrak{g} < \mathfrak{b}$ with $\mathfrak{g} \equiv \mathfrak{b} - \mathfrak{a}$ ($\mathfrak{a} \in \mathfrak{p}, \mathfrak{q}$), let $\mathfrak{v}_{\mathfrak{a}} \in \mathfrak{V}^*$ be a basis of a vertex lattice $\mathfrak{V}^{[\mathfrak{a}]}$ of type $\mathfrak{a}$ and let $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}) = (\mathfrak{v}_{\mathfrak{a}}, \mathfrak{v}_{\mathfrak{a}})$ be its moment matrix. We consider the linear combination:

$$\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{o}, \mathfrak{v}^{[\mathfrak{b}]}) := \mathfrak{a}'(\mathfrak{o}, \mathfrak{v}^{[\mathfrak{b}]}) - \sum_{\substack{\mathfrak{o} \leq \mathfrak{g} < \mathfrak{b} \text{ and} \\ \mathfrak{g} \equiv \mathfrak{b} - \mathfrak{a} \text{ (}\mathfrak{a} \in \mathfrak{p}, \mathfrak{q}\text{)}}} \mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]}),$$

where $\mathfrak{v}^{[\mathfrak{a}]} \in \mathfrak{V}^{[\mathfrak{a}]}$ are admissible Schwartz functions of weight $\mathfrak{g}$ and type $\mathfrak{a}$ (while $\mathfrak{v}^{[\mathfrak{b}]} \in \mathfrak{V}^{[\mathfrak{b}]}$). The constants $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}) \in \mathfrak{V}^*$ are chosen to satisfy

$$\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) = \mathfrak{a} \quad \text{for all } \mathfrak{o} \leq \mathfrak{g} < \mathfrak{b} \text{ with } \mathfrak{g} \equiv \mathfrak{b} - \mathfrak{a} \text{ (}\mathfrak{a} \in \mathfrak{p}, \mathfrak{q}\text{)}.$$

Since $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) = \mathfrak{a}$ for any $\mathfrak{a} > \mathfrak{r}$ and $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) \neq \mathfrak{a}$ for any $\mathfrak{a}$, this forms a linear system with an invertible upper triangular coefficient matrix, which uniquely determines the constants $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a})$.

Example 1.2.4. (1) When $\mathfrak{b} = \mathfrak{o}$ and $\mathfrak{v}^{[\mathfrak{b}]} = \mathfrak{v}^{[\mathfrak{b}], \mathfrak{r}}$ for a self-dual lattice $\mathfrak{V}$, there is no error term in (1.2.4). By Lemma 3.4.5, we have $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) = \mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})$, where $\mathfrak{a}$ is the identity matrix of rank $\mathfrak{r}$. By (1.2.3), equation (1.2.4) in this case is identical to

$$\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) \otimes \mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) \otimes \mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) = \frac{\mathfrak{a}'(\mathfrak{o}, \mathfrak{v}^{[\mathfrak{b}]})}{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})} = \frac{\mathfrak{a}'(\mathfrak{g}, \mathfrak{v}^{[\mathfrak{a}]})}{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})},$$

which reproves the Kudla-Rapoport conjecture at hyperspecial level [LZ22b, Theorem 3.4.1].

(2) When $\mathfrak{b} = \mathfrak{a}$ and $\mathfrak{v}^{[\mathfrak{b}]} = \mathfrak{v}^{[\mathfrak{b}], \mathfrak{r}}$ for an almost self-dual lattice $\mathfrak{V}^{[\mathfrak{a}]}$, we have $\mathfrak{v}^{[\mathfrak{a}]} = \mathfrak{v}^{[\mathfrak{a}], \mathfrak{r}}$ for a self-dual lattice $\mathfrak{V}^{[\mathfrak{a}]}$. By Lemma 3.4.5, we have

$$\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) = \mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]}) \quad \text{where } \mathfrak{a} = (\mathfrak{a}, \dots, \mathfrak{a}, \mathfrak{a}).$$

By relation (1.2.3), equation (1.2.4) in this case is identical to (after modifying $\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a})$ by constant factors in (1.2.3))

$$\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) \otimes \mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) \otimes \mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) = \frac{\mathfrak{a}}{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})} \left(\mathfrak{a}'(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]}) - \mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]}) \right), \quad (1.2.5)$$

As explained in Remark 1.2.3, we have

$$\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}) = \frac{\mathfrak{a}'(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})}{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})} = \frac{\mathfrak{a}'(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})}{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})} = \frac{\mathfrak{a}}{\mathfrak{a} - (-\mathfrak{a}) - \mathfrak{r}},$$

where the last equality is due to [Cho22, Proposition A.5]. By [LZ22b, after (13.6.1.1)], we have

$$\frac{\mathfrak{a}}{\mathfrak{a} - (-\mathfrak{a}) - \mathfrak{r}} \frac{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})}{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})} = \frac{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})}{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})} = \mathfrak{a} + \mathfrak{a}.$$

Hence (1.2.5) is equivalent to

$$\mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) \otimes \mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) \otimes \mathfrak{b}_{\mathfrak{g}, \mathfrak{g}}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{b}]}) = \frac{\mathfrak{a}}{\mathfrak{a} + \mathfrak{a}} \left(\frac{\mathfrak{a}'(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})}{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})} - \frac{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})}{\mathfrak{a}(\mathfrak{a}, \mathfrak{v}^{[\mathfrak{a}]})} \right).$$

This recovers the Kudla-Rapoport conjecture at almost self-dual level for pure $\mathfrak{p}$ -cycles proved in [LZ22b, Theorem 10.5.1]. Note that all the other cases are new in Theorem 1.2.2.

1.3. The arithmetic Siegel–Weil formula. Next, we describe the global applications of our local theorem to the arithmetic Siegel–Weil formula for semi-global integral models.

Let us now switch to global notations. Let $\mathfrak{F}/\mathfrak{K}$ be a CM extension with $\mathfrak{F} \neq \mathfrak{K}$. We fix a CM type Φ for $\mathfrak{F}$ with a distinguished element $\mathfrak{p} \in \Phi$. Let $\mathfrak{H}$ be a hermitian space over $\mathfrak{F}$ of dimension n with signature $(n - r, r)$ at $\mathfrak{p}$ and signature $(r, n - r)$ for $\mathfrak{p} \in \Phi \setminus \{\mathfrak{p}\}$. Consider the following group data:

$$\begin{aligned} \mathfrak{U} &:= \{g \in \mathrm{GL}_n(\mathfrak{F}) \mid g \cdot \mathfrak{p} \in \mathfrak{p}\}, \\ \mathfrak{V} &:= \{g \in \mathrm{GL}_n(\mathfrak{F}) \mid g \cdot \mathfrak{p} \in \mathfrak{p}\}, \\ \mathfrak{W} &:= \mathfrak{U} \times \mathfrak{V} \xrightarrow{\sim} \mathfrak{U} \times \mathrm{GL}_n(\mathfrak{F}). \end{aligned}$$

Associated to $\mathfrak{W}$ is a natural Shimura datum $(\mathfrak{W}, \{\mathfrak{h}_\mathfrak{W}\})$ of PEL-type.

Let $\mathfrak{U}^\sim = \mathfrak{U} \times \mathfrak{V}$ be a compact open subgroup. Then the associated Shimura variety $\mathfrak{U}^\sim = \mathfrak{U}^\sim(\mathfrak{W}, \{\mathfrak{h}_\mathfrak{W}\})$ is of dimension $n - r$ and has a canonical model over the reflex field $\mathfrak{F}$.

Let $\mathfrak{L}$ be an $\mathfrak{F}$ -lattice and let $\mathfrak{a} \in \mathfrak{L}_{>0}$ be chosen to satisfy the following conditions:

- We require that $\mathfrak{a} \mid \mathfrak{a}^2$.
- If $\mathfrak{p}$ is a finite place of $\mathfrak{F}$ that is ramified in $\mathfrak{K}$, then $\mathfrak{p} \mid \mathfrak{a}$.
- For any finite place $\mathfrak{p} \nmid \mathfrak{a}$ of $\mathfrak{F}$, the localization $\mathfrak{L}_{\mathfrak{p}}$ in $\mathfrak{U}_{\mathfrak{p}} = \mathfrak{U} \otimes_{\mathfrak{F}} \mathfrak{K}_{\mathfrak{p}}$ is a vertex lattice.

We make the following choice of level structure:

- The level $\mathfrak{U}^\sim \subset \mathfrak{U}(\mathfrak{W}_{\mathfrak{p}})$ factors as

$$\mathfrak{U}^\sim = \mathfrak{U}^\sim_{\mathfrak{p}} \times \mathfrak{U}^\sim_{\mathfrak{p}^\infty}$$

where $\mathfrak{U}^\sim_{\mathfrak{p}} \subset \mathfrak{U}(\mathfrak{W}_{\mathfrak{p}})$ is any open compact subgroup and where $\mathfrak{U}^\sim_{\mathfrak{p}^\infty} = \mathfrak{p}(\widehat{\mathfrak{U}}_{\mathfrak{p}^\infty})$ is the stabilizer of the completion $\widehat{\mathfrak{U}}_{\mathfrak{p}^\infty}$ in $\mathfrak{U}(\mathfrak{W}_{\mathfrak{p}^\infty})$;

- The level $\mathfrak{U}^\sim \subset \mathfrak{U}(\mathfrak{W}_{\mathfrak{p}^\infty})$ factors similarly as

$$\mathfrak{U}^\sim = \mathfrak{U}^\sim_{\mathfrak{p}^\infty} \times \mathfrak{U}^\sim_{\mathfrak{p}}$$

where $\mathfrak{U}^\sim_{\mathfrak{p}^\infty} \subset \mathfrak{U}(\mathfrak{W}_{\mathfrak{p}^\infty})$ is any open compact subgroup and where $\mathfrak{U}^\sim_{\mathfrak{p}}$ is the maximal compact subgroup in $\mathfrak{U}(\mathfrak{W}_{\mathfrak{p}})$.

We construct a *punctured integral model* $\mathfrak{U}^\sim \rightarrow \mathfrak{U}^\sim[\mathfrak{a}^{-1}]$ of $\mathfrak{U}^\sim$ (§5.3). In general, $\mathfrak{U}^\sim$ is not smooth nor regular, but is a locally complete intersection, see [LMZ25, Remark 6.10].

Let $\mathfrak{W} \in \mathfrak{W}(\mathfrak{U}^\sim(\mathfrak{W}_{\mathfrak{p}}))$ be an *admissible Schwartz function*. Here, admissible means that

$$\mathfrak{W} = \mathfrak{W}_{\mathfrak{p}} \otimes \mathfrak{W}_{\mathfrak{p}^\infty} \quad \text{and} \quad \mathfrak{W}_{\mathfrak{p}^\infty} = \bigotimes_{\mathfrak{p} \in \Phi} \mathfrak{W}_{\mathfrak{p}^\infty}$$

where each $\mathfrak{W}_{\mathfrak{p}^\infty}$ is either $\mathfrak{p}_{\mathfrak{p}^\infty}$ or $\mathfrak{p}_{\mathfrak{p}^\infty}^\vee$.

For any $\mathfrak{g} \in \mathfrak{O}_{\mathfrak{F}}^\times(\mathfrak{p})$, we define the *weighted Kudla-Rapoport special cycle* (§6.1) $\mathfrak{U}^\sim(\mathfrak{g}, \mathfrak{W}) \rightarrow \mathfrak{U}^\sim$. Furthermore, for any hermitian matrix $\mathfrak{g} \in \mathfrak{O}_{\mathfrak{F}}^\times(\mathfrak{p})$ with totally positive diagonal entries, we define the *derived special cycles* $\mathfrak{U}^\sim(\mathfrak{g}, \mathfrak{W}) \in \mathfrak{U}^\sim(\mathfrak{U}^\sim)$ (§6.2).

On the other hand, let $\mathfrak{H}$ be the incoherent $\mathfrak{F}/\mathfrak{K}$ -hermitian space nearby $\mathfrak{H}$, namely $\mathfrak{H}$ is positive definite for $\mathfrak{p} \in \Phi$ and $\mathfrak{H}_{\mathfrak{p}} \cong \mathfrak{H}_{\mathfrak{p}}$ for all finite places $\mathfrak{p}$ over $\mathfrak{K}$. Let $\mathfrak{W} \otimes \mathfrak{W}_{\infty} \in \mathfrak{W}(\mathfrak{H})$, where $\mathfrak{W}_{\infty}$ is the standard Gaussian function (4.3.1), then there is a *incoherent Eisenstein series* $\mathfrak{E}(\mathfrak{g}, \mathfrak{W} \otimes \mathfrak{W}_{\infty})$ (see §7) on the hermitian upper half space

$$\mathfrak{E}(\mathfrak{g}, \mathfrak{W}_{\infty}) = \{ \mathfrak{g} = \mathfrak{g}_1 + \mathfrak{g}_2 \mid \mathfrak{g}_1 \in \mathfrak{O}_{\mathfrak{F}}^\times(\mathfrak{p}), \mathfrak{g}_2 \in \mathfrak{O}_{\mathfrak{F}}^\times(\mathfrak{p}_{\infty}) \}, \quad \text{where } \mathfrak{p}_{\infty} := \mathfrak{p} \otimes_{\mathfrak{F}} \mathfrak{K} \cong \mathfrak{p}^\vee.$$

This is the Siegel Eisenstein series associated to a standard Siegel–Weil section of the degenerate principal series. Since we will always fix the standard Gaussian function at archimedean places, we write the Eisenstein series as $\mathfrak{E}(\mathfrak{g}, \mathfrak{W})$ for saving notation.

²While this assumption may be unnecessary, we do not pursue it further in this paper.

The Eisenstein series has a meromorphic continuation and a functional equation relating $\mathbb{E} \leftrightarrow -\mathbb{E}$. The central value $\mathbb{E}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}) = 0$ due to incoherence. We thus consider its *central derivative*

$$\partial \mathbb{E}(\frac{1}{2}, \frac{1}{2}) := \left. \frac{\mathbb{E}(s, \frac{1}{2})}{s - \frac{1}{2}} \right|_{s=\frac{1}{2}}.$$

Let $\mathcal{P}(\mathcal{G}, \mathbb{A})$ be the set of places $\mathfrak{p}$ over $\mathbb{Q}$ where $\mathcal{G}$ does not represent $\mathbb{A}_{\mathfrak{p}}$. Let $\mathfrak{a}(\mathfrak{p}) = \#(\mathcal{G}(\mathbb{F}_{\mathfrak{p}}) \backslash \mathcal{G}(\mathbb{F}_{\mathfrak{p}})/\mathcal{G}(\mathbb{F}_{\mathfrak{p}}))$.

Theorem 1.3.1 (Arithmetic Siegel-Weil formula). *There exists an explicit coherent Eisenstein series $\mathbb{E}(\frac{1}{2}, \frac{1}{2})$ associated with the correction terms in Theorem 1.2.2 (see §10.1), such that for any positive definite $\mathcal{G} \in \mathcal{O}^+(\mathbb{Q})$ satisfying $\mathcal{P}(\mathcal{G}, \mathbb{A}) = \{\mathfrak{p}\}$ with $\mathfrak{p} \nmid \mathfrak{f}$, we have*

$$\frac{\mathfrak{a}(\mathfrak{p})}{\mathfrak{a}(\mathfrak{p})!} \cdot \frac{\mathfrak{a}(\mathfrak{p})}{\mathfrak{a}(\mathfrak{p})!} \cdot \mathfrak{a}(\mathfrak{p}) \cdot \mathfrak{a}(\mathfrak{p}) = \mathfrak{a}(\mathfrak{p}) \cdot (\partial \mathbb{E}(\frac{1}{2}, \frac{1}{2}) + \mathbb{E}(\frac{1}{2}, \frac{1}{2})),$$

where $\mathfrak{a}(\mathfrak{p})$ is an explicit constant determined by local factors at archimedean places (see (10.1.3)).

1.4. Proof strategy. We present the essential idea for proving Theorem 1.2.2 in the unramified hyperspecial level case. It should be noted that the generating series $\mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c})$ presented below are formal constructions - the genuine generating series requires a careful definition in an appropriate function space, see §10.4.

The proof of the local Kudla-Rapoport conjecture proceeds by induction. For $\mathfrak{f} = 1$, this follows directly from calculation, see §3.5. Let us assume the conjecture holds for all unramified $\mathfrak{f}$ -adic field ($\mathfrak{f} \neq 1$) extensions $\mathbb{Q}/\mathbb{Q}$ and all $\mathbb{Q}/\mathbb{Q}$ -hermitian spaces of dimension $\mathfrak{f} - 1$. Our goal is to establish the conjecture for a specific unramified extension $\mathbb{Q}/\mathbb{Q}$ of $\mathfrak{f}$ -adic fields and a hermitian space $\mathbb{H}$ of dimension $\mathfrak{f}$, and a collection of linearly independent vectors $\mathfrak{v} \in \mathbb{H}$.

We embed our local data into a global data in §1.3. By the $\mathfrak{f}$ -adic uniformization of derived special cycles (§6.3), the local constancy of both the geometric and analytic sides, we reduce the problem to proving the arithmetic Siegel-Weil formula for

$$\mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}), \quad \text{where } \mathfrak{a} = \begin{pmatrix} \mathfrak{a}_1 & \mathfrak{a}_2 \\ \mathfrak{a}_3 & \mathfrak{a}_4 \end{pmatrix} \quad \text{with } \mathcal{P}(\mathfrak{a}, \mathbb{A}) = \{\mathfrak{p}\}, \quad (\star)$$

where the matrices and Schwartz functions are chosen as follows:

- $\mathfrak{a}$ is a $(\mathfrak{f} - 1) \times (\mathfrak{f} - 1)$ matrix.
- $\mathfrak{b} = \mathfrak{a} \mathfrak{c}$ with a self-dual lattice $\mathfrak{c}$.
- $\mathfrak{c} = \mathfrak{a} \otimes \mathfrak{b}$ takes rational values.

We consider the following generating series

$$\mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) = \sum_{\mathfrak{c} \in \mathbb{Q}} \sum_{\mathfrak{d} \in \mathbb{Q}} \sum_{\substack{\mathfrak{e} = \begin{pmatrix} \mathfrak{e}_1 & \mathfrak{e}_2 \\ \mathfrak{e}_3 & \mathfrak{e}_4 \end{pmatrix} \\ \mathcal{P}(\mathfrak{e}, \mathbb{A}) = \{\mathfrak{p}\}}} \left(\frac{\mathfrak{a}(\mathfrak{p})}{\mathfrak{a}(\mathfrak{p})!} \cdot \frac{\mathfrak{a}(\mathfrak{p})}{\mathfrak{a}(\mathfrak{p})!} \cdot \mathfrak{a}(\mathfrak{p}) \cdot \mathfrak{a}(\mathfrak{p}) \right) \cdot \mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{e}), \quad (1.4.1)$$

where $\mathfrak{E}_{\mathfrak{p}} := \mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})$. This is a Hilbert modular form of weight $\mathfrak{f}$ involving two parts:

- The geometric part is the explicit expansion of the following intersection product

$$\langle \mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}), \mathfrak{E}_{\mathfrak{p}}(\mathfrak{c}, \mathfrak{d}) \rangle, \quad \text{where } \mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) = \sum_{\mathfrak{c} \in \mathbb{Q}} \mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}) \in \widehat{\mathfrak{E}}_{\mathfrak{p}}(\mathbb{A}).$$

The modularity of the intersection product was proved by [BHK⁺20] combined with [ES18] when $\mathfrak{f} = 2$, and by [Zha25] and [LMZ25] when $\mathfrak{f} \neq 2$.

- The analytic part is a Fourier-Jacobi type series of the Siegel Eisenstein series, which is also modular from definition.

When $\mathfrak{a} \in \mathfrak{a}(\mathbb{Q})$ is *regular* at two places (see Theorem 7.2.1), only non-singular terms contribute to the summation (1.4.1), allowing us to compute the local contributions term by term. Since $\mathfrak{a}$ is totally positive definite, by archimedean Siegel-Weil formula (Theorem 4.3.1(1)), the hermitian matrices $\mathfrak{a}$ must either be totally positive definite or have signature $(\mathfrak{f} - 1, 1)$. In the latter case, the corresponding terms in $\mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})$ vanish by the archimedean arithmetic Siegel-Weil formula [Liu11a], see also Theorem 4.3.1. Consequently, the summation in (1.4.1) involves only totally positive definite terms.

By induction and applying the cancellation laws on both the geometric side ([LZ22a, Lemma 2.11.1]) and analytic side ([LZ22a, (3.2.1.1)]), almost all Fourier coefficients in the summation vanish. By a careful choice of the global embedding, we can apply the vanishing result from [Zha21, Lemma 13.6] to conclude that $\mathfrak{g}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}) \equiv \mathfrak{c}$. Taking $\mathfrak{c}$'s Fourier coefficient, we get

$$\sum_{\substack{\mathfrak{c} = \begin{pmatrix} \mathfrak{a} & \mathfrak{b} \\ * & * \end{pmatrix} > \mathfrak{c} \\ \mathfrak{c}(\mathfrak{a}, \mathfrak{b}) = \{\mathfrak{a}, \mathfrak{b}\}}} \left(\frac{\mathfrak{g}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})}{\mathfrak{g}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})} \right) \widehat{\mathfrak{g}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})}(\mathfrak{c}) - \mathfrak{c} \partial \mathfrak{g}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) = \mathfrak{c}.$$

This is a finite sum which includes the index $\mathfrak{c}$ in $(\star)$. By modifying the Schwartz function (a step that should have been included in our choice of global data), we can ensure that only the terms corresponding to $\mathfrak{c}$ are nonzero, thus completing our induction step.

To extend the result to arbitrary maximal parahoric level, two main modifications are required. First, additional error terms must be incorporated into the analytic side. Second, the vanishing result needs to be replaced by the double induction method developed in [Zha25, Lemma 15.1]. A key local ingredient in this generalization is the proof of the cancellation law on the analytic side (Corollary 3.4.8).

Remark 1.4.1. Theorem 3.3.2 shows that the cancellation law on the analytic side holds term by term when $\mathfrak{b} \neq \mathfrak{c}$. The case $\mathfrak{b} = \mathfrak{c}$ is exceptional: in this case, the cancellation law holds only for the entire analytic function, not for individual terms (see the proof of Theorem 3.4.7). This is the unique pathological case where term-wise cancellation fails.

1.5. Open problems. We conclude this introduction by discussing several open problems. First, the formulation of the Kudla-Rapoport conjecture at deeper levels (e.g., Iwahori) or for weighted special cycles remain open problems. At these levels, additional classes of special divisors emerge, and the almost modularity results become harder and more interesting. Another promising direction is to establish a connection between our results and the Hecke algebra formulation of the Kudla-Rapoport conjecture, similar to [LR24a] and [LR24b].

Second, in the global proof, we eliminate all singular terms through the regularity assumption on $\mathfrak{g}_{\mathfrak{p}}$. This condition is fragile and fails to be preserved under the Weil representation $\mathfrak{g}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})$ in the adelic setting. A natural open problem is to relax this regularity assumption and obtain results for singular coefficients. This relates to the works of Chen [Che24a, Che24b, Che24c, Che24d].

Third, in contrast to the local proof in [LZ22a], we employ global modularity rather than local modularity. An open problem is to elucidate the relationship between these two forms of modularity, or to utilize global modularity to establish local modularity conjectures such as [Zha25, Conjecture 8.2].

Next, the global proof does not directly extend to ramified cases, as we lack a suitable definition of the weighted integral special divisors under Fourier transform - a prerequisite for applying the double induction method. A natural question is to formulate the correct special divisors in the Krämer model (and more broadly, in splitting models [HLS24]) that arise as Fourier transforms of special $\mathfrak{p}$ -divisors and $\mathfrak{c}$ -divisors³. See also the very recent work of He, Z. Zhang and Zhu [HZZ25].

Finally, the global method appears applicable to the orthogonal case. However, this extension requires a precise definition of relative GSpin Rapoport-Zink spaces and corresponding comparison results in the totally real case, or a more careful treatment with compactification for $\mathfrak{p} = \mathfrak{c}$ case.

1.6. The structure of the paper. In Part 1, we review the local geometric construction in §2. In §3, we define and study the weighted local density functions, we then formulate the Kudla-Rapoport conjecture and prove the cancellation law on the analytic side. In §4, we review the local result in the archimedean places. In Part 2, we define integral models of Shimura varieties and its special cycles in §5 and §6. In §7, we review results on Eisenstein series. In Part 3, we define the arithmetic Picard group and its arithmetic intersection pairing in §8. In §9, we compute and compare the local contribution of the intersection numbers. In §10, we construct generating series (1.4.1) and prove its modularity. In §11, we prove the Kudla-Rapoport conjecture.

³In fact, in this case, even the geometric cancellation law becomes more subtle, see [LRZ25, §6]

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1.8. Notations and Conventions.

1.8.1. Hermitian space. Let $\mathbb{A}/\mathbb{A}_f$ be a quadratic extension of local fields or ring of adeles and let $(\mathbb{A}, \mathfrak{h})$ be a $\mathbb{A}/\mathbb{A}_f$ -hermitian space of dimension n . Let $\mathfrak{h}^\times = \mathfrak{h}(\mathbb{A})$ be the corresponding unitary group.

H1 Let $\mathfrak{h}(\mathbb{A})$ be the set of $n \times n$ hermitian matrices and $\mathfrak{h}^\times(\mathbb{A})$ be the subset of all regular (non-singular) hermitian matrices.

H2 We define the map

$$\mathfrak{h} : \mathbb{A}^n \longrightarrow \mathfrak{h}^\times(\mathbb{A}), \quad \mathfrak{h} = (\mathfrak{h}_1, \dots, \mathfrak{h}_n) \longmapsto (\mathfrak{h}(\mathfrak{h}_i, \mathfrak{h}_j))_{i,j}.$$

H3 For any $\mathfrak{g} \in \mathfrak{h}^\times(\mathbb{A})$, we define the inverse image

$$\mathbb{A}^\mathfrak{g} := \{\mathfrak{h} \in \mathbb{A}^n : \mathfrak{h}(\mathfrak{h}) = \mathfrak{g}\} = \mathfrak{h}^{-1}(\mathfrak{g}).$$

H4 We define the regular subset

$$\mathbb{A}^\times := \{\mathfrak{h} \in \mathbb{A}^n \mid \mathfrak{h}(\mathfrak{h}) \neq 0\}.$$

For any $\mathfrak{h} \in \mathbb{A}^\times$ with $\mathfrak{h}(\mathfrak{h}) = \mathfrak{g}$, the map

$$\mathfrak{h} : \mathfrak{h}^\times(\mathbb{A}) \xrightarrow{\sim} \mathbb{A}^\mathfrak{g} = \mathfrak{h}^{-1}(\mathfrak{g}), \quad \mathfrak{h} \longmapsto \mathfrak{h} \cdot \mathfrak{h},$$

is an isomorphism.

H5 Fix basis elements $\mathfrak{e}_i \in \wedge^{\mathfrak{h}_i}(\mathbb{A}^\times)^*$ and $\mathfrak{f}_i \in \wedge^{\mathfrak{f}_i}(\mathfrak{h}^\times(\mathbb{A}))^*$ and write $\mathfrak{e}$ and $\mathfrak{f}$ for the translation invariant forms they define, where $(-)^*$ is the dual vector space. By [KR14, after Corollary 10.4]⁴, there is a form $\mathfrak{g}$ of degree n on $\mathbb{A}^\times$ such that

- $\mathfrak{e} = \mathfrak{h}^* \mathfrak{f} \wedge \mathfrak{g}$,
- $\mathfrak{g}$ is invariant under the action of $\mathfrak{h}^\times(\mathbb{A}) \times \mathfrak{h}^\times(\mathbb{A})$ on $\mathbb{A}^\times$,
- for all points $\mathfrak{h} \in \mathbb{A}^\times$, the restriction of $\mathfrak{g}$ to $\mathfrak{h}^\times(\mathbb{A})$ is non-zero. In particular, the pullback $\mathfrak{g}_\mathfrak{h} = \mathfrak{h}^* \mathfrak{g}$ is a gauge form on $\mathfrak{h}^\times(\mathbb{A})$, which is independent of the choice of $\mathfrak{h}$.

H6 We define a constant

$$\mathfrak{e}_\mathfrak{h} = \mathfrak{e}(\mathfrak{e}, \mathfrak{f}) \mathfrak{g}_\mathfrak{h}.$$

Here $\mathfrak{e}_\mathfrak{h}$ is the measure on $\mathbb{A}^\times$ determined by the gauge form $\mathfrak{e}$ and $\mathfrak{g}_\mathfrak{h}$ is the self-dual measure with respect to $\langle \mathfrak{h}, \mathfrak{h} \rangle = \mathfrak{f}(\mathfrak{h}(\mathfrak{h}), \mathfrak{h}(\mathfrak{h}))$.

H7 Similarly, we define a constant:

$$\mathfrak{f}_\mathfrak{h} = \mathfrak{f}(\mathfrak{f}, \mathfrak{e}) \mathfrak{g}_\mathfrak{h}.$$

Here $\mathfrak{f}_\mathfrak{h}$ is the measure on $\mathfrak{h}^\times(\mathbb{A})$ determined by the gauge form $\mathfrak{f}$ and $\mathfrak{g}_\mathfrak{h}$ is the self-dual measure with respect to $\langle \mathfrak{h}, \mathfrak{h} \rangle = \mathfrak{f}(\mathfrak{h}(\mathfrak{h}), \mathfrak{h}(\mathfrak{h}))$.

1.8.2. Local data.

L1 In the local setup, we fix an unramified quadratic extension of $\mathbb{A}$ -adic fields $\mathbb{A}/\mathbb{A}_f$ with uniformizer ϖ . Assume $\varpi \neq \bar{\varpi}$.

L2 Let $\mathbb{A}$ be a hermitian space and let $\mathfrak{h} \subseteq \mathbb{A}$ be a hermitian lattice. The hermitian dual $\mathfrak{h}^\#$ is defined by $\mathfrak{h}^\# := \{\mathfrak{h} \in \mathbb{A} \mid (\mathfrak{h}, \mathfrak{h}) \in \mathfrak{h}\}$. A *vertex lattice* is a hermitian lattice $\mathfrak{h} \subseteq \mathbb{A}$ satisfying $\mathfrak{h} \subseteq \mathfrak{h}^\# \subseteq \mathfrak{h}$. The *type* of a vertex lattice $\mathfrak{h}$ is $\mathfrak{t}(\mathfrak{h}) := \mathfrak{h}^\#/\mathfrak{h}$.

L3 Let $\mathfrak{h} \leq \mathfrak{h}^\# \leq \mathfrak{h}$ and let $\mathfrak{h} \subseteq \mathbb{A}$ be a vertex lattice. A Schwartz function $\mathfrak{h} \in \mathfrak{h}(\mathbb{A}^\times)$ is called *admissible* if $\mathfrak{h} = \bigotimes_{\mathfrak{h}} \mathfrak{h}_\mathfrak{h}$ such that $\mathfrak{h}_\mathfrak{h} = \mathfrak{h}_\mathfrak{h}$ or $\mathfrak{h}_\mathfrak{h}^\#$. It is of *type* $\mathfrak{h}$ if the vertex lattice $\mathfrak{h}$ has type $\mathfrak{h}$. A *weight vector* is an element $\mathfrak{h} \in \{\mathfrak{h}, \mathfrak{h}^\#\}^\mathfrak{h}$. An *admissible Schwartz function of weight* $\mathfrak{h}$ and *type* $\mathfrak{h}$ is a function $\mathfrak{h} := \mathfrak{h}_\mathfrak{h} \in \mathfrak{h}(\mathbb{A}^\times)$ such that

$$\mathfrak{h} = \bigotimes_{\mathfrak{h}} \mathfrak{h}_\mathfrak{h}, \quad \mathfrak{h}_\mathfrak{h} = \begin{cases} \mathfrak{h}_\mathfrak{h} & \text{if } \mathfrak{h}_\mathfrak{h} = \mathfrak{h}; \\ \mathfrak{h}_\mathfrak{h}^\# & \text{if } \mathfrak{h}_\mathfrak{h} = \mathfrak{h}^\#, \end{cases}$$

⁴Note that the construction is purely algebraic hence also carries over to the local setup, see [KRY06, Lemma 5.3.1 and Lemma 5.3.2].

where $\mathbb{H} = \mathbb{H}^{[a]}$ is a vertex lattice of type $\star$. An admissible Schwartz function is an admissible function of some weight $\star$ and type $\star$.

- L4 Two admissible Schwartz functions $\psi \in \mathcal{S}(\mathbb{H}^{[a]})$ ($\star \in \{\star, \star\}$) are *matched* if they share the same weight vector. For an admissible Schwartz function ψ of type $\star$, we consider it matched with any other admissible Schwartz function.

1.8.3. Global data.

- G1 For a global number field $\mathbb{K}$, let $\mathbb{A}_{\mathbb{K}}$ be the set of all places. Let $\mathbb{A}_{\mathbb{K}, \infty}$ (resp. $\mathbb{A}_{\mathbb{K}, \infty}$) be the set of finite (resp. infinite) places. Let $\mathfrak{a} \subset \mathbb{A}_{\mathbb{K}}$ be an ideal and let $\mathbb{A}_{\mathbb{K}, \square} := \mathbb{A}_{\mathbb{K}, \square} \setminus \{\star \in \mathbb{A}_{\mathbb{K}} : \star \mid \mathfrak{a}\}$ for $\square \in \{\text{empty}, \star, \infty\}$.
- G2 Let $\mathbb{K}/\mathbb{F}$ be a CM extension with fixed CM type Φ and an distinguished element $\star \in \Phi$. Let $\mathbb{H}$ be a hermitian space of signature $(\star - \star, \star)$ at $\star$ and $(\star, \star)$ for $\star \in \Phi \setminus \{\star\}$. Let $\star$ be a place of $\mathbb{F}$. If $\star$ is non-split or archimedean, let $\mathbb{H}^{(\star)}$ be the $\star$ -nearby hermitian space of $\mathbb{H}$. We recall that $\mathbb{H}^{(\star)} \simeq \mathbb{H}$ if $\star \in \Phi$ is the unique archimedean place over $\star$, and otherwise $\mathbb{H}^{(\star)}$ is characterized by the following conditions:
- For all $\star$ other than $\star$ and $\star$, we have $\mathbb{H}^{(\star)} \simeq \mathbb{H}$.
 - $\mathbb{H}^{(\star)}$ is positive definite.
 - $\mathbb{H}^{(\star)}$ has signature $(\star - \star, \star)$ if $\star$ is archimedean and $\star$ is not over $\star$.
- G3 For a CM extension $\mathbb{K}/\mathbb{F}$ with CM type Φ , let $\mathbb{O}^+(\mathbb{K})$ be the set of all totally positive definite matrices of rank $\star$, and $\mathbb{O}^+(\mathbb{K})$ be the set of hermitian matrices with signature $(\star - \star, \star)$ for one $\star \in \Phi$, and positive definite for all $\star' \in \Phi \setminus \{\star\}$.
- G4 Let $\mathbb{K} = \mathbb{K}_{\mathbb{F}}$ and let $\mathbb{K} = \mathbb{K}_{\mathbb{F}}$. We fix the non-trivial additive character $\chi = \chi_{\infty} : \mathbb{K}_{\infty}/\mathbb{K}_{\infty} \rightarrow \mathbb{C}^{\times}$ such that χ is unramified outside $\star$ ($\mathbb{K}/\mathbb{F}$) (the set of finite places of $\mathbb{F}$ split in $\mathbb{K}$). See [LZ22a, §12.2].
- G5 We let $\mathbb{G} = \mathbb{G}_{\mathbb{F}}$ as an algebraic group over $\mathbb{F}$. Let $\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty}), \mathbb{A}_{\mathbb{K}, \infty}, \star)$ be the space of smooth functions on $\mathbb{G}(\mathbb{K}_{\infty})$, invariant under left $\mathbb{G}(\mathbb{F})$ and right $\mathbb{A}_{\mathbb{K}, \infty} \subset \mathbb{G}(\mathbb{K}_{\infty})$ translation, and of parallel weight $\star$ under the action of $\prod_{\star \in \Phi} \mathbb{G}(\mathbb{K}_{\star})$ (we do not impose any finiteness condition under the center of the universal enveloping algebra).
- G6 In this paper, we will restrict to the classical modular form notation in the *hermitian upper half space*

$$\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty})) = \{f = f + f \in \mathcal{S}_{\infty} : \frac{\star}{\star}(f - \bar{f}) > \star\}, \quad \text{where } \mathbb{K}_{\infty} := \mathbb{K} \otimes_{\mathbb{F}} \mathbb{C} \cong \mathbb{C}.$$

For any $\star \in \mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty}), \mathbb{A}_{\mathbb{K}, \infty}, \star)$, define the classical modular function

$$\star(a) : \mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty})) \longrightarrow \mathbb{C}, \quad a \longmapsto \mathbb{A}_{\mathbb{K}, \infty}(\star(a))^{-\star} \mathbb{O}^+(\mathbb{K})^{-\star/\star} \cdot \star(a),$$

where $a = f + f$ and $\star \in \mathbb{K}_{\infty}^{\times}$ such that $\star = \star$, and

$$\star = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix} \cdot \begin{pmatrix} \star & \\ & \star \end{pmatrix} \cdot \star \in \mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty})).$$

Thanks to the strong approximation theorem $\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty})) = \mathcal{S}_{\infty}(\mathbb{G}(\mathbb{F}))\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty}), \mathbb{A}_{\mathbb{K}, \infty}, \star)$, we will not lose information. The Fourier expansion becomes

$$\star(a) = \sum_{\star \in \mathbb{K}_{\infty}} \star(\star) \star, \quad \star = \chi_{\infty}(\star(a)).$$

- G7 $\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty}), \mathbb{A}_{\mathbb{K}, \infty}, \star)$: the space of automorphic forms (with moderate growth) on $\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty}))$, invariant under $\mathbb{A}_{\mathbb{K}, \infty} \subset \mathbb{G}(\mathbb{K}_{\infty})$, of parallel weight $\star$ under the action of $\prod_{\star \in \Phi} \mathbb{G}(\mathbb{K}_{\star})$, and holomorphic (i.e., annihilated by the element $\frac{\star}{\star}(\star - \star)$ in the complexified Lie algebra of $\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty})) = \mathcal{S}_{\infty}(\mathbb{G}(\mathbb{F}))$ for every $\star \in \mathbb{K}_{\infty}$). This is a finite dimensional vector space over $\mathbb{C}$, and it has a $\mathbb{C}$ -structure via Fourier expansion. For any subfield $\mathbb{H} \subset \mathbb{C}$ (for our application, it suffices to assume $\mathbb{H} \supset \mathbb{Q}$), we define $\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty}), \mathbb{A}_{\mathbb{K}, \infty}, \star)_{\mathbb{H}}$ as the subspace of $\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty}), \mathbb{A}_{\mathbb{K}, \infty}, \star)$ consisting of functions with Fourier coefficients all in $\mathbb{H}$. Then for any $\mathbb{H}$ -vector space $\mathbb{H}$, we have an $\mathbb{H}$ -vector space $\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty}), \mathbb{A}_{\mathbb{K}, \infty}, \star)_{\mathbb{H}} \otimes_{\mathbb{H}} \mathbb{H}$. Since the differential operator commutes with the right $\mathcal{S}_{\infty}(\mathbb{G}(\mathbb{K}_{\infty}))$ -action, by strong approximation, we may verify the holomorphicity when $\star = \star$.

Part 1. Local Theory

2. RAPOPORT-ZINK SPACES AND SPECIAL CYCLES

In this section, we define unramified basic relative Rapoport-Zink spaces with maximal parahoric level, and weighted special cycles on them. Throughout the section, we fix $\mathbb{F}/\mathbb{F}_0$, a unramified quadratic extension of $\mathbb{F}_0$ -adic fields ($\mathbb{F} \neq \mathbb{F}_0$) with uniformizer $\varpi \in \mathbb{F}_0$. We assume that the residue field of $\mathbb{F}_0$ has cardinality q .

2.1. Strict $\mathbb{F}_0$ -modules. We begin with a review of $\mathbb{F}_0$ -strict modules. For a comprehensive treatment, we refer to [Mih22, KRZ24, LMZ25].

For an $\mathbb{F}_0$ -scheme S , a *strict $\mathbb{F}_0$ -module* over S is a pair $(\mathcal{A}, \ast)$ where $\mathcal{A}$ is a $\mathbb{F}_0$ -divisible group over S and $\ast : \mathbb{F}_0^\times \rightarrow \text{Aut}(\mathcal{A})$ is an action such that $\mathbb{F}_0^\times$ acts on $\mathcal{A}$ via the structure morphism $\mathbb{F}_0 \rightarrow \mathbb{F}_0$. A strict $\mathbb{F}_0$ -module $(\mathcal{A}, \ast)$ is called *formal* if its underlying $\mathbb{F}_0$ -divisible group $\mathcal{A}$ is formal. The dimension of a strict $\mathbb{F}_0$ -module is the dimension of its underlying $\mathbb{F}_0$ -divisible group. For the notion of (relative) *height* of strict $\mathbb{F}_0$ -modules, we direct the reader to the aforementioned references.

Let $\mathcal{H}(\mathcal{A})$ be the (relative) de Rham homology of $\mathcal{A}$, which is a locally free $\mathbb{F}_0$ -module whose rank equals the height of $\mathcal{A}$. It fits into a canonical short exact sequence of $\mathbb{F}_0$ -modules:

$$0 \rightarrow \mathcal{H}(\mathcal{A}) \rightarrow \mathcal{H}(\mathcal{A}) \rightarrow \mathcal{H}(\mathcal{A}) \rightarrow 0.$$

We now specialize to the case where $\mathcal{A} = (\mathcal{A}, \ast)$ is *biformal*, a notion introduced in [Mih22, Definition 11.9]. For biformal strict $\mathbb{F}_0$ -modules, we can define the *relative dual* $\mathcal{A}^\vee$ and consequently the notion of *relative polarization* (see [Mih22, Definition 11.9]). The notation of relative dual defined here identifies with the one defined in [Fal02], see [LMZ25, Appendix].

2.2. Rapoport-Zink space. For any $\mathbb{F}_0$ -scheme S , a *hermitian $\mathbb{F}_0$ -module of dimension n and type $\mathfrak{b}$* (and signature $(n - \mathfrak{b}, \mathfrak{b})$) over S is a triple $(\mathcal{A}, \ast, \eta)$, where:

- $\mathcal{A}$ is a strict biformal $\mathbb{F}_0$ -module over S of relative height n and dimension n .
- $\ast : \mathbb{F}_0^\times \rightarrow \text{Aut}(\mathcal{A})$ is an action of $\mathbb{F}_0^\times$ on $\mathcal{A}$ that extends the action of $\mathbb{F}_0^\times$. We require that the *Kottwitz signature condition* of signature $(n - \mathfrak{b}, \mathfrak{b})$ holds for all $\mathfrak{b} \in \mathbb{F}_0^\times$:

$$\text{sgn}(\ast(\mathfrak{b}) | \mathcal{H}(\mathcal{A})) = (\ast - \mathfrak{b})^{n - \mathfrak{b}} (\ast - \overline{\mathfrak{b}}) \in \mathbb{F}_0^\times.$$

- $\eta : \mathcal{A} \rightarrow \mathcal{A}^\vee$ is a polarization on $\mathcal{A}$ as strict $\mathbb{F}_0$ -module, which is $\mathbb{F}_0/\mathbb{F}_0$ semi-linear in the sense that the Rosati involution $\eta^\ast$ induces the non-trivial involution $\overline{(\quad)} \in \text{Aut}(\mathbb{F}_0/\mathbb{F}_0)$ on $\ast : \mathbb{F}_0^\times \rightarrow \text{Aut}(\mathcal{A})$.
- We require that the finite flat group scheme $\mathcal{G}$ over S is contained in $\mathcal{A}[\mathfrak{p}]$ and is of order $\mathfrak{p}^{\mathfrak{b}}$.

An isomorphism $(\mathcal{A}_1, \ast_1, \eta_1) \xrightarrow{\sim} (\mathcal{A}_2, \ast_2, \eta_2)$ between two such triples is an $\mathbb{F}_0$ -linear isomorphism $\mathfrak{f} : \mathcal{A}_1 \xrightarrow{\sim} \mathcal{A}_2$ such that $\mathfrak{f}^\ast(\ast_2) = \ast_1$.

Up to $\mathbb{F}_0$ -linear quasi-isogeny compatible with the polarization, there exists a unique hermitian $\mathbb{F}_0$ -module $(\mathcal{A}^{[b]}, \ast^{[b]}, \eta^{[b]})$ over $S = \overline{S}$ of dimension n and type $\mathfrak{b}$, within the supersingular quasi-isogeny class. Fix one choice of $(\mathcal{A}^{[b]}, \ast^{[b]}, \eta^{[b]})$ as the *(basic) framing object*.

Definition 2.2.1. The (relative) Rapoport-Zink space of dimension n and type $\mathfrak{b}$ is the functor

$$\mathcal{RZ}^{[b]} \longrightarrow \mathbb{F}_0$$

sending S to the set of isomorphism classes of tuples $(\mathcal{A}, \ast, \eta, \mathcal{G})$, where:

- $(\mathcal{A}, \ast, \eta)$ is a hermitian $\mathbb{F}_0$ -module of dimension n and type $\mathfrak{b}$ over S .
- $\mathcal{G} : \mathcal{A} \times_S \mathbb{F}_0 \rightarrow \mathcal{A}^{[b]} \times_S \mathbb{F}_0$ an $\mathbb{F}_0$ -linear quasi-isogeny of height $\mathfrak{b}$ over the reduction $\overline{S} := S \times_{\mathbb{F}_0} \mathbb{F}_0$ such that $\mathfrak{G}^\ast(\ast^{[b]}, \eta^{[b]}) = \ast, \eta$.

By [RZ96, Theorem 2.16], the functor $\mathcal{RZ}^{[b]}$ is representable by a formal scheme that is locally formally of finite type over $\mathbb{F}_0$ dimension n .

2.3. Special cycles. Let $\mathcal{C}$ be a formal $\mathcal{O}_K$ -module over $\mathcal{O}_K$ of relative height h and dimension d . We can complete it into a triple $(\mathcal{C}, \mathcal{C}^\vee, \mathcal{C}^\vee)$ of *signature* $(\mathfrak{f}, \mathfrak{g})$, dimension d and type $\mathfrak{g}$ over $\mathcal{O}_K$, which we still denote by $\mathcal{C}$. By modifying the $\mathcal{O}_K$ -action on $\mathcal{C}$ via the Galois involution $\overline{(-)}$, we obtain a framing object $(\mathcal{C}, \mathcal{C}^\vee \circ \mathfrak{g}, \mathcal{C}^\vee)$ for $\mathcal{C}^{[\mathfrak{g}]}$, which we denote by $\mathcal{C}^\flat$. The theory of canonical liftings [Gro86] yields an isomorphism $\mathcal{C}^{[\mathfrak{g}]} \simeq \mathcal{C}^\flat \times_{\mathcal{O}_K} \mathcal{O}_K$, where the universal triple $\mathcal{C}^\flat$ over $\mathcal{C}^{[\mathfrak{g}]} \simeq \mathcal{C}^\flat \times_{\mathcal{O}_K} \mathcal{O}_K$ is the canonical lifting of $\mathcal{C}$.

Let the space of special quasi-homomorphisms be the $\mathcal{O}_K$ -vector space

$$\mathcal{S} = \mathcal{S}^{[\mathfrak{g}]} = \mathcal{S}^\circ(\mathcal{C}, \mathcal{C}^{[\mathfrak{g}]}) := \mathcal{S}^\circ(\mathcal{C}, \mathcal{C}^\flat) \otimes \mathcal{O}_K \quad (2.3.1)$$

equipped with the hermitian form such that for any $\mathfrak{f}, \mathfrak{g} \in \mathcal{S}$

$$(\mathfrak{f}, \mathfrak{g})_{\mathcal{S}} := \mathfrak{f}^\vee \circ \mathfrak{g}^\vee \circ \mathfrak{g} \circ \mathfrak{f} \in \mathcal{S}^\circ(\mathcal{C}, \mathcal{C}) \cong \mathcal{O}_K. \quad (2.3.2)$$

The automorphism group of quasi-isogenies $\mathcal{S}^\circ(\mathcal{C}, \mathcal{C})$ acts on $\mathcal{S}$ naturally. By Dieudonne theory, we have $\mathcal{S}^\circ(\mathcal{C}, \mathcal{C}) \cong \mathcal{V}(\mathcal{C})(\mathfrak{g})$. The group $\mathcal{V}(\mathcal{C})(\mathfrak{g})$ acts on $\mathcal{C}^{[\mathfrak{g}]}$ by

$$\mathfrak{g} \cdot (\mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}) = (\mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i} \circ \mathfrak{g}). \quad (2.3.3)$$

Definition 2.3.1. For any $\mathfrak{f} \in \mathcal{S}^{[\mathfrak{g}]}$, the *Kudla–Rapoport special divisor* $\mathcal{S}(\mathfrak{f}) \subseteq \mathcal{C}^{[\mathfrak{g}]}$ is the closed formal subscheme of $\mathcal{C}^{[\mathfrak{g}]}$ characterized as follows: for each $\mathcal{O}_K$ -scheme $\mathcal{S}$, the set $\mathcal{S}(\mathfrak{f})(\mathcal{S})$ consists of those points $(\mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}) \in \mathcal{C}^{[\mathfrak{g}]}(\mathcal{S})$ for which the quasi-homomorphism

$$\mathfrak{f}^\vee \circ \mathfrak{f} : \mathcal{C} \times_{\mathcal{O}_K} \mathcal{S} \longrightarrow \mathcal{C} \times_{\mathcal{O}_K} \mathcal{S}$$

extends to a homomorphism $\mathcal{C} \rightarrow \mathcal{C}$ over $\mathcal{S}$.

Similarly, the *Kudla–Rapoport special divisor* $\mathcal{S}^\vee(\mathfrak{f}) \subseteq \mathcal{C}^{[\mathfrak{g}]}$ is the closed formal subscheme of $\mathcal{C}^{[\mathfrak{g}]}$ characterized as follows: for each $\mathcal{O}_K$ -scheme $\mathcal{S}$, the set $\mathcal{S}^\vee(\mathfrak{f})(\mathcal{S})$ consists of those points $(\mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}) \in \mathcal{C}^{[\mathfrak{g}]}(\mathcal{S})$ for which the quasi-homomorphism

$$\mathfrak{f}^\vee \circ \mathfrak{f} \circ \mathfrak{g} \circ \mathfrak{g}^\vee : \mathcal{C} \times_{\mathcal{O}_K} \mathcal{S} \longrightarrow \mathcal{C}^\vee \times_{\mathcal{O}_K} \mathcal{S}$$

extends to a homomorphism $\mathcal{C} \rightarrow \mathcal{C}^\vee$ over $\mathcal{S}$.

Proposition 2.3.2. *The following properties hold for special divisors:*

- (1) For any nonzero $\mathfrak{f} \in \mathcal{S}$, the functors $\mathcal{S}(\mathfrak{f})$ and $\mathcal{S}^\vee(\mathfrak{f})$ are representable by Cartier divisors on $\mathcal{C}^{[\mathfrak{g}]}$.
- (2) For $\mathfrak{g} \leq \mathfrak{h} \leq \mathfrak{f} - \mathfrak{g}$, if $\mathfrak{g}((\mathfrak{f}, \mathfrak{g})) = \mathfrak{g}$, then $\mathcal{S}(\mathfrak{f}) \cong \mathcal{C}^{[\mathfrak{h}]}$.
- (3) For $\mathfrak{g} \leq \mathfrak{h} \leq \mathfrak{f}$, if $\mathfrak{g}((\mathfrak{f}, \mathfrak{g})) = -\mathfrak{g}$, then $\mathcal{S}^\vee(\mathfrak{f}) \cong \mathcal{C}^{[\mathfrak{h}-\mathfrak{g}]}$.
- (4) The following inclusion relations hold:

$$\mathcal{S}(\mathfrak{f}) \subseteq \mathcal{S}^\vee(\mathfrak{f}) \subseteq \mathcal{S}(\mathfrak{f}, \mathfrak{g}).$$

Moreover, for $\mathfrak{g} = \mathfrak{g}$, we have $\mathcal{S}(\mathfrak{f}) = \mathcal{S}^\vee(\mathfrak{f})$, and for $\mathfrak{g} = \mathfrak{f}$, we have $\mathcal{S}^\vee(\mathfrak{f}) = \mathcal{S}(\mathfrak{f}, \mathfrak{g})$.

Proof. Part (1) follows from the same argument as in [KR11, Proposition 5.9]. Parts (2) and (3) are established in [Cho18, Proposition 5.10]. Part (4) follows directly from the definition. $\square$

Next, we introduce weighted special cycles.

Definition 2.3.3. Let $\mathfrak{g} \leq \mathfrak{h} \leq \mathfrak{f}$ be an integer, and let $\mathcal{C}$ be the “nearby” hermitian space associated with $\mathcal{S} = \mathcal{S}^{[\mathfrak{g}]}$, i.e., the unique hermitian space of dimension $\mathfrak{f}$ containing a vertex lattice $\mathfrak{L}$ of type $\mathfrak{g}$.

Recall from (L3) that a Schwartz function $\mathfrak{w} \in \mathcal{W}(\mathcal{C})$ is *admissible of type $\mathfrak{g}$* if $\mathfrak{w} = \bigotimes_{\mathfrak{p}} \mathfrak{w}_{\mathfrak{p}}$, where each $\mathfrak{w}_{\mathfrak{p}}$ is either $\mathfrak{w}_{\mathfrak{p}, [\mathfrak{g}]}$ or $\mathfrak{w}_{\mathfrak{p}, [\mathfrak{g}], \#}$, where $\mathfrak{L}^{[\mathfrak{g}]}$ is a vertex lattice of type $\mathfrak{g}$.

- (1) For any nonzero $\mathfrak{f} \in \mathcal{S}$, we define

$$\mathcal{S}(\mathfrak{f}, \mathfrak{w}_{\mathfrak{p}, [\mathfrak{g}]}) := \mathcal{S}(\mathfrak{f}), \quad \text{and} \quad \mathcal{S}(\mathfrak{f}, \mathfrak{w}_{\mathfrak{p}, [\mathfrak{g}], \#}) := \mathcal{S}^\vee(\mathfrak{f}).$$

- (2) For nonzero vectors $\mathfrak{v} = (\mathfrak{v}_1, \dots, \mathfrak{v}_r) \in \mathcal{S}^{[\mathfrak{g}]}$ and an admissible Schwartz function $\mathfrak{w} \in \mathcal{W}(\mathcal{C})$ of type $\mathfrak{g}$, the *weighted special cycle* is defined as the scheme-theoretic intersection:

$$\mathcal{S}(\mathfrak{v}, \mathfrak{w}) := \mathcal{S}(\mathfrak{v}_1, \mathfrak{w}) \cap \dots \cap \mathcal{S}(\mathfrak{v}_r, \mathfrak{w}).$$

(3) The *derived weighted special cycle* $\mathbb{S}(\mathfrak{p}, \mathfrak{q})$ is defined as the image of

$$\mathfrak{p}(\mathfrak{q}, \mathfrak{q}) \otimes \mathbb{S} \dots \otimes \mathbb{S} \mathfrak{p}(\mathfrak{q}, \mathfrak{q})$$

in the $\mathfrak{q}$ -th graded piece $\mathfrak{p}(\mathfrak{q}, \mathfrak{q})^{[\mathfrak{q}]}$.

(4) When $\mathfrak{p} = \mathfrak{q}$, the *weighted arithmetic intersection number* is defined as

$$\mathbb{I}(\mathfrak{p}, \mathfrak{q}) := \mathbb{I}(\mathfrak{p}^{[\mathfrak{q}]}, \mathfrak{p}(\mathfrak{q}, \mathfrak{q})),$$

where $\mathbb{I}$ denotes the Euler–Poincaré characteristic.

Example 2.3.4. If $\mathfrak{p} = \mathfrak{q}$, then by the theory of canonical liftings [Gro86], we have:

(1) For any $\mathfrak{p}$ in the non-split hermitian space $\mathfrak{p}^{[\mathfrak{q}]}$,

$$\mathbb{I}(\mathfrak{p}, \mathfrak{p}^{[\mathfrak{q}]}) = \begin{cases} \frac{\star(\mathfrak{p}, \mathfrak{p}) + \mathfrak{q}}{\mathfrak{q}}, & \text{if } \star(\mathfrak{p}, \mathfrak{p}) \geq \mathfrak{q}; \\ \mathfrak{q}, & \text{otherwise.} \end{cases}$$

(2) For any $\mathfrak{p}$ in the split hermitian space $\mathfrak{p}^{[\mathfrak{q}]}$,

$$\mathbb{I}(\mathfrak{p}, \mathfrak{p}^{[\mathfrak{q}]}) = \begin{cases} \frac{\star(\mathfrak{p}, \mathfrak{p})}{\mathfrak{q}}, & \text{if } \star(\mathfrak{p}, \mathfrak{p}) \geq \mathfrak{q}; \\ \mathfrak{q}, & \text{otherwise.} \end{cases}$$

This follows from the observation that, after identifying the hermitian modules $\mathfrak{p}^{[\mathfrak{q}]}$ and $\mathfrak{p}^{[\mathfrak{q}], \vee}$ with $\mathfrak{q}$, the polarization $\mathfrak{p}^{[\mathfrak{q}]} \rightarrow \mathfrak{p}^{[\mathfrak{q}], \vee}$ corresponds to multiplication by $[\mathfrak{q}] : \mathfrak{q} \rightarrow \mathfrak{q}$.

(3) For any $\mathfrak{p}$ in the split hermitian space $\mathfrak{p}^{[\mathfrak{q}]}$,

$$\mathbb{I}(\mathfrak{p}, \mathfrak{p}^{[\mathfrak{q}]}_{\mathfrak{q}}) = \begin{cases} \frac{\star(\mathfrak{p}, \mathfrak{p})}{\mathfrak{q}} + \mathfrak{q}, & \text{if } \star(\mathfrak{p}, \mathfrak{p}) \geq -\mathfrak{q}; \\ \mathfrak{q}, & \text{otherwise.} \end{cases}$$

This follows from (2) and Proposition 2.3.2(4).

By Proposition 2.3.2(2), (3), and (4), we obtain the following geometric cancellation law.

Proposition 2.3.5 ([Cho18, Proposition 5.10]). *Let $\mathfrak{p}^{[\mathfrak{q}]} = \mathfrak{p}^{[\mathfrak{q}]} \in \mathfrak{p}(\mathfrak{q}, \mathfrak{q})$ be an admissible Schwartz function of weight $\mathfrak{q} \in \{\mathfrak{p}, \mathfrak{q}\}^{\mathfrak{q}}$ (L3), and let $\mathfrak{p} = (\mathfrak{p}_1, \dots, \mathfrak{p}_r) \in \mathfrak{p}^{\mathfrak{q}}$ a collection of linearly independent vectors. For any fixed $\mathfrak{q} \leq \mathfrak{p} \leq \mathfrak{q}$, define:*

- $\mathfrak{p}_b := \langle \mathfrak{p}_b \rangle^\perp$ and set $\mathfrak{p}_{b,b}$ being the orthogonal projection of $\mathfrak{p}_b$ for $b \neq \mathfrak{p}$.
- $\mathfrak{p}_b := (\mathfrak{p}_{b,b}, \dots, \widehat{\mathfrak{p}_{b,b}}, \dots, \mathfrak{p}_{b,b}) \in \mathfrak{p}_b^{\mathfrak{q}-\mathfrak{p}}$.
- $\mathfrak{p}_b := (\mathfrak{p}_{b,b}, \dots, \widehat{\mathfrak{p}_{b,b}}, \dots, \mathfrak{p}_{b,b}) \in \{\mathfrak{p}, \mathfrak{q}\}^{\mathfrak{q}-\mathfrak{p}}$.

Suppose we are in one of the following situations:

- When $\mathfrak{p}_b = \mathfrak{p}$ and $\star(\mathfrak{p}_b, \mathfrak{p}_b) = \begin{cases} \mathfrak{q} & \text{if } \mathfrak{p}_b \neq \mathfrak{p}; \\ \mathfrak{p} & \text{if } \mathfrak{p}_b = \mathfrak{p}. \end{cases}$
- When $\mathfrak{p}_b = \mathfrak{q}$ and $\star(\mathfrak{p}_b, \mathfrak{p}_b) = \begin{cases} -\mathfrak{q} & \text{if } \mathfrak{p}_b \neq \mathfrak{q}; \\ \mathfrak{q} & \text{if } \mathfrak{p}_b = \mathfrak{q}. \end{cases}$

Then we have $\mathbb{I}(\mathfrak{p}, \mathfrak{p}^{[\mathfrak{q}]}) = \mathbb{I}(\mathfrak{p}_b, \mathfrak{p}_b^{[\mathfrak{q}-\mathfrak{p}]})$. □

The following result plays a crucial role in the globalization procedure.

Theorem 2.3.6 ([Mih22]). *For any collection of linearly independent vectors $\mathfrak{p} \in \mathfrak{p}^{\mathfrak{q}}$ and any admissible Schwartz function $\mathfrak{p} \in \mathfrak{p}(\mathfrak{q}, \mathfrak{q})$ of type $\mathfrak{p}$, the weighted arithmetic intersection number $\mathbb{I}(\mathfrak{p}, \mathfrak{p})$ is locally constant as $\mathfrak{p}$ varies in $\mathfrak{p}^{\mathfrak{q}}$.*

Proof. By [Mih22, Proposition 4.2], the special cycles $\mathfrak{p}(\mathfrak{p})$ and $\mathfrak{q}(\mathfrak{p})$ satisfy the local constancy axiom in *loc. cit.* The result then follows from [Mih22, Theorem 1.2]. □

3. WEIGHTED LOCAL DENSITY FUNCTIONS AND KUDLA-RAPOPORT CONJECTURE

In this section, we study the weighted local density function introduced by [Kud97] and formulate the Kudla–Rapoport conjecture for weighted special cycles. We show that the weighted local density function satisfies the *cancellation law*, mirroring the corresponding geometric cancellation law (Proposition 2.3.5). Finally, we recall local Whittaker functions and the local Siegel–Weil formula, and their relation to the weighted local density function.

3.1. Weighted local density function. Let $\mathbb{F}/\mathbb{F}_0$ be an unramified quadratic extension of $\mathbb{F}_0$ -adic fields, equipped with a fixed unramified additive character ψ . Let $(\mathbb{H}, \mathbb{H}')$ be a hermitian space of dimension $2n$. When no confusion arises, we denote $\mathbb{H}(\mathbb{F}, \mathbb{F}_0)$ simply by $(\mathbb{H}, \mathbb{F}_0)$.

This choice uniquely determines self-dual Haar measures $\mu_{\mathbb{H}}^{\mathbb{F}_0}$ on $\mathbb{H}$ and $\mu_{\mathbb{H}'}^{\mathbb{F}_0}$ on $\mathbb{H}'$. Let $\mu_{\mathbb{H}}^{\mathbb{F}} = \mu_{\mathbb{H}}^{\mathbb{F}_0} \otimes \mu_{\mathbb{H}'}^{\mathbb{F}_0}$ be the induced Haar measure on $\mathbb{H}(\mathbb{F})$ as given in (H5).

Definition 3.1.1. For any natural number $k \geq 0$, define $\mathbb{H}^{(k)} := \mathbb{H} \oplus \mathbb{H}_{k, \mathbb{F}_0}$, where $\mathbb{H}_{k, \mathbb{F}_0}$ is the split hermitian space of dimension $2k$, i.e., a hermitian space of dimension $2k$ with basis $\{e_i, e_i' \mid 1 \leq i \leq k\}$ satisfying $(e_i, e_i') = 1$.

(1) For any Schwartz function $\varphi \in \mathcal{S}(\mathbb{H}^{(k)}(\mathbb{F}))$, define

$$\varphi^{(k)} = \varphi \otimes \chi_{\mathbb{H}_{k, \mathbb{F}_0}} \in \mathcal{S}((\mathbb{H}^{(k)})^{\mathbb{F}}),$$

where $\chi_{\mathbb{H}_{k, \mathbb{F}_0}}$ is the characteristic function of $(\mathbb{H}_{k, \mathbb{F}_0})^{\mathbb{F}}$ for a self-dual lattice $\mathbb{H}_{k, \mathbb{F}_0}$ in $\mathbb{H}_{k, \mathbb{F}_0}$.

(2) For any $\psi \in \mathcal{O}^{\times}(\mathbb{F}_0)$, define the complex-valued *weighted local density function* $\mathcal{D}_{\psi}^k(\varphi, \psi)$ as the integral

$$\mathcal{D}_{\psi}^k(\varphi, \psi) := \int_{\mathcal{O}^{\times}(\mathbb{F}_0)} \int_{(\mathbb{H}^{(k)})^{\mathbb{F}}} \psi\left(\varphi((\mathbb{H}, \mathbb{H}') - \psi)\right) \varphi^{(k)}(\psi) \mu_{\mathbb{H}}^{\mathbb{F}}.$$

This notion was introduced by Kudla [Kud97, Appendix].

When φ is the characteristic function of $\mathbb{H}^{\mathbb{F}}$ for some lattice $\mathbb{H}$, it coincides with the representation density function; see [Kud97, Proposition A.6] or §3.5. The definition is independent of the choice of the self-dual lattice $\mathbb{H}_{k, \mathbb{F}_0}$.

In this paper, we focus on admissible functions $\varphi \in \mathcal{S}(\mathbb{H}^{(k)}(\mathbb{F}))$ as defined in (L3). For clarity, throughout this section, we restrict our attention to Schwartz functions of the form:

$$\varphi^{[a]} := (\chi_{\mathbb{H}^{[a]}})^{\otimes a} \otimes (\chi_{\mathbb{H}^{[a], \mathbb{F}_0}})^{\otimes a}, \quad (3.1.1)$$

where $\mathbb{H}^{[a]}$ is a vertex lattice of type a . The results presented in this section extend naturally to general admissible Schwartz functions.

Recall that in the unramified case, the type of a hermitian lattice uniquely determines the Hasse invariant of the hermitian space. By a direct change of variables, we obtain the following result.

Lemma 3.1.2. *Let φ be an admissible Schwartz function of type a , and let $\psi \in \mathcal{O}^{\times}(\mathbb{F}_0)$. The integral $\mathcal{D}_{\psi}^k(\varphi, \psi)$ is independent of the choice of type a vertex lattices $\mathbb{H}^{[a]}$. $\square$*

3.2. A Fubini-type result. In this subsection, we state and prove a Fubini-type result, which was suggested and proven by T. Yang.

Recall from (H3) that for any $\psi \in \mathcal{O}^{\times}(\mathbb{F}_0)$, the inverse image $\mathbb{H}_{\psi} := \mathbb{H}^{-1}(\psi)$ consisting of vectors $\mathbb{H} \in \mathbb{H}^{\mathbb{F}}$ satisfying $(\mathbb{H}, \mathbb{H}') = \psi$.

Theorem 3.2.1. *Let $\psi \in \mathcal{O}^{\times}(\mathbb{F}_0)$ be a nonsingular matrix. For any admissible Schwartz function $\varphi \in \mathcal{S}(\mathbb{H}^{(k)}(\mathbb{F}))$, we have the equality:*

$$\mathcal{D}_{\psi}^k(\varphi, \psi) = \int_{\mathcal{O}^{\times}(\mathbb{F}_0)} \int_{(\mathbb{H}^{(k)})^{\mathbb{F}}} \psi\left(\varphi((\mathbb{H}, \mathbb{H}') - \psi)\right) \varphi^{(k)}(\psi) \mu_{\mathbb{H}}^{\mathbb{F}} = \int_{\mathbb{H}_{\psi}^{(k)}} \varphi^{(k)}(\psi) \mu_{\mathbb{H}}^{\mathbb{F}}.$$

Proof. The degenerate subspace

$$\{\mathbb{H} \in (\mathbb{H}^{(k)})^{\mathbb{F}} \mid \varphi(\mathbb{H}, \mathbb{H}') = 0\}$$

is Zariski closed and has measure zero. Similarly, the degenerate subspace $\mathcal{O}^{\times}(\mathbb{F}_0) \setminus \mathcal{O}^{\times}(\mathbb{F}_0)$ has measure zero. Therefore, we have

$$\begin{aligned} \mathcal{D}_{\psi}^k(\varphi, \psi) &= \int_{\mathcal{O}^{\times}(\mathbb{F}_0)} \int_{(\mathbb{H}^{(k)})^{\mathbb{F}}} \psi\left(\varphi((\mathbb{H}, \mathbb{H}') - \psi)\right) \varphi^{(k)}(\psi) \mu_{\mathbb{H}}^{\mathbb{F}}, \\ &= \int_{\mathcal{O}^{\times}(\mathbb{F}_0)} \int_{\mathcal{O}^{\times}(\mathbb{F}_0)} \int_{\mathbb{H}_{\psi}^{(k)}} \psi\left(\varphi(\tilde{\psi} - \psi)\right) \varphi^{(k)}(\tilde{\psi}) \mu_{\mathbb{H}}^{\mathbb{F}} \mu_{\mathbb{H}}^{\mathbb{F}}. \end{aligned}$$

For any $\psi \in \mathcal{O}^{\times}(\mathbb{F}_0)$, define

$$\mathcal{D}_{\psi}(\psi) = \mathcal{D}_{\psi}(\psi, \psi) := \int_{\mathbb{H}_{\psi}^{(k)}} \varphi^{(k)}(\psi) \mu_{\mathbb{H}}^{\mathbb{F}} = \int_{\mathbb{H}(\mathbb{F})} \varphi^{(k)}(\psi) \mu_{\mathbb{H}}^{\mathbb{F}},$$

where $\mathfrak{v}_{\mathfrak{g}}$ is any vector in $\mathfrak{g}$, and $\mathfrak{v}_{\mathfrak{g}} = \mathfrak{v}_{\mathfrak{g}} \cdot \mathfrak{v}_{\mathfrak{g}}$.

The function $\mathfrak{f}(\mathfrak{g}, \mathfrak{v})$ is continuous on $\mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g})$ and extends to $\mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g})$ (as a $\mathfrak{H}$ -function or a distribution). Indeed, for any $\tilde{\mathfrak{g}}$ and $\tilde{\mathfrak{v}}$ in a small neighborhood $\mathfrak{U} \subset \mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g})$, the continuity of the map $\mathfrak{f} : (\mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g})) \rightarrow \mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g})$ ensures that we can find $\mathfrak{v}_{\mathfrak{g}}$ and $\mathfrak{v}_{\tilde{\mathfrak{g}}}$ in a small neighborhood of the preimage $\mathfrak{f}^{-1}(\mathfrak{U})$. Since the function

$$\mathfrak{v}_{\mathfrak{g}} \mapsto \int_{\mathfrak{U}(\mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g}))} \mathfrak{f}(\mathfrak{g}, \mathfrak{v}_{\mathfrak{g}})(\mathfrak{v}_{\tilde{\mathfrak{g}}}) \mathfrak{v}_{\tilde{\mathfrak{g}}}$$

is continuous, it follows that $\mathfrak{f}(\mathfrak{g}, \mathfrak{v})$ is continuous.

Since $\mathfrak{f}(\mathfrak{g}, \mathfrak{v})$ has compact support, $\mathfrak{f}(\mathfrak{g}, \mathfrak{v})$ is a Schwartz function on $\mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g})$. We have

$$\begin{aligned} \mathfrak{f}(\mathfrak{g}, \mathfrak{v}) &= \int_{\mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g})} \int_{\mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g})} \mathfrak{f}(\mathfrak{g}, \mathfrak{v}(\tilde{\mathfrak{g}} - \mathfrak{g})) \mathfrak{f}(\tilde{\mathfrak{g}}, \mathfrak{v}_{\tilde{\mathfrak{g}}}) \mathfrak{v}_{\tilde{\mathfrak{g}}}, \\ &= \int_{\mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g})} \mathfrak{f}(\mathfrak{g}, -\mathfrak{g}) \widehat{\mathfrak{f}}(-\mathfrak{g}) \mathfrak{v}_{\mathfrak{g}} = \widehat{\mathfrak{f}}(-\mathfrak{g}) = \mathfrak{f}(\mathfrak{g}, \mathfrak{v}), \end{aligned}$$

where in the last line, $\widehat{\mathfrak{f}}(\mathfrak{g})$ denotes the Fourier transform of $\mathfrak{f}(\mathfrak{g}, \mathfrak{v})$, and the final equality follows from the Fourier inversion formula. The proof is completed by evaluating these integrals. $\square$

3.3. A recursion formula. In this subsection, we establish a recursion formula for weighted local density functions, which will play a crucial role in proving the analytic cancellation law. We begin with a lemma that is well known to experts.

Lemma 3.3.1. *Let $\mathfrak{H} \subset \mathfrak{H} \subset \mathfrak{H}^{\sharp}$ be a vertex lattice, and let $\mathfrak{v} \in \mathfrak{g}$ be a vector. Then the following statements hold:*

(1) *If $\star(\mathfrak{v}, \mathfrak{v}) = \mathfrak{v}$ and $\mathfrak{v} \in \mathfrak{H}$, then*

$$\mathfrak{H} = \langle \mathfrak{v} \rangle \oplus \mathfrak{H}_{\mathfrak{v}} \quad \text{and} \quad \mathfrak{H}^{\sharp} = \langle \mathfrak{v} \rangle \oplus \mathfrak{H}_{\mathfrak{v}}^{\sharp}.$$

(2) *If $\star(\mathfrak{v}, \mathfrak{v}) = -\mathfrak{v}$ and $\mathfrak{v} \in \mathfrak{H}^{\sharp}$, then*

$$\mathfrak{H} = \langle \mathfrak{v} \rangle \oplus \mathfrak{H}_{\mathfrak{v}} \quad \text{and} \quad \mathfrak{H}^{\sharp} = \langle \mathfrak{v} \rangle \oplus \mathfrak{H}_{\mathfrak{v}}^{\sharp}.$$

Proof. The proof follows from the Gram–Schmidt process. For case (1), let $\mathfrak{v} \in \mathfrak{H}$ be any vector. Then we can write

$$\mathfrak{v} = \frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \mathfrak{v} + \left(\mathfrak{v} - \frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \mathfrak{v} \right).$$

Since $\mathfrak{v} \in \mathfrak{H} \subseteq \mathfrak{H}^{\sharp}$ and $\star(\mathfrak{v}, \mathfrak{v}) = \mathfrak{v}$, we have $\frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \in \mathfrak{H}$. Thus, applying the Gram–Schmidt process, we obtain the orthogonal complement $\langle \mathfrak{v} \rangle^{\perp} \subset \mathfrak{H}$. The same argument applies to $\mathfrak{H}^{\sharp}$.

For case (2), one can directly use the dual hermitian space (see [Zha25, after Proposition 5.3]), but we provide a direct proof here. Since $\mathfrak{v} \in \mathfrak{H}^{\sharp}$, it follows that $\mathfrak{v} \in \mathfrak{H}^{\sharp}$. Moreover, since $(\mathfrak{v}, \mathfrak{v})$ has unit length, we conclude that $\mathfrak{v} \in \mathfrak{H}^{\sharp}$ and $\mathfrak{v} \in \mathfrak{H}$ are primitive vectors.

For any $\mathfrak{v} \in \mathfrak{H}$, consider the decomposition

$$\mathfrak{v} = \frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \mathfrak{v} + \left(\mathfrak{v} - \frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \mathfrak{v} \right) = \frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \mathfrak{v} + \left(\mathfrak{v} - \frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \mathfrak{v} \right).$$

Since $\mathfrak{v} \in \mathfrak{H}$ and $\mathfrak{v} \in \mathfrak{H}^{\sharp}$, we have $(\mathfrak{v}, \mathfrak{v}) \in \mathfrak{H}$. Given that $\star(\mathfrak{v}, \mathfrak{v}) = -\mathfrak{v}$, we obtain $\frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \in \mathfrak{H}$, and hence the Gram–Schmidt process applies.

For any $\mathfrak{v} \in \mathfrak{H}^{\sharp}$, consider the decomposition

$$\mathfrak{v} = \frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \mathfrak{v} + \left(\mathfrak{v} - \frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \mathfrak{v} \right) = \frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \mathfrak{v} + \left(\mathfrak{v} - \frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \mathfrak{v} \right).$$

Since $\mathfrak{v} \in \mathfrak{H}$ and $\mathfrak{v} \in \mathfrak{H}^{\sharp}$, we have $(\mathfrak{v}, \mathfrak{v}) \in \mathfrak{H}$. As $\star(\mathfrak{v}, \mathfrak{v}) = -\mathfrak{v}$, we conclude that $\frac{(\mathfrak{v}, \mathfrak{v})}{(\mathfrak{v}, \mathfrak{v})} \in \mathfrak{H}$, so the Gram–Schmidt process applies. $\square$

Theorem 3.3.2. *Fix $\mathfrak{v} + \mathfrak{v} = \mathfrak{v}$. Let $\mathfrak{g} \in \mathfrak{O} \setminus \mathfrak{v}(\mathfrak{g})$ be a hermitian matrix such that the weighted local density function $\mathfrak{f}(\mathfrak{g}, \mathfrak{v}^{[.]})$, where $\mathfrak{v}^{[.]}$ is defined in (3.1.1), is not identically zero.*

(1) Define

$$\star_b := \star - \overline{\star} \quad \text{where} \quad \star = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix},$$

where $\star$ is an $(\star - \star) \times (\star - \star)$ matrix. Suppose $\star \neq \star$ and $\star(\star) = \star$. Then

$$\star(\star, \star) = \star(\star, \star) \star(\star, \star),$$

where $\star \in \star(\star_b)$ is a Schwartz function in the hermitian space $\star_b$ of dimension $\star - \star$. The space $\star_b$ is split (resp. non-split) if $\star$ is split (resp. non-split).

(2) Define

$$\star_b := \star - \overline{\star} \quad \text{where} \quad \star = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix},$$

where $\star$ is an $(\star - \star) \times (\star - \star)$ matrix. Suppose $\star \neq \star$ and $\star(\star) = -\star$. Then

$$\star(\star, \star) = \star(\star, \star) \star(\star, \star),$$

where $\star \in \star(\star_b)$ is a Schwartz function in the hermitian space $\star_b$ of dimension $\star - \star$. The space $\star_b$ is split (resp. non-split) if $\star$ is non-split (resp. split).

Proof. We first consider the case where $\star \in \star(\star)$ is of the form

$$\star = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix},$$

where $\star \in \star(\star)$ is an $(\star - \star) \times (\star - \star)$ matrix. It is straightforward to verify that:

$$\begin{aligned} \star(\star, \star) &= \int_{\star} \star(\star, \star) = \int_{\star} (\star(\star, \star))^{\otimes \star} \otimes (\star(\star, \star))^{\otimes \star} \\ &= \int_{\star \in \langle \star \rangle} \star(\star, \star) \int_{\star \in \langle \star \rangle^\perp} (\star(\star, \star))^{\otimes \star} \otimes (\star(\star, \star))^{\otimes \star} \\ &= \star(\star, \star) \star(\star, \star). \end{aligned}$$

The last equality follows from Lemma 3.1.2. A similar computation applies to case (2).

Now we prove the general case. For case (1), by assumption, we can find infinitely many values of $\star$ such that $\star(\star, \star) \neq \star$. Fix one such $\star$, then there exists a collection of vectors $\star = (\star) \in \star(\star)$ such that $\star(\star, \star) = \star$.

Define

$$\star = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix} \in \star(\star),$$

so that

$$\overline{\star} = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix} = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix}.$$

We claim that $\star(\star, \star) = \star(\star, \star)$ for any $\star \in \star(\star)$. Checking each factor, we reduce to showing that for $\star = \star$ or $\star = \star$ and $\star = (\star) \in \star$, the map

$$\star \mapsto \star, \quad \star \mapsto \star - \frac{(\star, \star)}{(\star, \star)} \star, \quad \star \in \{\star, \dots, \star\}$$

are well-defined and are automorphisms of $\star$. In fact, we only need to prove that the vector

$$\frac{(\star, \star)}{(\star, \star)} \star \in \star, \quad \star \in \{\star, \dots, \star\}$$

which follows from the proof of Lemma 3.3.1. A similar argument holds for case (2).

Thus, substituting $\star$ by $\star$, and noting that $\star(\star) = \star$, we obtain

$$\star(\star, \star) = \int_{\star} \star(\star) \star = \int_{\star} \star(\star) \star.$$

Theorem 3.3.2 now follows from the diagonal case. $\square$

3.4. Kudla–Rapoport conjecture. We now formulate the Kudla–Rapoport conjecture for mixed special cycles in $\mathcal{M}_g^{[h]}$. For $\mathfrak{a} + \mathfrak{b} = \mathfrak{c}$, we write

$$\mathfrak{a}^{[h]} := \begin{cases} \text{moon}(\text{apple}(\mathfrak{a}-\mathfrak{b}), \text{orange}(\mathfrak{b}-\mathfrak{a}), (\text{orange}-\text{apple})^{\mathfrak{a}}) & \text{if } \mathfrak{b} \geq \mathfrak{a}, \\ \text{moon}(\text{apple}(\mathfrak{a}-\mathfrak{b}), (\text{orange}-\text{apple})^{\mathfrak{b}}) & \text{if } \mathfrak{b} < \mathfrak{a}. \end{cases}$$

For any admissible Schwartz function $\mathfrak{w}^{[h]} \in \mathfrak{W}(\mathfrak{a}, \mathfrak{b})$, it is known that there exists a polynomial $\mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[h]}) \in \mathbb{Z}[\mathfrak{a}]$ such that for each integer $\mathfrak{c} \geq \mathfrak{a}$,

$$\mathfrak{g}(\mathfrak{c}, \mathfrak{w}^{[h]}) = \mathfrak{g}(\mathfrak{a}, ((-\mathfrak{c})^{-\mathfrak{a}})^{\mathfrak{b}}, \mathfrak{w}^{[h]}).$$

Let $\mathfrak{g} = (\mathfrak{g}_1, \mathfrak{g}_2)$ be the moment matrix of a vector $\mathfrak{v} \in \mathfrak{V}^{\mathfrak{a}}$. Since $\mathfrak{g}$ is not represented in $\mathfrak{V}$, we have $\mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[h]}) = \mathfrak{a}$. In this case, we define the derivative of the weighted local density function as

$$\mathfrak{g}'(\mathfrak{a}, \mathfrak{w}^{[h]}) := -\frac{\mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[h]})}{\mathfrak{a}} \Big|_{\mathfrak{a}=\mathfrak{a}}. \quad (3.4.1)$$

Recall that $\mathfrak{V}$ is the unique $\mathfrak{g}/\mathfrak{g}$ -hermitian space of dimension $\mathfrak{a}$ that is not isomorphic to $\mathfrak{V}$.

Proposition 3.4.1. *For any $\mathfrak{a}$ and $\mathfrak{a} \leq \mathfrak{b} \leq \mathfrak{c}$, let $\mathfrak{a}, \mathfrak{b}$ be fixed such that $\mathfrak{a} + \mathfrak{b} = \mathfrak{c}$. Let $\mathfrak{w}^{[h]} \in \mathfrak{W}(\mathfrak{a}, \mathfrak{b})$ be an admissible Schwartz function of type $\mathfrak{b}$. For $\mathfrak{a} \leq \mathfrak{c} < \mathfrak{b}$, $\mathfrak{c} \equiv \mathfrak{b} - \mathfrak{a} \pmod{\mathfrak{c}}$, let $\mathfrak{w}^{[\mathfrak{c}]} \in \mathfrak{W}(\mathfrak{a}, \mathfrak{c})$ be admissible functions matching with $\mathfrak{w}^{[h]}$, see (L4). Then there exist unique constants $\mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[\mathfrak{c}]})$ for $\mathfrak{a} \leq \mathfrak{c} < \mathfrak{b}$ such that*

$$\mathfrak{g}'(\mathfrak{a}, \mathfrak{w}^{[h]}) - \sum_{\substack{\mathfrak{a} \leq \mathfrak{c} < \mathfrak{b} \\ \mathfrak{c} \equiv \mathfrak{b} - \mathfrak{a} \pmod{\mathfrak{c}}}} \mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[\mathfrak{c}]}) \mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[\mathfrak{c}]}) = \mathfrak{a} \quad (3.4.2)$$

for all $\mathfrak{g} = \mathfrak{g}^{[\mathfrak{c}]}$ such that $\mathfrak{a} \leq \mathfrak{b}$ and $\mathfrak{a} \equiv \mathfrak{b} - \mathfrak{a} \pmod{\mathfrak{c}}$.

Proof. By definition or Theorem 3.3.2, we observe that for any $\mathfrak{a} < \mathfrak{c} < \mathfrak{b}$ with $\mathfrak{c} \equiv \mathfrak{b} - \mathfrak{a} \pmod{\mathfrak{c}}$ ($\mathfrak{a} \equiv \mathfrak{b} - \mathfrak{a} \pmod{\mathfrak{c}}$), we have

$$\mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[\mathfrak{c}]}) = \mathfrak{a}.$$

This implies that equation (3.4.2) forms a linear system with an upper triangular coefficient matrix.

By direct computation or by applying Lemma 3.4.5, we see that the diagonal entries of the coefficient matrix are nonzero. Since an upper triangular matrix with nonzero diagonal entries is invertible, the corresponding linear system has a unique solution. $\square$

Definition 3.4.2. For any weight vector $\mathfrak{g}$ of rank $\mathfrak{a}$, let $\mathfrak{a}$ be the number of $\mathfrak{a}$ -entries in $\mathfrak{g}$ and $\mathfrak{b}$ the number of $\mathfrak{b}$ -entries in $\mathfrak{g}$. Define $\mathfrak{g}^{[h]} := \mathfrak{g}(\mathfrak{a}, \mathfrak{b})$.

Corollary 3.4.3. *For any weight vector $\mathfrak{g}$ of rank $\mathfrak{a}$, let $\mathfrak{a} \in \mathfrak{V}$ be a fixed permutation such that $\mathfrak{a} \cdot \mathfrak{g} = \mathfrak{g}^{[\mathfrak{a}]}$, $\mathfrak{a} := (\mathfrak{a}_1, \mathfrak{a}_2)$ for some $\mathfrak{a} + \mathfrak{b} = \mathfrak{c}$. Let $\mathfrak{g}^{[h]} \in \mathfrak{W}(\mathfrak{a}, \mathfrak{b})$ be as in Definition 3.4.2, and set $\mathfrak{g}^{[\mathfrak{c}]} := \mathfrak{a} \cdot \mathfrak{g}^{[h]} \mathfrak{a}^{-1}$, where we view $\mathfrak{a} \in \mathfrak{V} = \mathfrak{V}(\mathfrak{a})/\mathfrak{g}$ as an element in the Weyl group⁵. Then, we have*

$$\mathfrak{g}'(\mathfrak{a}, \mathfrak{w}^{[h]}) - \sum_{\substack{\mathfrak{a} \leq \mathfrak{c} < \mathfrak{b} \\ \mathfrak{c} \equiv \mathfrak{b} - \mathfrak{a} \pmod{\mathfrak{c}}}} \mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[\mathfrak{c}]}) \mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[\mathfrak{c}]}) = \mathfrak{a}$$

for all $\mathfrak{g} = \mathfrak{g}^{[\mathfrak{c}]}$ such that $\mathfrak{a} \leq \mathfrak{b}$ and $\mathfrak{a} \equiv \mathfrak{b} - \mathfrak{a} \pmod{\mathfrak{c}}$. $\square$

We can now state the main theorem in the $\mathfrak{a}$ -adic setting.

Theorem 3.4.4 (Kudla–Rapoport conjecture for unramified maximal parahorics). *Let $\mathfrak{v} \in \mathfrak{V}^{\mathfrak{a}}$ be a collection of vectors that are linearly independent, and let $\mathfrak{g} = (\mathfrak{g}_1, \mathfrak{g}_2) \in \mathfrak{V}(\mathfrak{a})$ be its moment matrix. Let $\mathfrak{w}^{[h]} = \mathfrak{w}^{[h]} \in \mathfrak{W}(\mathfrak{a}, \mathfrak{b})$ be an admissible Schwartz function, and let $\mathfrak{w}^{[\mathfrak{c}]} \in \mathfrak{W}(\mathfrak{a}, \mathfrak{c})$ be the corresponding matched admissible Schwartz functions of type $\mathfrak{c}$ for $\mathfrak{a} < \mathfrak{b}$ and $\mathfrak{c} \equiv \mathfrak{b} - \mathfrak{a} \pmod{\mathfrak{c}}$ ($\mathfrak{a} \equiv \mathfrak{b} - \mathfrak{a} \pmod{\mathfrak{c}}$), see (L4). Then, we have:*

$$\mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[h]}) = \frac{\mathfrak{a}}{\mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[h]})} \left(\mathfrak{g}'(\mathfrak{a}, \mathfrak{w}^{[h]}) - \sum_{\substack{\mathfrak{a} \leq \mathfrak{c} < \mathfrak{b} \\ \mathfrak{c} \equiv \mathfrak{b} - \mathfrak{a} \pmod{\mathfrak{c}}}} \mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[\mathfrak{c}]}) \mathfrak{g}(\mathfrak{a}, \mathfrak{w}^{[\mathfrak{c}]}) \right).$$

⁵ $\mathfrak{g}^{[\mathfrak{c}]}$ depends on the choice of $\mathfrak{a} \in \mathfrak{V}$.

Here, $\mathcal{H}^{[b]} = \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) \subset \mathcal{H}(\mathcal{H})$ is the parahoric subgroup that defines the level structure, and the measure $\mathcal{H}^{[b]} = \mathcal{H}^{[b]}(\mathcal{H}^{[b]})$ is defined at the beginning of §3.1. We denote the right-hand side by $\partial \mathcal{H}^{[b]}(\mathcal{H}^{[b]})$ or $\partial \mathcal{H}^{[b]}(\mathcal{H}^{[b]})$.

We will prove Theorem 3.4.4 in §11. Next, we reformulate the analytic side of Theorem 3.4.4 to establish connections with previous works [LZ22a, CHZ23] and to prove the analytic cancellation law.

Lemma 3.4.5. *For any $\mathfrak{b} + \mathfrak{c} = \mathfrak{d}$, we have*

$$\mathcal{H}^{[b]}(\mathcal{H}^{[b]}) = \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) \cdot \mathcal{H}^{[c]}(\mathcal{H}^{[c]}).$$

Proof. We prove the case when $\mathfrak{b} \geq \mathfrak{c}$ and leave the case $\mathfrak{b} < \mathfrak{c}$ to the reader. Let $\mathcal{H}^{[b]} = (\mathcal{H}^{[b]}, \dots, \mathcal{H}^{[b]}) \in \mathcal{H}^{[b]} \oplus \mathcal{H}^{[b]}$ be a collection of vectors with moment matrix

$$(\mathcal{H}^{[b]}, \mathcal{H}^{[b]}) = \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) = \begin{pmatrix} \mathcal{H}^{[b]} & & \\ & \mathcal{H}^{[b]} & \\ & & \mathcal{H}^{[b]} \end{pmatrix}.$$

By Lemma 3.3.1(2), we have $\mathcal{H}^{[b]} + \mathcal{H}^{[b]} \in \mathcal{H}^{[b]}$, hence

$$\mathcal{H}^{[b]} := (\mathcal{H}^{[b]}, \dots, \mathcal{H}^{[b]}, \mathcal{H}^{[b]} + \mathcal{H}^{[b]}, \dots, \mathcal{H}^{[b]} + \mathcal{H}^{[b]}) \in \mathcal{H}^{[b]}.$$

Furthermore, $\mathcal{H}^{[b]}$ forms a basis of $\mathcal{H}^{[b]}$ as it spans a type $\mathfrak{b}$ sublattice in $\mathcal{H}^{[b]}$. By Theorem 3.2.1, we can rewrite the integral as

$$\mathcal{H}^{[b]}(\mathcal{H}^{[b]}) = \int_{\mathcal{H}^{[b]}(\mathcal{H}^{[b]})} \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) \mathcal{H}^{[b]} = \int_{\mathcal{H}^{[b]}(\mathcal{H}^{[b]})} \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) \mathcal{H}^{[b]}.$$

We claim that the support of the function $\mathcal{H}^{[b]} \mapsto \mathcal{H}^{[b]}(\mathcal{H}^{[b]})$ is exactly $\mathcal{H}^{[b]}(\mathcal{H}^{[b]})$, which will complete the proof. Indeed, for any $\mathcal{H}^{[b]} \in \mathcal{H}^{[b]}(\mathcal{H}^{[b]})$, we clearly have $\mathcal{H}^{[b]} \in \mathcal{H}^{[b]} \oplus \mathcal{H}^{[b]}$. Conversely, suppose $\mathcal{H}^{[b]} \in \mathcal{H}^{[b]}(\mathcal{H}^{[b]})$ satisfies

$$\mathcal{H}^{[b]} = (\mathcal{H}^{[b]}, \dots, \mathcal{H}^{[b]}) \in \mathcal{H}^{[b]} \oplus \mathcal{H}^{[b]}.$$

By Lemma 3.3.1(2), we see that

$$\mathcal{H}^{[b]} = (\mathcal{H}^{[b]}, \dots, \mathcal{H}^{[b]}, \mathcal{H}^{[b]} + \mathcal{H}^{[b]}, \dots, \mathcal{H}^{[b]} + \mathcal{H}^{[b]}) \in \mathcal{H}^{[b]}$$

forms a basis of $\mathcal{H}^{[b]}$. Therefore, we have

$$\mathcal{H}^{[b]} \mathcal{H}^{[b]} = \mathcal{H}^{[b]} \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) = \mathcal{H}^{[b]} \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) = \mathcal{H}^{[b]},$$

which completes the proof. $\square$

Corollary 3.4.6. *The right hand side of Theorem 3.4.4 can be rewritten as:*

$$\partial \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) = \frac{\mathcal{H}^{[b]}(\mathcal{H}^{[b]})}{\mathcal{H}^{[b]}(\mathcal{H}^{[b]})} \left(\mathcal{H}^{[b]}(\mathcal{H}^{[b]}) - \sum_{\substack{\mathfrak{a} \leq \mathfrak{b} < \mathfrak{c} \text{ and} \\ \mathfrak{a} \equiv \mathfrak{b} - \mathfrak{c} \pmod{\mathfrak{d}}}} \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) \mathcal{H}^{[c]}(\mathcal{H}^{[c]}) \right). \quad (3.4.3)$$

In particular, for $\mathcal{H}^{[b]}$, Theorem 3.4.4 is identical to [Cho22, Conjecture 1.6]. For any $\mathcal{H}^{[b]}$ with $\mathcal{H}^{[b]} = \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) = \mathcal{H}^{[b]}(\mathcal{H}^{[b]})$, Theorem 3.4.4 is identical to [CHZ23, Conjecture 1.3]. $\square$

By combining Lemma 3.4.5 with Theorem 3.3.2, we deduce the following analytic cancellation law.

Theorem 3.4.7. *Let $\mathcal{H}^{[b]} \in \mathcal{H}^{[b]}(\mathcal{H}^{[b]})$ be a hermitian matrix that represents the hermitian form of $\mathcal{H}^{[b]}$.*

(1) *Consider the following partition, where $\mathcal{H}^{[b]}$ is a $(\mathfrak{b} - \mathfrak{c}) \times (\mathfrak{b} - \mathfrak{c})$ matrix:*

$$\mathcal{H}^{[b]} = \begin{pmatrix} \mathcal{H}^{[b]} & \mathcal{H}^{[b]} \\ \mathcal{H}^{[b]} & \mathcal{H}^{[b]} \end{pmatrix}, \quad \mathcal{H}^{[b]} = \mathcal{H}^{[b]} - \mathcal{H}^{[b]}.$$

(i) *If $\mathfrak{b} \neq \mathfrak{c}$ and $\mathcal{H}^{[b]}(\mathcal{H}^{[b]}) = \mathcal{H}^{[b]}(\mathcal{H}^{[b]})$, then*

$$\partial \mathcal{H}^{[b]}(\mathcal{H}^{[b]}) = \partial \mathcal{H}^{[b]}(\mathcal{H}^{[b]}).$$

(ii) If $\mathfrak{b} = \text{red}$ and $\star(\text{red}) = \text{red}$, then

$$\partial \text{blue}(\text{green}, \text{blue}^{[\text{red}]}) = \partial \text{blue}(\text{green}_b, \text{blue}^{[\text{red}-\text{red}]}).$$

(2) Consider the following partition, where green is a $(\text{red} - \text{red}) \times (\text{red} - \text{red})$ matrix:

$$\text{green} = \begin{pmatrix} \text{green} & \text{green} \\ \text{green} & \text{green} \end{pmatrix}, \quad \text{green}_b = \text{green} - \text{green}.$$

(i) If $\mathfrak{b} \neq \text{red}$ and $\star(\text{red}) = -\text{red}$, then

$$\partial \text{blue}(\text{green}, \text{blue}^{[\mathfrak{b}]}) = \partial \text{blue}(\text{green}_b, \text{blue}^{[\mathfrak{b}-\text{red}]}).$$

(ii) If $\mathfrak{b} = \text{red}$ and $\star(\text{red}) = \text{red}$, then

$$\partial \text{blue}(\text{green}, \text{blue}^{[\text{red}]}) = \partial \text{blue}(\text{green}_b, \text{blue}^{[\text{red}]}).$$

Proof. By Theorem 3.3.2, for case (1)(i), we have for any $\text{red} \leq \text{red} \leq \text{red}$,

$$\text{blue}(\text{green}, \text{blue}^{[\text{red}]}) = \text{blue}(\text{green}, \text{blue}^{[\text{red}]}_{\text{blue}}) \text{blue}(\text{green}_b, \text{blue}^{[\text{red}]}_{\text{blue}}), \quad \text{and} \quad \text{blue}'(\text{green}, \text{blue}^{[\text{red}]}) = \text{blue}'(\text{green}, \text{blue}^{[\text{red}]}_{\text{blue}}) \text{blue}'(\text{green}_b, \text{blue}^{[\text{red}]}_{\text{blue}}).$$

Therefore, we obtain

$$\begin{aligned} \partial \text{blue}(\text{green}, \text{blue}^{[\mathfrak{b}]}) &= \frac{\text{red}}{\text{blue}(\text{green}, \text{blue}^{[\mathfrak{b}]}_{\text{blue}})} \left(\text{blue}'(\text{green}, \text{blue}^{[\mathfrak{b}]}_{\text{blue}}) - \sum_{\substack{\text{red} \leq \text{red} < \mathfrak{b} \\ \text{red} \equiv \mathfrak{b} - \text{red}}} \text{blue}^{[\mathfrak{b}]}(\text{green}) \text{blue}(\text{green}_b, \text{blue}^{[\mathfrak{b}]}_{\text{blue}}) \right) \\ &= \frac{\text{blue}(\text{green}, \text{blue}^{[\text{red}]}_{\text{blue}})}{\text{blue}(\text{green}, \text{blue}^{[\mathfrak{b}]}_{\text{blue}}) \text{blue}(\text{green}_b, \text{blue}^{[\mathfrak{b}]}_{\text{blue}})} \left(\text{blue}'(\text{green}, \text{blue}^{[\mathfrak{b}]}_{\text{blue}}) - \sum_{\substack{\text{red} \leq \text{red} < \mathfrak{b} \\ \text{red} \equiv \mathfrak{b} - \text{red}}} \frac{\text{blue}(\text{green}, \text{blue}^{[\text{red}]}_{\text{blue}})}{\text{blue}(\text{green}, \text{blue}^{[\mathfrak{b}]}_{\text{blue}})} \text{blue}^{[\mathfrak{b}]}(\text{green}) \text{blue}(\text{green}_b, \text{blue}^{[\mathfrak{b}]}_{\text{blue}}) \right) \\ &= \frac{\text{red}}{\text{blue}(\text{green}, \text{blue}^{[\mathfrak{b}]}_{\text{blue}})} \left(\text{blue}'(\text{green}, \text{blue}^{[\mathfrak{b}]}_{\text{blue}}) - \sum_{\substack{\text{red} \leq \text{red} < \mathfrak{b} \\ \text{red} \equiv \mathfrak{b} - \text{red}}} \frac{\text{blue}(\text{green}, \text{blue}^{[\text{red}]}_{\text{blue}})}{\text{blue}(\text{green}, \text{blue}^{[\mathfrak{b}]}_{\text{blue}})} \text{blue}^{[\mathfrak{b}]}(\text{green}) \text{blue}(\text{green}_b, \text{blue}^{[\mathfrak{b}]}_{\text{blue}}) \right). \end{aligned} \quad (3.4.4)$$

Notice that (3.4.4) has the same form as $\text{blue}(\text{green}_b, \text{blue}^{[\mathfrak{b}]}_{\text{blue}})$ in (3.4.3), differing only in the unknown coefficients. When $\text{green}_b = \text{blue}^{[\text{red}]}_{\text{blue}}$ and $\text{green} = \text{green} = \text{red}$, we have $\text{green} = \text{blue}^{[\text{red}]}_{\text{blue}}$, hence the unknown coefficients align according to Proposition 3.4.1. Case (2)(ii) follows from case (1)(i) since $\text{blue} = \text{blue}^\#$ when $\mathfrak{b} = \text{red}$ and there is no error term to consider.

For case (2)(i), by a similar computation, we obtain

$$\partial \text{blue}(\text{green}, \text{blue}^{[\mathfrak{b}]}) = \frac{\text{red}}{\text{blue}(\text{green}_b, \text{blue}^{[\mathfrak{b}]}_{\text{blue}})} \left(\text{blue}'(\text{green}, \text{blue}^{[\mathfrak{b}]}_{\text{blue}}) - \sum_{\substack{\text{red} \leq \text{red} < \mathfrak{b} - \text{red} \\ \text{red} \equiv \mathfrak{b} - \text{red}}} \text{blue}^{[\mathfrak{b}]}(\text{green}) \text{blue}(\text{green}_b, \text{blue}^{[\mathfrak{b}]}_{\text{blue}}) \right). \quad (3.4.5)$$

For the term $\text{red} = \text{red}$, there is no vector with norm $-\text{red}$, hence the corresponding term vanishes.

It remains to verify case (1)(ii). Let

$$\text{green} = \begin{pmatrix} \text{green} & \text{green} & \text{green} \\ \text{green} & \text{green} & \text{green} \\ \text{green} & \text{green} & \text{green} \end{pmatrix}, \quad \text{green}^\vee = \begin{pmatrix} \text{green} & \text{green} & \text{green} \\ \text{green} & \text{green} & \text{green} \\ \text{green} & \text{green} & \text{green} \end{pmatrix},$$

where green is an $(\text{red} - \text{red}) \times (\text{red} - \text{red})$ matrix, green is a $\text{red} \times \text{red}$ matrix, and green is an $(\text{red} - \text{red}) \times \text{red}$ matrix with $\text{green} = \text{green}$.

Applying [Cho22, Theorem 3.16]⁶ and rewriting the summation using ∂blue , we obtain the equality

$$\partial \text{blue}(\text{green}, \text{blue}^{[\text{red}]}_{\text{blue}}) \cdot \text{blue}^{[\text{red}]}_{\text{blue}}(\text{green}, \text{blue}^{[\text{red}]}_{\text{blue}}) = \partial \text{blue}(\text{green}^\vee, \text{blue}^{[\text{red}]}_{\text{blue}}) \cdot \text{blue}^{[\text{red}]}_{\text{blue}}(\text{green}, \text{blue}^{[\text{red}]}_{\text{blue}}).$$

⁶The lattice $\text{blue}^\vee$ in [Cho22, Theorem 3.16] corresponds to our lattice blue .

Applying case (1)(i) and (2)(i) to (3.4), we obtain the dual relation for $\mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}]}$:

$$\partial \left(\left(\begin{pmatrix} \text{key} & \text{apple} \\ \text{apple} & \text{apple} \end{pmatrix}, \mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}]} \right) = \frac{\text{key} \cdot \text{apple} \cdot \text{apple} \cdot \text{apple}}{\text{apple} \cdot \text{apple} \cdot \text{apple} \cdot \text{apple}} \partial \left(\left(\begin{pmatrix} \text{key} & \text{apple} \\ \text{apple} & \text{apple} \end{pmatrix}, \mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}]} \right). \quad (3.4.6)$$

Applying Theorem 3.3.2, we have that

$$\begin{aligned} \frac{\text{key} \cdot \text{apple} \cdot \text{apple} \cdot \text{apple}}{\text{apple} \cdot \text{apple} \cdot \text{apple} \cdot \text{apple}} &= \frac{\text{key} \cdot \text{apple} \cdot \text{apple} \cdot \text{apple}}{\text{apple} \cdot \text{apple} \cdot \text{apple} \cdot \text{apple}} \\ &= \frac{\text{key} \cdot \text{apple} \cdot \text{apple} \cdot \text{apple}}{\text{apple} \cdot \text{apple} \cdot \text{apple} \cdot \text{apple}}. \end{aligned}$$

Applying this to (3.4.6), we have

$$\partial \left(\left(\begin{pmatrix} \text{key} & \text{apple} \\ \text{apple} & \text{apple} \end{pmatrix}, \mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}]} \right) = \frac{\text{key} \cdot \text{apple} \cdot \text{apple} \cdot \text{apple}}{\text{apple} \cdot \text{apple} \cdot \text{apple} \cdot \text{apple}} \partial \left(\left(\begin{pmatrix} \text{key} & \text{apple} \\ \text{apple} & \text{apple} \end{pmatrix}, \mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}]} \right). \quad (3.4.7)$$

In our case, we take (where key and apple are not necessary identical)

$$\text{key} = \begin{pmatrix} \text{key} & \text{apple} \\ \text{apple} & \text{apple} \end{pmatrix} = \begin{pmatrix} \text{apple} & \text{apple} \\ \text{apple} & \text{apple} \end{pmatrix}.$$

We compare equation (3.4.7) for $\mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}]}$ to equation (3.4.6) for $\mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}-\mathfrak{a}]}$. Suppose $\star(\text{apple}) = \text{apple}$, then $\star(\text{key} \cdot \text{apple}) = \text{apple}$. Applying case (1)(i) to the right-hand side of (3.4.6), we obtain

$$\partial \left(\begin{pmatrix} \text{key} & \text{apple} \\ \text{apple} & \text{apple} \end{pmatrix}, \mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}]} \right) = \partial \left(\begin{pmatrix} \text{key} & \text{apple} \\ \text{apple} & \text{apple} \end{pmatrix}, \mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}-\mathfrak{a}]} \right).$$

This concludes the proof of case (1)(ii). $\square$

We reformulate Theorem 3.4.7 to align with the geometric cancellation law Proposition 2.3.5.

Corollary 3.4.8. Let $\mathfrak{w}^{[\mathfrak{h}]} = \mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}]} \in \mathfrak{w}(\mathfrak{e}^{[\mathfrak{h}]})$ be an admissible Schwartz function and let $\mathfrak{v} = (\mathfrak{v}_1, \dots, \mathfrak{v}_r) \in \mathfrak{V}^{\mathfrak{h}}$ be vectors that are linearly independent. For any fixed $\mathfrak{a} \leq \mathfrak{b} \leq \mathfrak{h}$, define:

- $\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}} := \langle \mathfrak{v}_{\mathfrak{a}} \rangle^{\perp}$ and set $\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}$ being the orthogonal projection of $\mathfrak{v}_{\mathfrak{a}}$ for $\mathfrak{a} \neq \mathfrak{b}$.
- $\mathfrak{v}_{\mathfrak{b}} := (\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}, \dots, \widehat{\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}}, \dots, \mathfrak{v}_{\mathfrak{h}, \mathfrak{b}}) \in \mathfrak{V}_{\mathfrak{b}}^{\mathfrak{h}-\mathfrak{a}}$.
- $\mathfrak{v}_{\mathfrak{b}} := (\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}, \dots, \widehat{\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}}, \dots, \mathfrak{v}_{\mathfrak{h}, \mathfrak{b}}) \in \{\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}, \mathfrak{v}_{\mathfrak{h}, \mathfrak{b}}\}^{\mathfrak{h}-\mathfrak{a}}$.

Suppose we are in one of the following situations:

- When $\mathfrak{v}_{\mathfrak{a}} = \mathfrak{v}_{\mathfrak{h}}$ and $\star(\mathfrak{v}_{\mathfrak{a}}, \mathfrak{v}_{\mathfrak{h}}) = \begin{cases} \text{apple} & \text{if } \mathfrak{a} \neq \mathfrak{h}; \\ \text{apple} & \text{if } \mathfrak{a} = \mathfrak{h}. \end{cases}$
- When $\mathfrak{v}_{\mathfrak{a}} = \mathfrak{v}_{\mathfrak{h}}$ and $\star(\mathfrak{v}_{\mathfrak{a}}, \mathfrak{v}_{\mathfrak{h}}) = \begin{cases} -\text{apple} & \text{if } \mathfrak{a} \neq \mathfrak{h}; \\ \text{apple} & \text{if } \mathfrak{a} = \mathfrak{h}. \end{cases}$

Then we have $\partial \left(\begin{pmatrix} \mathfrak{v}_{\mathfrak{a}} & \mathfrak{v}_{\mathfrak{h}} \end{pmatrix}, \mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}]} \right) = \partial \left(\begin{pmatrix} \mathfrak{v}_{\mathfrak{a}} & \mathfrak{v}_{\mathfrak{h}} \end{pmatrix}, \mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}-\mathfrak{a}]} \right)$. $\square$

Combining Proposition 2.3.5 and Corollary 3.4.8, we obtain the following cancellation law.

Proposition 3.4.9. Let $\mathfrak{w}^{[\mathfrak{h}]} = \mathfrak{w}_{\mathfrak{a}, \mathfrak{b}}^{[\mathfrak{h}]} \in \mathfrak{w}(\mathfrak{e}^{[\mathfrak{h}]})$ be an admissible Schwartz function, and let $\mathfrak{v} = (\mathfrak{v}_1, \dots, \mathfrak{v}_r) \in \mathfrak{V}^{\mathfrak{h}}$ be vectors that are linearly independent. For any fixed $\mathfrak{a} \leq \mathfrak{b} \leq \mathfrak{h}$, define:

- $\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}} := \langle \mathfrak{v}_{\mathfrak{a}} \rangle^{\perp}$ and set $\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}$ being the orthogonal projection of $\mathfrak{v}_{\mathfrak{a}}$ for $\mathfrak{a} \neq \mathfrak{b}$.
- $\mathfrak{v}_{\mathfrak{b}} := (\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}, \dots, \widehat{\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}}, \dots, \mathfrak{v}_{\mathfrak{h}, \mathfrak{b}}) \in \mathfrak{V}_{\mathfrak{b}}^{\mathfrak{h}-\mathfrak{a}}$.
- $\mathfrak{v}_{\mathfrak{b}} := (\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}, \dots, \widehat{\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}}, \dots, \mathfrak{v}_{\mathfrak{h}, \mathfrak{b}}) \in \{\mathfrak{v}_{\mathfrak{a}, \mathfrak{b}}, \mathfrak{v}_{\mathfrak{h}, \mathfrak{b}}\}^{\mathfrak{h}-\mathfrak{a}}$.

Suppose we are in one of the following situations:

- When $\mathfrak{v}_{\mathfrak{a}} = \mathfrak{v}_{\mathfrak{h}}$ and $\star(\mathfrak{v}_{\mathfrak{a}}, \mathfrak{v}_{\mathfrak{h}}) = \begin{cases} \text{apple} & \text{if } \mathfrak{a} \neq \mathfrak{h}; \\ \text{apple} & \text{if } \mathfrak{a} = \mathfrak{h}. \end{cases}$
- When $\mathfrak{v}_{\mathfrak{a}} = \mathfrak{v}_{\mathfrak{h}}$ and $\star(\mathfrak{v}_{\mathfrak{a}}, \mathfrak{v}_{\mathfrak{h}}) = \begin{cases} -\text{apple} & \text{if } \mathfrak{a} \neq \mathfrak{h}; \\ \text{apple} & \text{if } \mathfrak{a} = \mathfrak{h}. \end{cases}$

Then the equality $\mathbb{E}(\mathbb{E}, \mathbb{E}^{[b]}) = \partial \mathbb{E}(\mathbb{E}, \mathbb{E}^{[b]})$ follows from $\mathbb{E}(\mathbb{E}_b, \mathbb{E}^{[b]}) = \partial \mathbb{E}(\mathbb{E}_b, \mathbb{E}^{[b]})$. $\square$

The following results will be used in Proposition 11.0.2, where we analyze the error terms after applying the Fourier transform.

Proposition 3.4.10. *For any integers $\bullet \leq \flat \leq \spadesuit$ and $\bullet \leq \clubsuit \leq \spadesuit - \heartsuit$, we have the equality:*

$$\mathbb{E}_{\clubsuit+\heartsuit, \spadesuit-\heartsuit}^{[b]}(\heartsuit) = \frac{\mathbb{E}(\mathbb{E}^{[a]})}{\mathbb{E}(\mathbb{E}^{[b]})} \cdot \mathbb{E}_{\clubsuit, \spadesuit}^{[b]},$$

where $\mathbb{E}(\mathbb{E}) = |\mathbb{E}^\vee : \mathbb{E}|^{\bullet/\spadesuit}$ denotes the volume of the vertex lattice L in the hermitian space $\mathbb{E}$ or $\mathbb{E}^\vee$, computed with respect to the self-dual Haar measure.

Proof. There is nothing to prove when $\flat = \bullet$. To proceed, assume $\flat \neq \bullet$. We prove the claim by induction. For the base case $\spadesuit = \heartsuit$ and $\flat = \heartsuit$, the statement follows directly from Proposition 3.5.1 in the next subsection. Assuming the statement holds for all dimension $\leq \spadesuit - \heartsuit$, we want to prove it for dimension $\spadesuit$. By (3.4.4) and (3.4.5), we see that:

(1) When $\flat \neq \spadesuit$, and $\heartsuit \leq \clubsuit \leq \spadesuit$, we have

$$\mathbb{E}_{\clubsuit, \spadesuit}^{[b]}(\heartsuit) = \frac{\mathbb{E}_{\clubsuit, \spadesuit}(\heartsuit, \mathbb{E}^{[a]})}{\mathbb{E}_{\clubsuit, \spadesuit}(\heartsuit, \mathbb{E}^{[b]})} \mathbb{E}_{\clubsuit-\heartsuit, \spadesuit}^{[b]}(\heartsuit).$$

(2) When $\flat \neq \bullet$, and $\bullet \leq \clubsuit \leq \spadesuit - \heartsuit$, we have

$$\mathbb{E}_{\clubsuit, \spadesuit}^{[b]}(\heartsuit) = \frac{\mathbb{E}_{\clubsuit, \spadesuit}(\heartsuit, \mathbb{E}^{[a]})}{\mathbb{E}_{\clubsuit, \spadesuit}(\heartsuit, \mathbb{E}^{[b]})} \mathbb{E}_{\clubsuit-\heartsuit, \spadesuit-\heartsuit}^{[b]}(\heartsuit - \heartsuit).$$

In particular, by induction, the assertion holds in all cases except when $\flat = \spadesuit$ and $\clubsuit = \spadesuit - \heartsuit$. In this exceptional case, we have the identity $\mathbb{E}_{\spadesuit-\heartsuit, \spadesuit}^{[b]}(\heartsuit) = \mathbb{E}_{\spadesuit-\heartsuit, \spadesuit}^{[b]}(\heartsuit - \heartsuit)$, so it suffices to compare $\mathbb{E}_{\spadesuit-\heartsuit, \spadesuit}^{[b]}(\heartsuit - \heartsuit)$ with $\mathbb{E}_{\spadesuit, \spadesuit}^{[b]}(\heartsuit)$. By [CHZ23, Corollary 6.7], we have

$$\mathbb{E}_{\spadesuit, \spadesuit}^{[b]}(\heartsuit) = \frac{(-\heartsuit)^{\frac{(\spadesuit-\heartsuit)(\spadesuit+\heartsuit)}{2}}}{\heartsuit - (-\heartsuit)^{-(\spadesuit-\heartsuit)}} \cdot \frac{\mathbb{E}(\mathbb{E}^{[a]})}{\mathbb{E}(\mathbb{E}^{[b]})}$$

where the representation density functions relate to weighted local density functions via (3.5.1). The assertion then follows from the identity $\mathbb{E}(\mathbb{E}^{[a]}) = \mathbb{E}(\mathbb{E}^{[b]})$. $\square$

3.5. Base Case. We verify Theorem 3.4.4 for the case $\spadesuit = \heartsuit$ as the starting point for the induction in §11. To apply the result from [Hir98], we recall and use the representation density function from [KR14].

For hermitian matrices $\mathbb{E}$ and $\mathbb{F}$ of sizes $\spadesuit$ and $\heartsuit$, respectively, the *representation density function* is defined as

$$\mathbb{E}(\mathbb{E}, \mathbb{F}) = \lim_{\substack{\rightarrow \\ \mathfrak{p}}} \frac{1}{\mathfrak{p}} \sum_{\substack{\mathfrak{a} \in \mathfrak{p} \\ \mathfrak{a} \equiv \mathbb{E} \pmod{\mathfrak{p}}}} \mathbb{E}(\mathbb{E}, \mathbb{F})_{\mathfrak{p}},$$

where $\mathfrak{p}$ is the cardinality of the residue field of $\mathfrak{p}$, and the term in the limit is defined as

$$\mathbb{E}(\mathbb{E}, \mathbb{F})_{\mathfrak{p}} := \left\{ \mathfrak{a} \in \mathfrak{p} \mid \mathfrak{a} \equiv \mathbb{E} \pmod{\mathfrak{p}} \mid \mathbb{E}[\mathfrak{a}] \equiv \mathbb{F} \pmod{\mathfrak{p}} \right\}.$$

By [Kud97, Proposition A.6], the representation density function relates to the weighted local density function defined in Definition 3.1.1 via the relation

$$\mathbb{E}(\mathbb{E}, \mathbb{F})_{\mathfrak{p}} = \frac{1}{\mathfrak{p}} \sum_{\substack{\mathfrak{a} \in \mathfrak{p} \\ \mathfrak{a} \equiv \mathbb{E} \pmod{\mathfrak{p}}}} \mathbb{E}(\mathbb{E}, \mathbb{F})_{\mathfrak{p}}. \quad (3.5.1)$$

We define the derivative of the representation density function $\mathbb{E}'(\mathbb{E}, \mathbb{F})$ analogously to (3.4.1), see [KR11, §9]. By direct computation using the formula in [Hir98], we obtain the following result:

Proposition 3.5.1. (1) *For any integer $\spadesuit \geq \heartsuit$, we have*

$$\mathbb{E}((\heartsuit)_{\heartsuit \times \heartsuit}, (\heartsuit)_{\heartsuit \times \heartsuit}) = \begin{cases} \heartsuit - (-\heartsuit)^{-\heartsuit} & \spadesuit \text{ even}; \\ \heartsuit & \spadesuit \text{ odd}. \end{cases}$$

Moreover, for any odd $\spadesuit \geq \heartsuit$, we have $\mathbb{E}'((\heartsuit)_{\heartsuit \times \heartsuit}, (\heartsuit)_{\heartsuit \times \heartsuit}) = (\heartsuit - (-\heartsuit)^{-\heartsuit}) \frac{\heartsuit+\heartsuit}{2}$.

(2) *For any integer $\spadesuit \geq \heartsuit$, we have*

$$\mathbb{E}((\heartsuit)_{\heartsuit \times \heartsuit}, (\heartsuit)_{\heartsuit \times \heartsuit}) = \begin{cases} \heartsuit & \spadesuit \text{ even}; \\ \heartsuit + \heartsuit & \spadesuit \text{ odd}. \end{cases}$$

Moreover, for any even $\spadesuit \geq \heartsuit$, we have $\mathbb{E}'((\heartsuit)_{\heartsuit \times \heartsuit}, (\heartsuit)_{\heartsuit \times \heartsuit}) = (\heartsuit + \heartsuit) \frac{\heartsuit}{2} + \heartsuit$.

(3) For any integer $\mathfrak{w} \geq -\bullet$, we have

$$\mathfrak{L}((\mathfrak{L}^{-\bullet})_{\bullet \times \bullet}, (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}) = \begin{cases} \bullet & \mathfrak{w} \text{ even}; \\ \bullet + \bullet^{-\bullet} & \mathfrak{w} \text{ odd}. \end{cases}$$

Moreover, for any even $\mathfrak{w} \geq \bullet$, we have $\mathfrak{L}'((\mathfrak{L}^{-\bullet})_{\bullet \times \bullet}, (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}) = (\bullet + \bullet^{-\bullet})^{\frac{\mathfrak{w}}{2}} + \bullet^{-\bullet}$.

Proof. We refer the reader to [San17, Theorem 3.2] for the relevant notation and statements. A straightforward computation shows that

$$\prod_{\mathfrak{w}} ((\mathfrak{L}), (\mathfrak{L}^{\mathfrak{w}})) = \begin{cases} \bullet & \mathfrak{L} = \bullet; \\ (-\bullet)^{\mathfrak{L}-\bullet} (\bullet + (-\bullet)) & \bullet \leq \mathfrak{L} \leq \mathfrak{w}; \\ (-\bullet)^{\mathfrak{w}} & \mathfrak{L} = \mathfrak{w} + \bullet. \end{cases}$$

By [San17, Theorem 3.2], for $\mathfrak{w} = (\mathfrak{w}) \in \mathbb{N}_{\geq \bullet}^{\bullet}$ and $\mathfrak{v} = (\mathfrak{v}) \in \mathbb{N}_{\geq \bullet}^{\bullet+\bullet}$, we have

$$\begin{aligned} & \mathfrak{L}((\mathfrak{L}^{-\bullet}, \dots, \mathfrak{L}^{\mathfrak{v}+\bullet}), (\mathfrak{L}^{\mathfrak{w}})) \\ &= \bullet + (\bullet + (-\bullet)^{-\bullet}) \cdot \sum_{\bullet \leq \mathfrak{L} \leq \mathfrak{w}} \left((-\bullet)^{\mathfrak{L}-\bullet} (-\bullet)^{-\mathfrak{L}+\langle \mathfrak{v}', \mathfrak{L} \rangle} \right) + (-\bullet)^{\mathfrak{w}+\bullet} (-\bullet)^{-\bullet} (-\bullet)^{-\mathfrak{w}+\langle \mathfrak{v}', \mathfrak{w}+\bullet \rangle}, \end{aligned}$$

where $\mathfrak{v}'$ denotes the conjugate of $\mathfrak{v}$ as Young tableau, and $\langle \mathfrak{v}', \mathfrak{w}' \rangle := \sum_{\mathfrak{w} \geq \mathfrak{v}} \mathfrak{v}'_{\mathfrak{w}} \mathfrak{w}'_{\mathfrak{w}}$.

In the case (1), we choose $\mathfrak{v} = (\bullet^{\bullet+\bullet})$ and we have

$$\mathfrak{L}(\mathfrak{L}^{-\bullet}, (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}) = \bullet + (\bullet + (-\bullet)^{-\bullet}) \sum_{\bullet \leq \mathfrak{L} \leq \mathfrak{w}} \left((-\bullet)^{\mathfrak{L}-\bullet} (-\bullet)^{-\mathfrak{L}+\bullet} \right) + (-\bullet)^{\mathfrak{w}+\bullet} (-\bullet)^{-\bullet} (-\bullet)^{-\mathfrak{w}+(\mathfrak{w}+\bullet)+\mathfrak{w}}.$$

In the case (2), we choose $\mathfrak{v} = (\bullet, \bullet^{\bullet})$, and we have

$$\mathfrak{L}\left(\left(\mathfrak{L}^{-\bullet}\right), (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}\right) = \bullet + (\bullet + (-\bullet)) \sum_{\bullet \leq \mathfrak{L} \leq \mathfrak{w}} \left((-\bullet)^{\mathfrak{L}-\bullet} (-\bullet)^{-\mathfrak{L}+\bullet} \right) + (-\bullet)^{\mathfrak{w}+\bullet} (-\bullet)^{-\bullet} (-\bullet)^{-\mathfrak{w}+(\mathfrak{w}+\bullet)}.$$

In the case (3), we choose $\mathfrak{v} = (\bullet^{\bullet}, \bullet)$, and we have

$$\mathfrak{L}\left(\left(\mathfrak{L}^{-\bullet}\right), (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}\right) = \bullet + (\bullet + (-\bullet)^{-\bullet}) \sum_{\bullet \leq \mathfrak{L} \leq \mathfrak{w}} \left((-\bullet)^{\mathfrak{L}-\bullet} (-\bullet)^{-\mathfrak{L}+(\mathfrak{L}-\bullet)} \right) + (-\bullet)^{\mathfrak{w}+\bullet} (-\bullet)^{-\bullet} (-\bullet)^{-\mathfrak{w}+\mathfrak{w}},$$

and we have

$$\mathfrak{L}\left(\left(\mathfrak{L}^{-\bullet}\right), (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}\right) = \mathfrak{L}\left(\left(\mathfrak{L}^{-\bullet}\right), (\mathfrak{L}^{\mathfrak{w}+\bullet})_{\bullet \times \bullet}\right).$$

All assertions now follow from direct computation. $\square$

Corollary 3.5.2. *Theorem 3.4.4 holds for $\mathfrak{w} = \bullet$.*

Proof. This follows from Example 2.3.4 and Proposition 3.5.1. For instance, let $\mathfrak{w} \geq -\bullet$ be any even integer. Then, by the definition given in (3.4.3), we have

$$\partial \mathfrak{L}((\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}, \mathfrak{L}^{\bullet}) = \frac{\bullet}{\mathfrak{L}((\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}, (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet})} \left(\mathfrak{L}'((\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}, (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}) - \mathfrak{L}^{\bullet} \mathfrak{L}((\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}, (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}) \right).$$

Here, $\mathfrak{L}^{\bullet} \mathfrak{L} = \mathfrak{L}^{\bullet} / \bullet$ is the modified term according to the relation in (3.5.1). By Proposition 3.4.1, we have

$$\mathfrak{L}^{\bullet} \mathfrak{L} = \frac{\mathfrak{L}'((\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}, (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet})}{\mathfrak{L}((\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}, (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet})} = \bullet - (-\bullet)^{-\bullet}.$$

Moreover, we compute

$$\partial \mathfrak{L}(\mathfrak{L}^{\mathfrak{w}}, \mathfrak{L}^{\bullet}) = \frac{\bullet}{\bullet + \bullet} \left(\mathfrak{L}'((\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}, (\mathfrak{L}^{\mathfrak{w}})_{\bullet \times \bullet}) - \bullet \right) = \frac{\mathfrak{w}}{\mathfrak{w} + \bullet}.$$

This matches the geometric side as described in Example 2.3.4. $\square$

3.6. Local Whittaker functions. In this subsection, we recall the local Whittaker function and Siegel-Weil formula. We change our local setup and let $\mathbb{F}/\mathbb{F}_0$ be a quadratic extension of local fields. If $\mathbb{F}_0$ is archimedean, we take $\mathbb{F}/\mathbb{F}_0 = \mathbb{C}/\mathbb{R}$. If $\mathbb{F}_0$ is non-archimedean, we assume $\mathbb{F}/\mathbb{F}_0$ is unramified.

Let $\mathbb{H}$ be a $2n$ -dimensional split hermitian space over $\mathbb{F}$, with unitary group $\mathbb{U} = \mathbb{U}(\mathbb{H})$. Let $\mathbb{V}$ be a hermitian space of dimension n , which we assume to be positive definite when $\mathbb{F}/\mathbb{F}_0 = \mathbb{C}/\mathbb{R}$. Let χ be the quadratic character associated to $\mathbb{F}$ and let ψ be a fixed additive character. Then we have a Weil representation ω of $\mathbb{U} \times \mathbb{U}(\mathbb{V})$ on $\mathcal{W}(\mathbb{V})$, such that for each $w \in \mathcal{W}(\mathbb{V})$ and $v \in \mathbb{V}$, we have:

$$\begin{aligned} \omega(u, v)(w) &= \chi(u) | \det(u) |^{1/2} \psi(\langle u \cdot v, w \rangle), & \mathbb{U}(v) &= \begin{pmatrix} v & \\ & -\bar{v} \end{pmatrix} \in \mathbb{U}(\mathbb{V}), \\ \omega(u, v)(w) &= \psi(\langle u \cdot v, w \rangle), & \mathbb{U}(v) &= \begin{pmatrix} v & \\ & -\bar{v} \end{pmatrix} \in \mathbb{U}(\mathbb{V}), \\ \omega(u, v)(w) &= \chi(u) \cdot \widehat{w}(v), & \mathbb{U}(v) &= \begin{pmatrix} v & \\ & -\bar{v} \end{pmatrix}, \\ \omega(u, v)(w) &= \psi(\langle u \cdot v, w \rangle), & \mathbb{U}(v) &= \begin{pmatrix} v & \\ & -\bar{v} \end{pmatrix}. \end{aligned}$$

Here χ is the local Weil constant (see [KR14, (10.3)]), and $\widehat{w}$ is the Fourier transform of w using the self-dual Haar measure on $\mathbb{V}$ with respect to $\psi \circ \langle \cdot, \cdot \rangle$.

There is an intertwining map

$$\mathcal{W} : \mathcal{W}(\mathbb{V}) \longrightarrow \mathcal{W}(\mathbb{V}), \quad \mathcal{W}(w)(v) = \omega(u, v)(w),$$

where $\mathcal{W}(v, \chi)$ denotes the principal series representation associated with χ . We extend $\mathcal{W}(w)$ to a Siegel-Weil section $\mathcal{W}(w)(v) \in \mathcal{W}(\mathbb{V})$.

For any $v \in \mathbb{V}$, we define the *local Whittaker function* for $\mathcal{W}$ as

$$\mathcal{W}(v, w) = \int_{\mathbb{V}} \omega(u, v)(w) \psi(\langle u \cdot v, w \rangle) \psi(-\langle u \cdot v, w \rangle) dv.$$

Fix basis elements $e_1 \in \wedge^2(\mathbb{V})^*$ and $e_2 \in \wedge^2(\mathbb{V})^*$. By (H5), we have a form $\mathfrak{w}$ of degree $2n$ on $\mathbb{V}$ along with two measures and constants:

$$\mathfrak{w}_1 = \mathfrak{w}(e_1, \mathfrak{w}), \quad \mathfrak{w}_2 = \mathfrak{w}(e_2, \mathfrak{w}).$$

Let $v \in \mathbb{V}$. For any Schwartz function $w \in \mathcal{W}(\mathbb{V})$, we define the *local orbital integral*

$$\mathcal{O}(v, w) := \int_{\mathbb{V}} \mathfrak{w}(v) \mathfrak{w}_1 dv,$$

where $\mathfrak{w}_1$ is the measure on $\mathbb{V}$ determined by the restriction of $\mathfrak{w}$ to $\mathbb{V}$.

3.7. Non-archimedean local Siegel-Weil formula. We recall the non-archimedean local Siegel-Weil formula, following the formalism of [KR14, p. 39]. While *loc. cit.* focuses on the case $\mathbb{F}_0 = \mathbb{C}$, all results extend directly to general $\mathbb{F}_0$ -adic fields $\mathbb{F}$.

Theorem 3.7.1 (Non-archimedean local Siegel-Weil formula). *Suppose $\mathbb{F}/\mathbb{F}_0$ is a quadratic extension of $\mathbb{F}_0$ -adic fields. Let $v \in \wedge^2(\mathbb{V})^*$ be a hermitian matrix, and let $w \in \mathcal{W}(\mathbb{V})$ be a Schwartz function. Then, we have the equality*

$$\mathcal{W}(v, w) = \mathfrak{w}(v, \mathfrak{w}, \mathfrak{w}_1, \mathfrak{w}_2) \cdot \mathcal{O}(v, w),$$

where $\mathfrak{w}(v, \mathfrak{w}, \mathfrak{w}_1, \mathfrak{w}_2)$ is a constant independent of v , given by

$$\mathfrak{w}(v, \mathfrak{w}, \mathfrak{w}_1, \mathfrak{w}_2) = \frac{\mathfrak{w}(v) \mathfrak{w}_1(v)}{\mathfrak{w}_2(v)}.$$

Proof. See [KR14, p. 39]. □

We also relate the local Whittaker function to weighted local density functions.

Proposition 3.7.2 ([KR14, Proposition 10.1]). *Suppose $\mathbb{F}/\mathbb{F}_0$ is a quadratic extension of $\mathbb{F}_0$ -adic fields. Let $v \in \wedge^2(\mathbb{V})^*$ be a nonsingular hermitian matrix, and let $w \in \mathcal{W}(\mathbb{V})$ be any Schwartz function. For any integer $k \geq 0$, we have*

$$\mathcal{W}(v, w) = \mathfrak{w}(v) \mathfrak{w}_1(w).$$

In particular,

$$\mathfrak{g}(\mathfrak{o}, \mathfrak{o}, \mathfrak{o}) = \mathfrak{g}(\mathfrak{o}) \otimes_{\mathfrak{o}} \mathfrak{g}(\mathfrak{o}, \mathfrak{o}) \quad \text{and} \quad \mathfrak{g}'(\mathfrak{o}, \mathfrak{o}, \mathfrak{o}) = \mathfrak{g}(\mathfrak{o}) \otimes_{\mathfrak{o}} \mathfrak{g}'(\mathfrak{o}, \mathfrak{o}) \otimes_{\mathfrak{o}} \mathfrak{g}(\mathfrak{o}).$$

Here, $\mathfrak{g}$ denotes the cardinality of the residue field of $\mathfrak{o}$. $\square$

4. SYMMETRIC SPACES AND SPECIAL CYCLES

In this section, we review the archimedean theory of hermitian symmetric spaces and their special cycles. We then introduce Kudla's Green currents and recall the local arithmetic Siegel–Weil formula.

4.1. Hermitian Symmetric spaces. Let $\mathfrak{g}/\mathfrak{g}_0 = \mathfrak{g}/\mathfrak{g}_0$, and let $\mathfrak{g} \geq \mathfrak{g}_0$ be a positive integer. Let $(\mathfrak{g}, \mathfrak{g}_0)$ be a non-degenerate hermitian space over $\mathfrak{g}_0$ of dimension $\mathfrak{g}$, with signature $(\mathfrak{g} - \mathfrak{g}_0, \mathfrak{g}_0)$. Let $\mathfrak{g}(\mathfrak{g})$ denote the unitary group.

We fix an orthogonal decomposition $\mathfrak{g} = \mathfrak{g}^+ \oplus \mathfrak{g}^-$, where $\mathfrak{g}^+$ and $\mathfrak{g}^-$ are positive and negative definite, respectively. Define

$$\mathfrak{g} = \mathfrak{g}(\mathfrak{g}^+) \times \mathfrak{g}(\mathfrak{g}^-)$$

to be a maximal compact subgroup of $\mathfrak{g}(\mathfrak{g})$.

Let $\mathfrak{g}(\mathfrak{g})$ denote the space of $\mathfrak{g}$ -dimensional complex subspaces of $\mathfrak{g}$. Define the *hermitian symmetric space*

$$\mathfrak{g} = \{\mathfrak{g} \in \mathfrak{g}(\mathfrak{g}) : \mathfrak{g}|_{\mathfrak{g}^+} < 0\} \quad (4.1.1)$$

as the open subset of $\mathfrak{g}(\mathfrak{g})$ consisting of negative definite lines. It is clear that $\mathfrak{g}$ carries a $\mathfrak{g}(\mathfrak{g})$ -invariant complex structure, and we have $\mathfrak{g} \simeq \mathfrak{g}(\mathfrak{g})/\mathfrak{g}$.

There is another symmetric space associated with the analytic side. Let $\mathfrak{g}$ be the standard split $\mathfrak{g}/\mathfrak{g}_0$ -skew-hermitian space of dimension $\mathfrak{g}$. The hermitian symmetric domain for $\mathfrak{g} = \mathfrak{g}(\mathfrak{g})$ is the *hermitian upper half-space*, given by

$$\begin{aligned} \mathfrak{g}(\mathfrak{g}) &= \left\{ \mathfrak{g} \in \mathfrak{g}(\mathfrak{g}) : \frac{\mathfrak{g} - \overline{\mathfrak{g}}}{\mathfrak{g}} > 0 \right\} \\ &= \left\{ \mathfrak{g} = \mathfrak{g} + \mathfrak{g} : \mathfrak{g} \in \mathfrak{g}, \mathfrak{g} \in \mathfrak{g}^+ \right\}. \end{aligned} \quad (4.1.2)$$

For any partition $\sum_{i=1}^{\mathfrak{g}} \mathfrak{g}_i = \mathfrak{g}$, we define an embedding

$$\prod_{i=1}^{\mathfrak{g}} \mathfrak{g}(\mathfrak{g}_i) \longrightarrow \mathfrak{g}(\mathfrak{g}), \quad (\mathfrak{g}_i) \longmapsto \mathfrak{g} = \begin{pmatrix} \mathfrak{g}_1 & & & \\ & \mathfrak{g}_2 & & \\ & & \ddots & \\ & & & \mathfrak{g}_{\mathfrak{g}} \end{pmatrix}.$$

4.2. Special cycles and Green currents. Let $\mathfrak{g} \in \mathfrak{g}$ be any nonzero vector. For any $\mathfrak{g} \in \mathfrak{g}$, we decompose $\mathfrak{g}$ orthogonally with respect to $\mathfrak{g}$ as

$$\mathfrak{g} = \mathfrak{g}_{\mathfrak{g}} + \mathfrak{g}_{\mathfrak{g}^\perp},$$

where $\mathfrak{g}_{\mathfrak{g}} \in \mathfrak{g}$ and $\mathfrak{g}_{\mathfrak{g}^\perp} \perp \mathfrak{g}$. Define the function

$$\mathfrak{g}(\mathfrak{g}, \mathfrak{g}) = -(\mathfrak{g}_{\mathfrak{g}}, \mathfrak{g}_{\mathfrak{g}}).$$

The special cycle (often denoted by $\mathfrak{g}_{\mathfrak{g}}$) is then defined as

$$\mathfrak{g}(\mathfrak{g}) := \{\mathfrak{g} \in \mathfrak{g} : \mathfrak{g} \perp \mathfrak{g}\} = \{\mathfrak{g} \in \mathfrak{g} : \mathfrak{g}(\mathfrak{g}, \mathfrak{g}) = 0\}.$$

The cycle $\mathfrak{g}(\mathfrak{g})$ is nonempty if and only if $(\mathfrak{g}, \mathfrak{g}) > 0$, in which case it forms an analytic divisor on $\mathfrak{g}$.

For $\mathfrak{g} = \mathfrak{g} + \mathfrak{g} \in \mathfrak{g}(\mathfrak{g})$, Kudla's Green function is defined as

$$\mathfrak{g}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}) = -\mathfrak{g}(-\mathfrak{g}_{\mathfrak{g}}(\mathfrak{g}, \mathfrak{g})), \quad \text{where} \quad \mathfrak{g}(\mathfrak{g}) = -\int_{\mathfrak{g}} \frac{\mathfrak{g}}{\mathfrak{g}}.$$

The function $\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{w}, -)$ is smooth on $\mathfrak{V} \setminus \mathfrak{P}(\mathfrak{v})$ and has a logarithmic singularity along $\mathfrak{P}(\mathfrak{v})$. If $\mathfrak{P}(\mathfrak{v})$ is empty, the function is then smooth on $\mathfrak{V}$. By [Liu11a, Proposition 4.9], it satisfies the $(\mathfrak{v}, \mathfrak{w})$ -current equation for $\mathfrak{P}(\mathfrak{v})$:

$$\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{w}) + \mathfrak{g}_{\mathfrak{m}}(\mathfrak{w}, \mathfrak{v}) = [\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{w})]^7, \quad (4.2.1)$$

where $\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, -) = \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{v}) \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, -)$, and $\mathfrak{g}_{\mathfrak{m}}(-, -) \in \mathfrak{H}(\mathfrak{V}) \otimes (\mathfrak{g}_{\mathfrak{m}}(\mathfrak{V}))^{\vee}(\mathfrak{V})$ is the *Kudla–Millson Schwartz form* ([KM86]). We emphasize that the relation (4.2.1) holds for *any* nonzero vectors $\mathfrak{v} \in \mathfrak{V}$, see [Kud97, Proposition 11.1]. When $\mathfrak{v} = \mathfrak{o}$, we set $\mathfrak{g}_{\mathfrak{m}}(\mathfrak{o}, \mathfrak{w}, \mathfrak{v}) = -|\mathfrak{w}|$.

Recall the following fact. Let $\mathfrak{v}_1$ and $\mathfrak{v}_2$ be analytic cycles in a smooth manifold $\mathfrak{M}$ of codimensions $\mathfrak{d}_1$ and $\mathfrak{d}_2$, respectively, that intersect properly. Let $\mathfrak{g}_{\mathfrak{m},1}$ and $\mathfrak{g}_{\mathfrak{m},2}$ be Green currents on $\mathfrak{M}$ satisfying

$$\mathfrak{g}_{\mathfrak{m},1} + \mathfrak{g}_{\mathfrak{m},2} = [\mathfrak{g}_{\mathfrak{m}}], \quad \mathfrak{v}_1 = \mathfrak{v}_2.$$

When $\mathfrak{g}_{\mathfrak{m},1} = [\mathfrak{g}_{\mathfrak{m}}]$ is the current defined by a smooth form on $\mathfrak{M} \setminus \mathfrak{v}_2$, the *star product* is given by

$$\mathfrak{g}_{\mathfrak{m},1} \star \mathfrak{g}_{\mathfrak{m},2} = \mathfrak{g}_{\mathfrak{m},1} \wedge \mathfrak{g}_{\mathfrak{m},2} + \mathfrak{g}_{\mathfrak{m},2} \wedge \mathfrak{g}_{\mathfrak{m},1}. \quad (4.2.2)$$

Let $\mathfrak{v} = (\mathfrak{v}_1, \dots, \mathfrak{v}_r) \in \mathfrak{V}^r$ be a set of linearly independent vectors. Define the intersection of special cycles as

$$\mathfrak{P}(\mathfrak{v}) = \mathfrak{P}(\mathfrak{v}_1) \cap \dots \cap \mathfrak{P}(\mathfrak{v}_r).$$

Let $\mathfrak{g}_{\mathfrak{m}} = \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}_1, \dots, \mathfrak{v}_r) = \mathfrak{g}_{\mathfrak{m}} + \mathfrak{g}_{\mathfrak{m}} \in \mathfrak{H}(\mathfrak{V})$. *Kudla’s Green current* is defined via the star product:

$$\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{w}) := [\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}_1, \mathfrak{w})] \star \dots \star [\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}_r, \mathfrak{w})].$$

It satisfies the $(\mathfrak{v}, \mathfrak{w})$ -current equation for $\mathfrak{P}(\mathfrak{v})$ (see also [GS19, Theorem 2.7.1]):

$$\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{w}) + \mathfrak{g}_{\mathfrak{m}}(\mathfrak{w}, \mathfrak{v}) = [\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{w}) \wedge \dots \wedge \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}_r, \mathfrak{w})].$$

Remark 4.2.1. In [GS19], Garcia and Sankaran generalize Kudla’s Green function to take values for all $\mathfrak{v} \in \mathfrak{H}(\mathfrak{V})$. However, this level of generality is not needed for our purposes.

4.3. Archimedean local Siegel–Weil formula. We now relate local Whittaker functions to their classical counterparts and recall the Siegel–Weil formula. Let $\mathfrak{g}_{\mathfrak{m}}$ denote the Weil representation, and let $\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{w}, \mathfrak{v})$ be the local Whittaker function defined in §3.6. The hermitian symmetric space for $\mathfrak{g}_{\mathfrak{m}} = \mathfrak{H}(\mathfrak{V})$ is the hermitian upper half-space given in (4.1.2).

For any $\mathfrak{v} = \mathfrak{v}_1 + \mathfrak{v}_2 \in \mathfrak{H}(\mathfrak{V})$, the *classical Whittaker function* is defined as

$$\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{v}, \mathfrak{v}) := \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}_1, \mathfrak{v}_2)^{-1} \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}_2, \mathfrak{v}_1)^{-1/2} \cdot \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}_1, \mathfrak{v}_2, \mathfrak{v}), \quad \mathfrak{g}_{\mathfrak{m}} := \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}) \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}) \in \mathfrak{H}(\mathfrak{V}),$$

where $\mathfrak{v} \in \mathfrak{H}(\mathfrak{V})$ satisfies $\mathfrak{v} = \mathfrak{v}_1 + \mathfrak{v}_2$.

For the remainder of this paper, we fix $\mathfrak{v} \in \mathfrak{H}(\mathfrak{V})$ (in the archimedean place) to be the *standard Gaussian function*:

$$\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}) = \mathfrak{g}_{\mathfrak{m}}(-\mathfrak{v}, \mathfrak{v}) \in \mathfrak{H}(\mathfrak{V}), \quad (4.3.1)$$

where $\mathfrak{v}$ is a positive definite hermitian space of dimension $\mathfrak{r}$. The archimedean local (arithmetic) Siegel–Weil formula is as follows:

Theorem 4.3.1. *Let $\mathfrak{g}_{\mathfrak{m}} \in \mathfrak{H}(\mathfrak{V})$ be a nonsingular hermitian matrix, and define $\mathfrak{g}_{\mathfrak{m}} := \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{v})$.*

(1) *For $\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}) = (\mathfrak{v}, \mathfrak{v})$, we have*

$$\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}) = \mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{v}) \geq \mathfrak{v}.$$

(2) *When $\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}) = (\mathfrak{v}, \mathfrak{v})$ is positive definite,*

$$\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v}, \mathfrak{v}, \mathfrak{v}) = \frac{(\mathfrak{v}, \mathfrak{v})^{\mathfrak{r}}}{\mathfrak{g}_{\mathfrak{m}}(\mathfrak{v})}.$$

⁷Here, we recall the standard differential operators:

$$\mathfrak{g}_{\mathfrak{m}} = \partial + \bar{\partial}, \quad \mathfrak{g}_{\mathfrak{m}} = \frac{\mathfrak{v}}{\mathfrak{g}_{\mathfrak{m}}}(\partial - \bar{\partial}), \quad \mathfrak{g}_{\mathfrak{m}} = -\frac{\mathfrak{v}}{\mathfrak{g}_{\mathfrak{m}}} \partial \bar{\partial}.$$

(3) When $\mathfrak{h}(\mathfrak{g}) = (\mathfrak{g} - \mathfrak{a}, \mathfrak{a})$, the Whittaker function vanishes identically, and

$$\mathfrak{h}'(\mathfrak{g}, \mathfrak{a}, \mathfrak{b}) = \mathfrak{h}(\mathfrak{g}, \mathfrak{b}) \cdot \frac{(\mathfrak{g} - \mathfrak{a})}{\mathfrak{h}(\mathfrak{g}, \mathfrak{a})}.$$

Here, $\mathfrak{h}(\mathfrak{g}, \mathfrak{b}) = \int_{\mathfrak{g}} \mathfrak{h}(\mathfrak{g}, \mathfrak{b})$ is the local height function, where $\mathfrak{h} \in \mathfrak{g}$ is a basis with $\mathfrak{g} = (\mathfrak{h}, \mathfrak{h})$. By [Liu11a, Proposition 4.10], this local height function is independent of the choice of $\mathfrak{h}$.

Proof. Parts (1) and (2) follow from [Liu11a, Proposition 4.5], while part (3) is given in [Liu11a, Theorem 4.17]. Garcia and Sankaran give a different proof in [GS19]. $\square$

Part 2. Global theory

5. SHIMURA VARIETIES AND THEIR INTEGRAL MODELS

We now turn to the global theory. In this section, we review the RSZ Shimura varieties introduced in [KR14, RSZ21, RSZ20] and generalized in [LMZ25], along with their archimedean and non-archimedean uniformizations.

5.1. Group-theoretic data. Let $\mathfrak{g}/\mathfrak{g}_0$ be a totally imaginary quadratic extension of a totally real number field with non-trivial Galois involution $\mathfrak{g} \mapsto \mathfrak{g}_0$. Assume $\mathfrak{g}_0 \neq \mathfrak{g}$. We write $\mathfrak{g}_0$, $\mathfrak{g}_0$ for the adele rings of $\mathfrak{g}$ and $\mathfrak{g}_0$, resp. We systematically use a subscript $\mathfrak{f}$ for the ring of finite adeles, and a superscript $\mathfrak{a}$ for the adeles away from the prime number $\mathfrak{p}$.

We fix a CM type $\mathfrak{c}$ for $\mathfrak{g}$ with a distinguished element $\mathfrak{c}_0 \in \mathfrak{c}$, and let $\mathfrak{g}_{\mathfrak{c}} = \mathfrak{g} \otimes_{\mathfrak{g}_0} \mathfrak{g} \simeq \mathfrak{g}_{\mathfrak{c}_0}$. We fix a hermitian $\mathfrak{g}_{\mathfrak{c}}$ -vector space $(\mathfrak{g}, \mathfrak{h})$ of dimension $\mathfrak{d}$ such that, for $\mathfrak{g} \in \mathfrak{g}$,

$$\mathfrak{h}(\mathfrak{g}) = \begin{cases} (\mathfrak{g} - \mathfrak{c}_0, \mathfrak{c}_0) & \text{if } \mathfrak{g} = \mathfrak{c}_0, \\ (\mathfrak{g}, \mathfrak{c}_0) & \text{if } \mathfrak{g} \neq \mathfrak{c}_0. \end{cases}$$

Let $\mathfrak{g}_{\mathfrak{c}} = \mathfrak{h}(\mathfrak{g})$ be the unitary group of $\mathfrak{g}$, viewed as an algebraic group over $\mathfrak{g}_0$. Following [RSZ21, §3], we consider the following reductive groups over $\mathfrak{g}_0$, where $\mathfrak{c}(\mathfrak{g})$ is the similitude factor of $\mathfrak{g}$:

$$\begin{aligned} \mathfrak{g}_{\mathfrak{c}} &:= \{ \mathfrak{g} \in \mathfrak{g}_{\mathfrak{c}} \mid \mathfrak{g} \mathfrak{g}_0(\mathfrak{g}) \in \mathfrak{c}(\mathfrak{g}) \}, \\ \mathfrak{g}_{\mathfrak{c}}^{\sim} &:= \{ \mathfrak{g} \in \mathfrak{g}_{\mathfrak{c}} \mid \mathfrak{c}(\mathfrak{g}) \in \mathfrak{c}(\mathfrak{g}) \}, \\ \mathfrak{g}_{\mathfrak{c}}^{\sim} &:= \mathfrak{g}_{\mathfrak{c}} \times_{\mathfrak{g}_0} \mathfrak{g}_{\mathfrak{c}}^{\sim} \xrightarrow{\sim} \mathfrak{g}_{\mathfrak{c}} \times \mathfrak{g}_{\mathfrak{c}}/\mathfrak{g}_0. \end{aligned}$$

The last isomorphism is given by $(\mathfrak{g}, \mathfrak{g}) \mapsto (\mathfrak{g}, \mathfrak{g}^{-1})$. Consider the RSZ Shimura datum ([RSZ21, §3.2]),

$$(\mathfrak{g}_{\mathfrak{c}}^{\sim}, \{ \mathfrak{g}_{\mathfrak{c}}^{\sim} \}) = (\mathfrak{g}_{\mathfrak{c}}^{\sim}, \mathfrak{g}_{\mathfrak{c}}) \times (\mathfrak{g}_{\mathfrak{c}}/\mathfrak{g}_0, \{ \mathfrak{g}_{\mathfrak{c}}/\mathfrak{g}_0 \}).$$

For any neat compact open subgroup $\mathfrak{g}_{\mathfrak{c}}^{\sim} = \mathfrak{g}_{\mathfrak{c}} \times \mathfrak{g}_{\mathfrak{c}} \leq \mathfrak{g}_{\mathfrak{c}}^{\sim}(\mathfrak{g}_{\mathfrak{c}})$, we consider the *RSZ unitary Shimura variety* associated to $(\mathfrak{g}_{\mathfrak{c}}^{\sim}, \{ \mathfrak{g}_{\mathfrak{c}}^{\sim} \})$ with level $\mathfrak{g}_{\mathfrak{c}}^{\sim}$

$$\mathfrak{g}_{\mathfrak{c}}^{\sim} := \mathfrak{g}_{\mathfrak{c}}^{\sim}(\mathfrak{g}_{\mathfrak{c}}) \longrightarrow \mathfrak{g}_{\mathfrak{c}}^{\sim},$$

where $\mathfrak{g}_{\mathfrak{c}}^{\sim}$ is the reflex field. By [RSZ21, (3.4)], $\mathfrak{g}_{\mathfrak{c}}^{\sim}$ can be described through

$$\mathfrak{g}_{\mathfrak{c}}^{\sim}(\overline{\mathfrak{g}}/\mathfrak{g}_{\mathfrak{c}}) = \{ \mathfrak{g} \in \mathfrak{g}_{\mathfrak{c}}^{\sim}(\overline{\mathfrak{g}}/\mathfrak{g}_{\mathfrak{c}}) \mid \mathfrak{g} \circ \mathfrak{c}_0 \in \mathfrak{c} \text{ and } \mathfrak{h}(\mathfrak{g}_{\mathfrak{c}}) = \mathfrak{h}(\mathfrak{g}_{\mathfrak{c}}) \text{ for all } \mathfrak{g} \in \mathfrak{g}_{\mathfrak{c}} \}.$$

The goal of this section is recall the definition of the (punctured) integral model $\mathfrak{g}_{\mathfrak{c}}^{\sim} \rightarrow \mathfrak{g}_{\mathfrak{c}}^{\sim}$ and its uniformization.

5.2. Integral models of toric Shimura varieties. We first recall the integral model for $\mathfrak{g}_{\mathfrak{c}}^{\sim}(\mathfrak{g}_{\mathfrak{c}})$, see [RSZ21, §3.1]. Let $\mathfrak{p} \in \mathfrak{g}_{\mathfrak{c}}^{\sim}$ be a product of primes such that the neat level $\mathfrak{g}_{\mathfrak{c}}^{\sim}$ factors as

$$\mathfrak{g}_{\mathfrak{c}}^{\sim} = \mathfrak{g}_{\mathfrak{c}}^{\sim} \times \mathfrak{g}_{\mathfrak{c}}^{\sim}$$

where $\mathfrak{g}_{\mathfrak{c}}^{\sim}$ is the maximal compact subgroup in $\mathfrak{g}_{\mathfrak{c}}^{\sim}(\mathfrak{g}_{\mathfrak{c}})$.

As an auxiliary datum, we also fix a hermitian $\mathfrak{g}_{\mathfrak{c}}/\mathfrak{g}_0$ -module $\mathfrak{g}_{\mathfrak{c}}$ of rank $\mathfrak{d}$. Following [RSZ20, §3.2] and [RSZ21, Definition 3.6, §4.1], fixing an ideal $\mathfrak{f}$ of $\mathfrak{g}_{\mathfrak{c}}$, we define

$$\mathfrak{g}_{\mathfrak{c}}^{\sim} \longrightarrow \mathfrak{g}_{\mathfrak{c}}^{\sim}[\mathfrak{f}^{-1}]$$

as the functor sending a scheme $\mathfrak{S}$ over $\mathfrak{g}_{\mathfrak{c}}^{\sim}[\mathfrak{f}^{-1}]$ to the set of isomorphism classes of tuples $(\mathfrak{g}_{\mathfrak{c}}, \mathfrak{g}_{\mathfrak{c}}, \mathfrak{g}_{\mathfrak{c}}, \mathfrak{g}_{\mathfrak{c}})$ where

- (1) $\mathcal{A}_{\mathfrak{p}}$ is an abelian scheme over $\mathcal{M}$ of dimension $[\mathfrak{p} : \mathbb{Q}]$.
- (2) $\mathfrak{p} : \mathcal{A}_{\mathfrak{p}}[\mathfrak{p}^{-\bullet}] \rightarrow \mathcal{A}_{\mathfrak{p}}[\mathfrak{p}^{-\bullet}]$ is an $\mathfrak{p}$ -action that is $(\mathfrak{p}, \mathfrak{p})$ -strict, see [LMZ25, §6.2].
- (3) $\mathfrak{p} \in \mathcal{P}(\mathcal{A}_{\mathfrak{p}}, \mathcal{A}_{\mathfrak{p}}^{\vee})[\mathfrak{p}^{-\bullet}]$ is a *prime-to- $\mathfrak{p}$ polarization*, meaning it is a quasi-isogeny such that locally on $\mathcal{M}$ some $\mathfrak{p}$ -multiple is a polarization.
- (4) We require that $\mathfrak{p}$ and $\mathfrak{p}$ are compatible in the sense that for all $\mathfrak{p} \in \mathcal{P}_{\mathfrak{p}}$, we have

$$\mathfrak{p}^{-\bullet} \circ \mathfrak{p}(\mathfrak{p})^{\vee} \circ \mathfrak{p} = \mathfrak{p}(\overline{\mathfrak{p}}).$$

- (5) We require that $\mathfrak{p}(\mathfrak{p})[\mathfrak{p}^{-\bullet}] = \mathfrak{p}[\mathfrak{p}][\mathfrak{p}^{-\bullet}]$, where $\mathfrak{p} \geq \mathfrak{p}$ is chosen (locally on $\mathcal{M}$) such that $\mathfrak{p} \in \mathcal{P}(\mathcal{A}_{\mathfrak{p}}, \mathcal{A}_{\mathfrak{p}}^{\vee})$ (and the space is independent of the choice of $\mathfrak{p}$).
- (6) $\overline{\mathfrak{p}}$ is a $\mathfrak{p}$ -level structure in the sense of [Liu21, C.3] using $\mathfrak{p}$.

One may always choose $\mathfrak{p}$ and $\mathfrak{p}$ such that $\mathfrak{p}'$ is non-empty, and we fix such choices. Then $\mathfrak{p}'$ is representable by a finite étale $\mathfrak{p}$ -scheme. As in [RSZ21, §3.4], its generic fiber $(\mathfrak{p}')_{\mathfrak{p}}$ is a disjoint union of copies of the Shimura variety $\mathfrak{p}(\mathfrak{p}, \mathfrak{p})$. We work with a fixed copy of this union, and let $\mathfrak{p} \subseteq \mathfrak{p}'$ be its closure. This is an open and closed subscheme of $\mathfrak{p}'$ which provides the finite, étale integral model of $\mathfrak{p}(\mathfrak{p}, \mathfrak{p})$.

5.3. Punctured integral model. Now we define the punctured integral model of $\mathfrak{p}(\mathfrak{p}, \mathfrak{p})$. Let $\mathfrak{p}$ be an $\mathfrak{p}$ -lattice in $\mathfrak{p}$ and let $\mathfrak{p} \in \mathfrak{p}_{\mathfrak{p}}$ be chosen such that:

- If $\mathfrak{p}$ is a finite place of $\mathfrak{p}$ that is ramified in $\mathfrak{p}$, then $\mathfrak{p} \mid \mathfrak{p}$.
- For any finite non-split place $\mathfrak{p} \nmid \mathfrak{p}$ of $\mathfrak{p}$, the localization $\mathfrak{p}_{\mathfrak{p}}$ in $\mathfrak{p}_{\mathfrak{p}} = \mathfrak{p} \otimes_{\mathfrak{p}} \mathfrak{p}_{\mathfrak{p}}$ is a vertex lattice, i.e. $\mathfrak{p}_{\mathfrak{p}} \subseteq \mathfrak{p}_{\mathfrak{p}} \subseteq \mathfrak{p}_{\mathfrak{p}}^{\vee}$, where $\mathfrak{p}_{\mathfrak{p}} \in \mathfrak{p}_{\mathfrak{p}}$ is a uniformizer.
- The level $\mathfrak{p} = \mathfrak{p}$ factors as

$$\mathfrak{p} = \mathfrak{p} \times \mathfrak{p}$$

where $\mathfrak{p} \subset \mathfrak{p}(\mathfrak{p}, \mathfrak{p})$ is any open compact subgroup and where $\mathfrak{p} = \mathfrak{p}(\widehat{\mathfrak{p}})$ is the stabilizer of the completion $\widehat{\mathfrak{p}}$ in $\mathfrak{p}(\mathfrak{p}, \mathfrak{p})$.

- The level $\mathfrak{p}$ factors similarly as

$$\mathfrak{p} = \mathfrak{p}_{\mathfrak{p}} \times \mathfrak{p}_{\mathfrak{p}}$$

where $\mathfrak{p}_{\mathfrak{p}} \subset \mathfrak{p}_{\mathfrak{p}}(\mathfrak{p})$ is any open compact subgroup and where $\mathfrak{p}_{\mathfrak{p}}$ is the maximal compact subgroup in $\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p})$.

Definition 5.3.1. The *punctured integral model* of the RSZ Shimura variety

$$\widetilde{\mathfrak{p}} \rightarrow \mathfrak{p}[\mathfrak{p}^{-\bullet}]$$

is the functor that takes an $\mathfrak{p}[\mathfrak{p}^{-\bullet}]$ -scheme $\mathcal{M}$ to the set of isomorphism classes of tuples

$$(\mathcal{A}_{\mathfrak{p}}, \mathfrak{p}, \mathfrak{p}, \overline{\mathfrak{p}}; \mathfrak{p}, \mathfrak{p}, \overline{\mathfrak{p}})$$

that are of the following form:

- (1) $(\mathcal{A}_{\mathfrak{p}}, \mathfrak{p}, \mathfrak{p}, \overline{\mathfrak{p}}) \in \mathfrak{p}(\mathcal{M})$.
- (2) $\mathcal{A}_{\mathfrak{p}}$ is an abelian scheme over $\mathcal{M}$ of dimension $[\mathfrak{p} : \mathbb{Q}]$.
- (3) $\mathfrak{p} : \mathcal{A}_{\mathfrak{p}}[\mathfrak{p}^{-\bullet}] \rightarrow \mathcal{A}_{\mathfrak{p}}[\mathfrak{p}^{-\bullet}]$ is an $\mathfrak{p}$ -action that is of signature $(\mathfrak{p} - \mathfrak{p}, \mathfrak{p})$ in the sense of [LMZ25, Definition 6.2].
- (4) $\mathfrak{p} \in \mathcal{P}(\mathcal{A}_{\mathfrak{p}}, \mathcal{A}_{\mathfrak{p}}^{\vee})[\mathfrak{p}^{-\bullet}]$ is a prime-to- $\mathfrak{p}$ polarization.
- (5) We require $\mathfrak{p}$ and $\mathfrak{p}$ to be compatible in the sense that for $\mathfrak{p} \in \mathcal{P}_{\mathfrak{p}}[\mathfrak{p}^{-\bullet}]$ we have $\mathfrak{p}^{-\bullet} \circ \mathfrak{p}(\mathfrak{p})^{\vee} \circ \mathfrak{p} = \mathfrak{p}(\overline{\mathfrak{p}})$. We also require that for every place $\mathfrak{p} \nmid \mathfrak{p}$ of $\mathfrak{p}$, corresponding to a prime ideal $\mathfrak{p} \subset \mathfrak{p}[\mathfrak{p}^{-\bullet}]$, that $\mathfrak{p}[\mathfrak{p}][\mathfrak{p}^{-\bullet}]$ is annihilated by $\mathfrak{p}_{\mathfrak{p}}$ and of order $|\mathfrak{p}_{\mathfrak{p}}^{\vee}/\mathfrak{p}_{\mathfrak{p}}|$.
- (6) $\overline{\mathfrak{p}}$ is a $\mathfrak{p}$ -orbit of isometries of hermitian $\prod_{\mathfrak{p} \mid \mathfrak{p}} \mathfrak{p}$ -sheaves over $\mathcal{M}$:

$$\mathfrak{p} : \mathfrak{p}(\mathcal{A}_{\mathfrak{p}}, \mathcal{A}_{\mathfrak{p}}) \xrightarrow{\sim} \mathfrak{p}(\mathfrak{p}, \mathfrak{p}).$$

Here, the notation is as follows. First,

$$\mathfrak{p}(\mathcal{A}_{\mathfrak{p}}, \mathcal{A}_{\mathfrak{p}}) := \prod_{\mathfrak{p} \mid \mathfrak{p}} \mathfrak{p}(\mathcal{A}_{\mathfrak{p}}, \mathcal{A}_{\mathfrak{p}}) \quad \text{and} \quad \mathfrak{p}(\mathcal{A}_{\mathfrak{p}}, \mathcal{A}_{\mathfrak{p}}) = \mathfrak{p} \otimes_{\mathfrak{p}} \mathfrak{p}(\mathfrak{p}(\mathcal{A}_{\mathfrak{p}}), \mathfrak{p}(\mathcal{A}_{\mathfrak{p}})),$$

where $\mathfrak{p}(\mathcal{A}_{\mathfrak{p}})$ and $\mathfrak{p}(\mathcal{A}_{\mathfrak{p}})$ denote the rational Tate modules at $\mathfrak{p}$; these are $\mathfrak{p}$ -sheaves on $\mathcal{M}$. Here, $\mathfrak{p}(\mathfrak{p}, \mathfrak{p})$ is made into a hermitian $\mathfrak{p}$ -vector space by

$$(\mathfrak{p}, \mathfrak{p}) := \mathfrak{p}^{-\bullet} \circ \mathfrak{p}^{\vee} \circ \mathfrak{p} \in \mathfrak{p}(\mathfrak{p}(\mathcal{A}_{\mathfrak{p}})) = \mathfrak{p}.$$

Second, we consider

$$\mathbb{G}(\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}) := \prod_{\mathfrak{p} \mid \mathfrak{p}} \mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \otimes_{\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}} \mathbb{G}_{\mathfrak{p}, \mathfrak{p}} = \prod_{\mathfrak{p} \mid \mathfrak{p}} \mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \otimes_{\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}} \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$$

as locally constant $\prod_{\mathfrak{p} \mid \mathfrak{p}} \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ sheaf on $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$.

An isomorphism between two such tuples $(\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}; \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}) \cong (\mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}; \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}})$ is given by an isomorphism $(\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}) \cong (\mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}})$, and a prime-to- $\mathfrak{p}$ isogeny $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \rightarrow \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}$ which induces isomorphisms $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}[\mathfrak{p}^{\infty}] \cong \mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}[\mathfrak{p}^{\infty}]$ ($\mathfrak{p} \nmid \mathfrak{p}$) compatible with $\mathfrak{p}$ and $\mathfrak{p}'$, $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ and $\mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}$, $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ and $\mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}$, $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ and $\mathbb{G}'_{\mathfrak{p}, \mathfrak{p}}$.

Proposition 5.3.2 ([LMZ25, Proposition 6.2]). $\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}$ is representable by a flat projective $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}[\mathfrak{p}^{\infty}]$ -scheme (recall that $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \neq \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$). $\square$

5.4. Basic uniformization. We now recall the $\mathfrak{p}$ -adic uniformization of $\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}$ along the basic locus.

Let $\mathfrak{p} \nmid \mathfrak{p}$ be a non-archimedean place of $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$, whose restriction to $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ (resp. $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$) is denoted by $\mathfrak{p}$ (resp. $\mathfrak{p}$). Assume that $\mathfrak{p}$ is *inert*. By [LMZ25, Theorem 7.3], we have the following non-archimedean uniformization along the basic locus:

$$\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}^{\wedge} = \widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) \setminus \left[\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}^{[b]} \times \widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) / \mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \right]. \quad (5.4.1)$$

Here, the left-hand side denotes the formal completion of the base change $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \otimes_{\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}} \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ along the basic locus of the geometric special fiber $\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}$. The basic locus consists of points $(\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}, \mathbb{G}_{\mathfrak{p}, \mathfrak{p}})$ such that $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}[\mathfrak{p}^{\infty}]$ is a supersingular $\mathfrak{p}$ -divisible group.

On the right-hand side, $\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p})$ is the analogue of $\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}$ for the nearby hermitian space $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p})$ (see (G2)), where $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ is fixed in §5.1. The formal scheme $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}^{[b]}$ is the (relative) Rapoport–Zink space for the quadratic extension $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}/\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$, where $\mathfrak{p}$ denotes the type of the vertex lattice $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$.

The uniformization (5.4.1) induces a projection onto a discrete set:

$$\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}^{\wedge} \longrightarrow \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) \setminus (\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) / \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}). \quad (5.4.2)$$

This provides a partition of the formal scheme $\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}^{\wedge}$, where each fiber is naturally isomorphic to

$$\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}^{\wedge} := \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) \setminus \left[\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}^{[b]} \times \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) / \mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \right]. \quad (5.4.3)$$

5.5. Complex uniformization. Let $\mathfrak{p} : \mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \hookrightarrow \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ be a complex place of the reflex field $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$. Its restriction to $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ (resp. $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$) is denoted by $\mathfrak{p}$ (resp. $\mathfrak{p}$). Define the complex manifold $\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}} := \mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \otimes_{\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}} \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ via $\mathfrak{p}$.

Let $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p})$ be the “nearby” hermitian space defined in (G2), i.e., the unique space that is positive definite at all archimedean places except at $\mathfrak{p}$, where it has signature $(\mathfrak{p} - \mathfrak{p}, \mathfrak{p})$, and is locally isomorphic to $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ at all non-archimedean places, where $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ is fixed in §5.1.

Let $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p})$ (resp. $\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p})$) be the unitary group (resp. unitary similitude group), viewed as a $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ -algebraic group (resp. $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$ -algebraic group), associated with $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p})$. Let $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p})$ be the hermitian symmetric space for $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p})$ defined in (4.1.1), which is the Grassmannian of negative definite $\mathfrak{p}$ -lines in $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) \otimes_{\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}} \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}$.

Then, we have the following complex uniformization:

$$\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}} = \widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) \setminus \left[\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}^{[b]} \times \widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) / \mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \right].$$

We obtain a partition via the projection

$$\widetilde{\mathbb{G}}_{\mathfrak{p}, \mathfrak{p}} \longrightarrow \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) \setminus (\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) / \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}), \quad (5.5.1)$$

where each fiber is naturally isomorphic to

$$\mathbb{G}_{\mathfrak{p}, \mathfrak{p}} := \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) \setminus \left[\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}^{[b]} \times \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) / \mathbb{G}_{\mathfrak{p}, \mathfrak{p}} \right].$$

Here, we fix an isomorphism $\mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p}) \simeq \mathbb{G}_{\mathfrak{p}, \mathfrak{p}}(\mathfrak{p})$.

6. KUDLA-RAPOPORT SPECIAL CYCLES

In this section, we define weighted Kudla–Rapoport special cycles and their derived variants. We establish their complex and $\mathfrak{p}$ -adic uniformizations and conclude by introducing their Green currents in the complex fibers.

6.1. Weighted special cycles. We keep the notation from §5. For any point $(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee, \mathfrak{z}_\bullet) \in \widetilde{\mathcal{H}}(\mathfrak{z}_\bullet)$, consider the hermitian form $\mathfrak{h}'(-, -)$ on $\mathfrak{H}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)[\mathfrak{p}^{-\bullet}]$ given by

$$\mathfrak{h}'(\mathfrak{v}_\bullet, \mathfrak{w}_\bullet) = \mathfrak{v}_\bullet \circ \mathfrak{w}_\bullet^\vee \circ \mathfrak{v}_\bullet \circ \mathfrak{w}_\bullet \in \mathfrak{H}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)[\mathfrak{p}^{-\bullet}] \simeq \mathfrak{H}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)[\mathfrak{p}^{-\bullet}].$$

By [KR14, Lemma 2.7], this is totally positive definition. The polarization $\mathfrak{h} : \mathfrak{z}_\bullet \rightarrow \mathfrak{z}_\bullet^\vee$ induces an injection

$$\mathfrak{H}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)[\mathfrak{p}^{-\bullet}] \hookrightarrow \mathfrak{H}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)[\mathfrak{p}^{-\bullet}].$$

We first define global admissible Schwartz functions and associated special cycles, extending the local definition given in Definition 2.3.3.

Definition 6.1.1. Recall from §5.3 that we have fixed an integral lattice $\mathfrak{H} \subset \mathfrak{E}$. For a weight vector $\mathfrak{w} \in \{\mathfrak{p}, \mathfrak{q}\}^\mathfrak{g}$, we construct the Schwartz function $\mathfrak{w}(\mathfrak{z}_\bullet) \in \mathfrak{W}(\mathfrak{E}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee))$ such that

$$\mathfrak{w}(\mathfrak{z}_\bullet) = \prod_{\mathfrak{v}} \mathfrak{w}(\mathfrak{z}_{\mathfrak{v}}), \quad \text{where} \quad \mathfrak{w}(\mathfrak{z}_{\mathfrak{v}}) = \begin{cases} \mathfrak{H}[\mathfrak{v}], & \text{if } \mathfrak{w} = \mathfrak{p}; \\ \mathfrak{H}[\mathfrak{v}], \mathfrak{H}[\mathfrak{v}], & \text{if } \mathfrak{w} = \mathfrak{q}. \end{cases}$$

A Schwartz function $\mathfrak{w} \in \mathfrak{W}(\mathfrak{E}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee))$ is called *admissible* if there exists a weight vector $\mathfrak{w}$ such that $\mathfrak{w} = \mathfrak{w}(\mathfrak{z}_\bullet)$, cf. Definition 2.3.3.

Definition 6.1.2. Fix an integer $\mathfrak{o} < \mathfrak{p} \leq \mathfrak{q}$. Let $\mathfrak{w} \in \mathfrak{W}(\mathfrak{E}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee))$ be an admissible Schwartz function. Let $\mathfrak{g} \in \mathfrak{O}(\mathfrak{z}_\bullet)$ be any hermitian matrix, and let $\mathfrak{r} \in \mathfrak{E}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)/\mathfrak{H}$ be a right $\mathfrak{H}$ -orbit. The *weighted Kudla–Rapoport cycle* $\widetilde{\mathfrak{w}}(\mathfrak{g}, \mathfrak{r}, \mathfrak{w})$ associates to each scheme $\mathfrak{M}$ over $\mathfrak{H}[\mathfrak{p}^{-\bullet}]$ the groupoid of tuples $(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee, \mathfrak{z}_\bullet, \mathfrak{r}, \mathfrak{g}, \mathfrak{w})$, where:

- $(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee, \mathfrak{z}_\bullet, \mathfrak{r}, \mathfrak{g}, \mathfrak{w}) \in \widetilde{\mathcal{H}}(\mathfrak{M})$.
- $\mathfrak{g} \in \mathfrak{H}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)[\mathfrak{p}^{-\bullet}] \otimes \mathfrak{H}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)[\mathfrak{p}^{-\bullet}]$ satisfies $\mathfrak{h}'(\mathfrak{g}, \mathfrak{g}) = \mathfrak{g}$ and $\mathfrak{r}(\mathfrak{g}) \in \mathfrak{r} \subset \mathfrak{E}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)$.
- For each $\mathfrak{a} \leq \mathfrak{b} \leq \mathfrak{p}$, we impose the following conditions:
 - If $\mathfrak{w}_{\mathfrak{a}} = \mathfrak{H}[\mathfrak{a}]$, then $\mathfrak{w} \in \mathfrak{H}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)[\mathfrak{p}^{-\mathfrak{a}}]$.
 - If $\mathfrak{w}_{\mathfrak{a}} = \mathfrak{H}[\mathfrak{a}], \mathfrak{H}[\mathfrak{a}]$, then $\mathfrak{w} \in \mathfrak{H}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)[\mathfrak{p}^{-\mathfrak{a}}]$.

A morphism between two objects $(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee, \mathfrak{z}_\bullet, \mathfrak{r}, \mathfrak{g}, \mathfrak{w})$ and $(\mathfrak{z}'_\bullet, \mathfrak{z}'_\bullet^\vee, \mathfrak{z}'_\bullet, \mathfrak{r}', \mathfrak{g}', \mathfrak{w}')$ consists of an isomorphism $(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee, \mathfrak{z}_\bullet, \mathfrak{r}) \xrightarrow{\sim} (\mathfrak{z}'_\bullet, \mathfrak{z}'_\bullet^\vee, \mathfrak{z}'_\bullet, \mathfrak{r}')$ in $\mathcal{H}(\mathfrak{M})$ and an $\mathfrak{H}$ -linear prime-to- $\mathfrak{p}$ isogeny $\mathfrak{h} : \mathfrak{z}_\bullet \rightarrow \mathfrak{z}'_\bullet$, compatible with $\mathfrak{w}$ and $\mathfrak{w}'$, with $\mathfrak{r}$ and $\mathfrak{r}'$, and satisfying $\mathfrak{g}' = \mathfrak{h} \circ \mathfrak{g}$.

We recall some basic properties of special cycles.

Proposition 6.1.3. (1) The morphism $\mathfrak{s} : \widetilde{\mathfrak{w}}(\mathfrak{g}, \mathfrak{r}, \mathfrak{w}) \rightarrow \mathcal{H}$ is representable, finite, and unramified.

(2) When $\mathfrak{p} = \mathfrak{p}$ and $\mathfrak{q} > \mathfrak{p}$ is totally positive, the morphism $\mathfrak{s}$ defines an étale locally Cartier divisor.

(3) Let $\mathfrak{g} \in \mathfrak{O}(\mathfrak{z}_\bullet)$ for $\mathfrak{p} = \mathfrak{p}$. Decompose $\mathfrak{r} \times \mathfrak{r} = \coprod_{\mathfrak{v}} \mathfrak{r}_{\mathfrak{v}} \subset \mathfrak{E}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)^{\mathfrak{p} + \mathfrak{q}}$ into $\mathfrak{H}$ -orbits. Then, the scheme-theoretic fiber product decomposes as

$$\widetilde{\mathfrak{w}}(\mathfrak{g}, \mathfrak{r}, \mathfrak{w}) \times_{\mathcal{H}} \widetilde{\mathfrak{w}}(\mathfrak{g}, \mathfrak{r}, \mathfrak{w}) = \prod_{\substack{\mathfrak{g} = \begin{pmatrix} \mathfrak{g} & * \\ * & \mathfrak{g} \end{pmatrix} \\ \mathfrak{g} \in \mathfrak{O}(\mathfrak{z}_\bullet) \\ \mathfrak{g} \in \mathfrak{H}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee)[\mathfrak{p}^{-\bullet}]}} \prod_{\mathfrak{v}} \mathfrak{w}(\mathfrak{z}_{\mathfrak{v}}, \mathfrak{r}_{\mathfrak{v}}, \mathfrak{w}_{\mathfrak{v}}),$$

where $\mathfrak{w}_{\mathfrak{v}} = \mathfrak{w}_{\mathfrak{p}} + \mathfrak{w}_{\mathfrak{q}}$ and $\mathfrak{w} = \mathfrak{w}_{\mathfrak{p}} \otimes \mathfrak{w}_{\mathfrak{q}} \in \mathfrak{W}(\mathfrak{E}(\mathfrak{z}_\bullet, \mathfrak{z}_\bullet^\vee))$.

Proof. Following the argument in [Che24c, Lemma A.2.2] (see also [AGHMP17, Proposition 2.7.2]), the morphism $\mathfrak{s}$ is quasi-compact. Then, part (1) follows from [KR14, Proposition 2.9], see also [Liu21, Proposition 5.22(1)].

For part (2), the key idea originates from [KR11, Proposition 3.5], while a complete proof using the line bundle of modular forms can be found in [Liu21, Proposition 5.22(3)]. In our setting, these line bundles are constructed in [LMZ25, §6.2] via the Eisenstein condition (see also equation

(6.2.2)). We note that proving $\widetilde{\text{Pink}}(\bullet, \text{tr}, \text{tr}, \text{tr})$ is a Cartier divisor requires the line bundle of modular forms for GL_2 , see [HLS24, §5].

Part (3) follows directly from the moduli description, observing that in the summation on the right-hand side, all but finitely many terms vanish. $\square$

6.2. Derived special cycle. In this subsection, we define derived special cycles over the RSZ integral model. Since the integral model $\widetilde{\text{Pink}} \rightarrow \text{GL}_2[\text{tr}]$ is generally not regular, it requires extra care in the definition. Instead of working with the Pink -group of perfect complexes, we consider the Pink -group of coherent sheaves (which we refer to as the Pink -group).

An alternative approach to constructing derived special cycles is through the higher Gysin map in the Chow group, as described in [Ful98] and [Sta18, OFBI]. For further discussion, see [LMZ25, Remark 6.16]. To prove the local Kudla–Rapoport conjecture, we may carefully choose the global embedding so that $\widetilde{\text{Pink}}$ is regular, see Remark 11.0.5.

Let tr be a noetherian scheme, and let $\text{Pink}(\text{tr})$ denote the Pink -group of coherent sheaves on tr . For each $n \geq 0$, define $\text{Pink}_n(\text{tr}) \subseteq \text{Pink}(\text{tr})$ as the subgroup generated by coherent Pink -modules whose support has dimension at most n .

Given a finite morphism $\text{tr} \rightarrow \text{tr}$, let $\text{Pink}_n(\text{tr})$ be the subgroup generated by coherent Pink -modules whose support is contained in the image of tr . For a comprehensive treatment of intersection pairing in the setting of finite morphisms, we refer the reader to [HM22, Appendix A]; see also [Zha21, Appendix B].

Definition 6.2.1. Let $\text{Pink}(\text{tr})$ denote the set of all hermitian matrices with totally definite diagonal entries.

(1) Let $n < \text{tr} \leq \text{tr}$ be an integer, and consider $\text{tr} \in \text{Pink}(\text{tr})$, an admissible Schwartz function $\text{tr} \in \text{Pink}(\text{tr})$, and $\text{tr} \in \text{Pink}(\text{tr})/\text{Pink}$. Suppose tr has main diagonal entries $\text{tr}, \dots, \text{tr}$, and choose a vector $(\text{tr}, \dots, \text{tr}) \in \text{tr}$. We define the *derived Kudla–Rapoport cycle* $\widetilde{\text{Pink}}(\text{tr}, \text{tr}, \text{tr}) \in \text{Pink}_n(\widetilde{\text{Pink}}(\text{tr}, \text{tr}, \text{tr}))$ as

$$\text{Pink}(\text{tr}, \text{tr}, \text{tr}) \otimes \text{Pink} \cdots \otimes \text{Pink} \text{Pink}(\text{tr}, \text{tr}, \text{tr}) \Big|_{\widetilde{\text{Pink}}(\text{tr}, \text{tr}, \text{tr})} \in \text{Pink}_n(\widetilde{\text{Pink}}(\text{tr}, \text{tr}, \text{tr})).$$

By Proposition 6.1.3 and [GW23, Proposition 23.58], together with the fact that all morphisms are finite and unramified, the derived tensor product is well-defined and lies in $\text{Pink}_n(\widetilde{\text{Pink}})_{\text{tr}}$. Furthermore, this derived tensor product naturally carries a module structure over

$$\text{Pink}(\text{tr}, \text{tr}, \text{tr}) \otimes \text{Pink} \cdots \otimes \text{Pink} \text{Pink}(\text{tr}, \text{tr}, \text{tr}).$$

We apply the idempotent corresponding to $\text{Pink}(\text{tr}, \text{tr}, \text{tr})$ (see Proposition 6.1.3(3)) to obtain the restriction of the complex to $\widetilde{\text{Pink}}(\text{tr}, \text{tr}, \text{tr})$. Hence we see that $\widetilde{\text{Pink}}(\text{tr}, \text{tr}, \text{tr}) \in \text{Pink}_n(\widetilde{\text{Pink}}(\text{tr}, \text{tr}, \text{tr}))_{\text{tr}}$.

(2) A Schwartz function $\text{tr} \in \text{Pink}(\text{tr})$ is called *admissible* if tr is admissible, i.e., $\text{tr} = \text{tr}(\text{tr})$ for some weighted vector $\text{tr} \in \{\text{tr}, \text{tr}\}$. For any $\text{tr} \in \text{Pink}(\text{tr})$, we define the *derived averaged weighted special cycle* as

$$\widetilde{\text{Pink}}(\text{tr}, \text{tr}) = \sum_{\text{tr} \in \text{Pink}(\text{tr})/\text{Pink}} \text{tr}(\text{tr}) \widetilde{\text{Pink}}(\text{tr}, \text{tr}) \in \bigoplus_{\substack{\text{tr} \in \text{Pink}(\text{tr})/\text{Pink} \\ \text{tr} \text{ admissible}}} \text{Pink}_n(\widetilde{\text{Pink}}(\text{tr}, \text{tr}))_{\text{tr}}. \quad (6.2.1)$$

Since the Schwartz function tr is Pink -invariant and has compact support, the summation in (6.2.1) is finite, so the right-hand side of (6.2.1) can be replaced by a finite direct sum.

Proposition 6.2.2. Let $\text{tr} \in \text{Pink}(\text{tr})$ be hermitian matrices for $\text{tr} \in \{\text{tr}, \text{tr}\}$, and set $\text{tr} = \text{tr} + \text{tr}$. For any cosets $\text{tr} \in \text{Pink}(\text{tr})/\text{Pink}$ for $\text{tr} \in \{\text{tr}, \text{tr}\}$, the derived intersection

$$\widetilde{\text{Pink}}(\text{tr}, \text{tr}) \cap \widetilde{\text{Pink}}(\text{tr}, \text{tr})$$

is defined as the restriction of the derived tensor product

$$\widetilde{\text{Pink}}(\text{tr}, \text{tr}) \otimes \text{Pink} \widetilde{\text{Pink}}(\text{tr}, \text{tr})$$

from $\text{Pink}_n(\widetilde{\text{Pink}})$ to $\text{Pink}_n(\widetilde{\text{Pink}}(\text{tr}, \text{tr}) \times \widetilde{\text{Pink}}(\text{tr}, \text{tr}))$. Alternatively, it is the restriction of

$$\text{Pink}(\text{tr}, \text{tr}) \otimes \text{Pink} \cdots \otimes \text{Pink} \text{Pink}(\text{tr}, \text{tr})$$

to $\widetilde{\mathfrak{p}}(\mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}) \times \widetilde{\mathfrak{p}}(\mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j})$, where $\mathfrak{g}$ are the diagonal entries of $\mathfrak{g}$ and $\mathfrak{h}$. These two constructions agrees in the derived category, then we pass it into $\mathfrak{g}$ -group. In particular, it lies in the $\mathfrak{g}$ -th filtration.

(1) Decompose $\mathfrak{g} \times \mathfrak{g} = \coprod_{\mathfrak{g}} \mathfrak{g} \subset \mathfrak{g}(\mathfrak{g}, \mathfrak{h})$ into $\mathfrak{g}$ -orbits, we obtain the decomposition:

$$\widetilde{\mathfrak{p}}(\mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}) \cap \widetilde{\mathfrak{p}}(\mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}) = \sum_{\substack{\mathfrak{g} = \begin{pmatrix} \mathfrak{g} & * \\ * & \mathfrak{g} \end{pmatrix} \\ \mathfrak{g} \in \mathfrak{g} \cap \mathfrak{h}(\mathfrak{g})}} \sum_{\mathfrak{g}} \widetilde{\mathfrak{p}}(\mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}).$$

(2) For $\mathfrak{g} \in \{\mathfrak{g}, \mathfrak{h}\}$, let $\mathfrak{g} \in \mathfrak{g}(\mathfrak{g}, \mathfrak{h})$ be an admissible Schwartz function. Then, we have

$$\widetilde{\mathfrak{p}}(\mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}) \cap \widetilde{\mathfrak{p}}(\mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}) = \sum_{\substack{\mathfrak{g} = \begin{pmatrix} \mathfrak{g} & * \\ * & \mathfrak{g} \end{pmatrix} \\ \mathfrak{g} \in \mathfrak{g} \cap \mathfrak{h}(\mathfrak{g})}} \widetilde{\mathfrak{p}}(\mathfrak{g}, \mathfrak{h}),$$

where $\mathfrak{g} = \mathfrak{g} \otimes \mathfrak{g} \in \mathfrak{g}(\mathfrak{g}, \mathfrak{h})$.

Proof. Part (1) follows directly from the definition by applying Proposition 6.1.3(2) repeatedly, along with [GW23, Proposition 23.58] (or equivalently, using [GW23, Definition 23.52 and Definition 23.57]). Moreover, all indices in the summation belong to $\mathfrak{g} \cap \mathfrak{h}(\mathfrak{g})$ since $\mathfrak{g} \in \mathfrak{g} \cap \mathfrak{h}(\mathfrak{g})$. Part (2) follows immediately from part (1). $\square$

Remark 6.2.3. For any Schwartz function $\mathfrak{g} \in \mathfrak{g}(\mathfrak{g}, \mathfrak{h})$, we define

$$\widetilde{\mathfrak{p}}(\mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}) := -\mathfrak{g}(\mathfrak{g}) \otimes \mathfrak{g}, \quad \text{and} \quad \widetilde{\mathfrak{p}}(\mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}) := -\mathfrak{g}(\mathfrak{g}) \otimes \mathfrak{g},$$

where

$$\mathfrak{g} \otimes \mathfrak{g} = (\mathfrak{g} \otimes \mathfrak{g}) \otimes (\mathfrak{g} \otimes \mathfrak{g})^\vee, \quad \text{and} \quad \mathfrak{g} \otimes \mathfrak{g} = (\mathfrak{g} \otimes \mathfrak{g}) \otimes (\mathfrak{g} \otimes \mathfrak{g}). \quad (6.2.2)$$

are line bundles of modular forms. Using these definitions, we can construct derived special cycles for hermitian matrices whose diagonal entries are either zero or totally positive definite (for all other hermitian matrices, the special cycles are empty).

An open problem remains to establish linear invariance relations analogous to those in [How19, §5], see also [Che24a, §3.6].

6.3. Special cycles in the formal neighborhood of the basic locus. We consider the restriction of the special cycles to the formal completion of $\widetilde{\mathfrak{p}}$ along the basic locus. Let $\mathfrak{g} \nmid \mathfrak{g}$ be a non-archimedean place of $\mathfrak{g}$, whose restriction to $\mathfrak{g}$ (resp. $\mathfrak{g}$) is denoted by $\mathfrak{g}$ (resp. $\mathfrak{g}$). Assume that $\mathfrak{g}$ is inert.

Let $\mathfrak{g}^{[b]} = \mathfrak{g}^{[b]} / \mathfrak{g}^{[b]} \rightarrow \mathfrak{g} \otimes \mathfrak{g}$ be the relative Rapoport-Zink space, where $\mathfrak{g}$ denotes the type of the vertex lattice $\mathfrak{g} \subset \mathfrak{g}(\mathfrak{g}, \mathfrak{h})$. Consider its base change

$$\mathfrak{g}^{[b]}_{\mathfrak{g}, \mathfrak{g}} = \mathfrak{g}^{[b]} \widehat{\otimes}_{\mathfrak{g} \otimes \mathfrak{g}} \mathfrak{g}.$$

Recalling from (5.4.1), we have the uniformization:

$$\mathfrak{g}^{[b]}_{\mathfrak{g}, \mathfrak{g}} = \mathfrak{g}^{[b]}(\mathfrak{g}) \setminus \left[\mathfrak{g}^{[b]}_{\mathfrak{g}, \mathfrak{g}} \times \mathfrak{g}(\mathfrak{g}, \mathfrak{h}) / \mathfrak{g} \right].$$

From §2.3, for each $\mathfrak{g} \in \mathfrak{g}(\mathfrak{g}, \mathfrak{h})$, we have the local weighted special cycle $\mathfrak{g}(\mathfrak{g}, \mathfrak{h})$ and the local derived special cycle $\mathfrak{g}(\mathfrak{g}, \mathfrak{h})$ on $\mathfrak{g}^{[b]}_{\mathfrak{g}, \mathfrak{g}}$.

For a pair $(\mathfrak{g}, \mathfrak{g}) \in \mathfrak{g}(\mathfrak{g}) \setminus \left[\mathfrak{g}^{[b]}_{\mathfrak{g}, \mathfrak{g}} \times \mathfrak{g}(\mathfrak{g}, \mathfrak{h}) / \mathfrak{g} \right]$ with $\mathfrak{g}'(\mathfrak{g}, \mathfrak{g}) \in \mathfrak{g} \otimes \mathfrak{g}(\mathfrak{g})$, we define the closed formal subscheme on $\mathfrak{g}^{[b]}_{\mathfrak{g}, \mathfrak{g}} \times \mathfrak{g}(\mathfrak{g}, \mathfrak{h}) / \mathfrak{g}$ by

$$\mathfrak{g}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g})_{\mathfrak{g}} = \mathfrak{g}(\mathfrak{g}, \mathfrak{g}) \times \mathfrak{g}(\mathfrak{g}, \mathfrak{h}). \quad (6.3.1)$$

We then consider the sum

$$\sum \mathfrak{g}(\mathfrak{g}', \mathfrak{g}', \mathfrak{g}, \mathfrak{g})_{\mathfrak{g}},$$

where the sum runs over $(\mathfrak{g}', \mathfrak{g}')$ in the $\mathfrak{g}(\mathfrak{g}) \setminus \left[\mathfrak{g}^{[b]}_{\mathfrak{g}, \mathfrak{g}} \times \mathfrak{g}(\mathfrak{g}, \mathfrak{h}) / \mathfrak{g} \right]$ -orbit of the pair $(\mathfrak{g}, \mathfrak{g})$ under the diagonal action of $\mathfrak{g}(\mathfrak{g}) \setminus \left[\mathfrak{g}^{[b]}_{\mathfrak{g}, \mathfrak{g}} \times \mathfrak{g}(\mathfrak{g}, \mathfrak{h}) / \mathfrak{g} \right]$.

$$\left[\mathfrak{g}(\mathfrak{g}) \setminus \left[\mathfrak{g}^{[b]}_{\mathfrak{g}, \mathfrak{g}} \times \mathfrak{g}(\mathfrak{g}, \mathfrak{h}) / \mathfrak{g} \right] \right] / \mathfrak{g}.$$

Since this sum is $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)$ -invariant, it descends to a cycle on $\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge$ in (5.4.1), which we denote by $[\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet, \mathfrak{g}_\bullet, \mathfrak{g}_\bullet)]_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}$. Similarly, we define $[\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet, \mathfrak{g}_\bullet, \mathfrak{g}_\bullet)]_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}$ in the $\mathfrak{g}$ -group of $[\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet, \mathfrak{g}_\bullet, \mathfrak{g}_\bullet)]_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}$.

Proposition 6.3.1. *Let $\mathfrak{g} \in \mathcal{O}^+(\mathfrak{g}_\bullet)$. The restriction of the special cycle $\tilde{\mathfrak{g}}^{(\mathfrak{a})}(\mathfrak{g}, \mathfrak{g})$ to each fiber $\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge$ of the projection (5.4.3) is given by the finite sum*

$$\sum_{(\mathfrak{g}, \mathfrak{g}) \in \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \setminus [\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \times \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)] / \mathbb{A}_{\mathfrak{g}_\bullet}^\wedge} \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}^{-1} \mathfrak{g}) \cdot [\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet, \mathfrak{g}_\bullet, \mathfrak{g}_\bullet)]_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}, \quad (6.3.2)$$

viewed as a cycle on (5.4.1).

Similarly, the restriction of the derived special cycles $\tilde{\mathfrak{g}}^{(\mathfrak{a})}(\mathfrak{g}, \mathfrak{g})$ is given by

$$\sum_{(\mathfrak{g}, \mathfrak{g}) \in \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \setminus [\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \times \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)] / \mathbb{A}_{\mathfrak{g}_\bullet}^\wedge} \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}^{-1} \mathfrak{g}) \cdot [\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet, \mathfrak{g}_\bullet, \mathfrak{g}_\bullet)]_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}. \quad (6.3.3)$$

Proof. This follows the same reasoning as the basic uniformization of special divisors in [Zha21, Proposition 7.4]. For the derived special cycles, see also Proposition 9.2.1. $\square$

6.4. Special cycles in the complex uniformization. Let $\mathfrak{g} : \Psi \rightarrow \mathfrak{g}$ be a complex place of the reflex field, and let $\mathfrak{g}^{(\mathfrak{a})}$ be the nearby hermitian space. Recall the notations for complex uniformization from §5.5.

For each $\mathfrak{g} \in \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)$ with totally positive norm, we define the special cycle $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}) \subset \mathfrak{g}^{(\mathfrak{a})}$ as the space of negative definite $\mathfrak{g}$ -lines perpendicular to $\mathfrak{g}$ (see §4.2).

For a pair $(\mathfrak{g}, \mathfrak{g}) \in \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \times \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) / \mathbb{A}_{\mathfrak{g}_\bullet}^\wedge$, we define the analytic submanifold

$$\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}, \mathfrak{g})_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge} = \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}) \times \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}) \subset \mathfrak{g}^{(\mathfrak{a})} \times \mathfrak{g}^{(\mathfrak{a})}.$$

We then consider the formal sum

$$\sum \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}', \mathfrak{g}')_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}$$

taken over $(\mathfrak{g}', \mathfrak{g}')$ in the $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)$ -orbit of the pair $(\mathfrak{g}, \mathfrak{g})$ under the diagonal action of $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)$ on $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \times \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) / \mathbb{A}_{\mathfrak{g}_\bullet}^\wedge$. Since this sum is $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)$ -invariant, it descends to a cycle on the quotient $\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge$, which we denote by $[\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}, \mathfrak{g})]_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}$.

Proposition 6.4.1. *Let $\mathfrak{g} \in \mathcal{O}^+(\mathfrak{g}_\bullet)$. The restriction of the special cycle $\tilde{\mathfrak{g}}^{(\mathfrak{a})}(\mathfrak{g}, \mathfrak{g}) := \tilde{\mathfrak{g}}^{(\mathfrak{a})}(\mathfrak{g}, \mathfrak{g}) \otimes_{\Psi, \mathfrak{g}}$ to each fiber $\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge$ of the projection (5.5.1) is given by*

$$\sum_{(\mathfrak{g}, \mathfrak{g}) \in \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \setminus [\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \times \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)] / \mathbb{A}_{\mathfrak{g}_\bullet}^\wedge} \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}^{-1} \mathfrak{g}) \cdot [\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}, \mathfrak{g})]_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}. \quad \square$$

6.5. Green currents of special cycles. Let $\mathfrak{g}$ be an archimedean place of Ψ , with its restriction to $\mathfrak{g}$ (resp. $\mathfrak{g}_\bullet$) denoted by $\mathfrak{g}$ (resp. $\mathfrak{g}_\bullet$). Since all constructions in this subsection depend only on $\mathfrak{g}_\bullet$, in §9.1, we will freely use the notation $\mathfrak{g}_\bullet^?$ rather than $\mathfrak{g}_\bullet^?$ for $? \in \{\mathfrak{g}, \mathfrak{g}_\bullet\}$.

For a pair $(\mathfrak{g}, \mathfrak{g}) \in \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \times \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) / \mathbb{A}_{\mathfrak{g}_\bullet}^\wedge$, we define

$$\mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g})_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge} = \mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g}) \times \mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}), \quad \mathfrak{g} \in \mathfrak{g}^{(\mathfrak{a})},$$

where $\mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g}) = \mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g}) + \mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g}) \in \mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g})$, with $\mathfrak{g}_\bullet^{\mathfrak{a}} \in \mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g})$. Here, $\mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g})$ is Kudla's Green function for the analytic special cycle $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g})$, as defined in §4.2.

Consider the sum

$$\sum \mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g})_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}$$

taken over $(\mathfrak{g}', \mathfrak{g}')$ in the $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)$ -orbit of the pair $(\mathfrak{g}, \mathfrak{g})$ under the diagonal action of $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)$ on $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \times \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) / \mathbb{A}_{\mathfrak{g}_\bullet}^\wedge$. Since this sum is $\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)$ -invariant, it descends to a current on the quotient, which we denote by $[\mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g})]_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}$. A parallel construction holds for the automorphic Green function $\mathfrak{g}_\bullet^{\mathfrak{a}}(\mathfrak{g})$, which we will discuss in §8.

For each $\mathfrak{g} \in \mathcal{O}^+(\mathfrak{g}_\bullet)$ and $\mathfrak{g} = \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}) \otimes_{\Psi, \mathfrak{g}} \in \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)$, where $\mathfrak{g} \in \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)$, we define a current $\tilde{\mathfrak{g}}_g^{(\mathfrak{a})}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g})$ on $\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge$ such that its restriction to each fiber $\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge$ of the projection (5.5.1) is

$$\mathfrak{g}_g^{(\mathfrak{a})}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g}) := \sum_{(\mathfrak{g}, \mathfrak{g}) \in \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \setminus [\mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet) \times \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}_\bullet)] / \mathbb{A}_{\mathfrak{g}_\bullet}^\wedge} \mathfrak{g}^{(\mathfrak{a})}(\mathfrak{g}^{-1} \mathfrak{g}) \cdot [\mathfrak{g}_g^{(\mathfrak{a})}(\mathfrak{g}, \mathfrak{g}, \mathfrak{g}, \mathfrak{g})]_{\mathbb{A}_{\mathfrak{g}_\bullet}^\wedge}. \quad (6.5.1)$$

For any linearly independent $\mathfrak{h} \in \mathbb{C}(\mathbb{R})^g$, recall from §4.2 that we have the Kudla-Millson Schwartz form $\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) \in (\mathbb{C}(\mathbb{R})^g)^{\otimes} \otimes \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g(\mathbb{R})^g$. Similarly, we construct $[\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h})]_{\mathfrak{h}}$ and define a form $\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h})$ on $\mathfrak{h}$ whose restriction to each fiber $\mathfrak{h}$ of the projection is

$$\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) := \sum_{(\mathfrak{h}, \mathfrak{h}) \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g \setminus (\mathbb{C}(\mathbb{R})^g(\mathbb{R})^g \times \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g) / \mathfrak{h}} \mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) \cdot [\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h})]_{\mathfrak{h}}.$$

From the construction, we obtain

$$\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) + \mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) = \mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}),$$

showing that $\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h})$ is a Green current for the special cycle $\mathfrak{h}(\mathfrak{h}, \mathfrak{h})$. The initial construction of this current is due to Kudla [Kud97].

Note that the form $\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h})$ is the $\mathfrak{h}$ -coefficient of the theta function $\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h})$ considered in [KM90]. For any $\mathfrak{h} \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g$, it is defined by

$$\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) := \sum_{\mathfrak{h} \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g} \mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) \cdot \mathfrak{h}(\mathfrak{h}) \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g(\mathfrak{h}),$$

where $\mathfrak{S}_{\mathfrak{h}} = \bigotimes_{\mathfrak{h} \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g} \mathfrak{S}_{\mathfrak{h}}$, such that $\mathfrak{S}_{\mathfrak{h}}$ is the Kudla-Millson Schwartz form and $\mathfrak{h}$ are standard Gaussian functions defined in (4.3.1) for $\mathfrak{h} \neq \mathfrak{h}$. It is a weight- $\mathfrak{h}$ automorphic function of $\mathfrak{h}(\mathfrak{h})$, with the level determined by $\mathfrak{h}$.

Theorem 6.5.1 ([GS19, Theorem 4.10]). *Let $\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h})$ and $\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h})$ be Green currents associated with the special cycles $\mathfrak{h}(\mathfrak{h}, \mathfrak{h})$ and $\mathfrak{h}(\mathfrak{h}, \mathfrak{h})$. Let $\mathfrak{h} = \mathfrak{h} \otimes \mathfrak{h}$ be a non-singular Schwartz function. Assume that $\mathfrak{h}$ and $\mathfrak{h}$ are non-degenerate and that $\mathfrak{h}(\mathfrak{h}, \mathfrak{h})$ and $\mathfrak{h}(\mathfrak{h}, \mathfrak{h})$ intersect properly. Recalling (4.2.2), we define the star product as follows:*

$$\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) \star \mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) := \mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) \wedge \mathfrak{h}(\mathfrak{h}, \mathfrak{h}) + \mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) \wedge \mathfrak{h}(\mathfrak{h}, \mathfrak{h}).$$

Under these assumptions, we obtain

$$\mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) \star \mathfrak{S}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h}) \equiv \sum_{\mathfrak{h} = \begin{pmatrix} \mathfrak{h} & \mathfrak{h} \\ \mathfrak{h} & \mathfrak{h} \end{pmatrix}} \mathfrak{S}_{\mathfrak{h}} \left(\mathfrak{h}, \begin{pmatrix} \mathfrak{h} & \mathfrak{h} \\ \mathfrak{h} & \mathfrak{h} \end{pmatrix}, \mathfrak{h} \right) \mathfrak{h} \partial + \mathfrak{h} \bar{\partial}.$$

Here, the summation ranges over all $\mathfrak{h} \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g$.

In our applications, we will always assume that $\mathfrak{h}$ is positive definite for $\mathfrak{h} \in \{\mathfrak{h}, \mathfrak{h}\}$ and that $\mathfrak{h}$ is regular at two places over $\mathfrak{h}$ (see Theorem 7.2.1). Under these conditions, the sum is taken over all $\mathfrak{h} \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g$, and Theorem 6.5.1 follows directly from the definitions.

Proposition 6.5.2. (1) *For any positive definite Hermitian matrix $\mathfrak{h}_b \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g$, the integral*

$$\int_{\mathfrak{h}_b} \mathfrak{S}_{\mathfrak{h}_b}(\mathfrak{h}_b, \mathfrak{h}_b) \wedge \mathfrak{S}_{\mathfrak{h}_b}(\mathfrak{h}_b, \mathfrak{h}_b)$$

converges absolutely.

(2) *Let $\mathfrak{h}_b \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g$ be invariant under $\mathfrak{h}_b \subset \mathfrak{h}_b(\mathfrak{h}_b, \mathfrak{h}_b)$ with respect to the Weil representation. Then, for any $\mathfrak{h}_b \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g$ and $\mathfrak{h}_b \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g$, the generating series*

$$\mathfrak{S}_{\mathfrak{h}_b}(\mathfrak{h}_b, \mathfrak{h}_b) = \sum_{\mathfrak{h}_b \in \mathbb{C}(\mathbb{R})^g(\mathbb{R})^g} \int_{\mathfrak{h}_b} \mathfrak{S}_{\mathfrak{h}_b}(\mathfrak{h}_b, \mathfrak{h}_b) \wedge \mathfrak{S}_{\mathfrak{h}_b}(\mathfrak{h}_b, \mathfrak{h}_b)$$

lies in $\mathbb{C}(\mathbb{R})^g(\mathbb{R})^g(\mathfrak{h}_b)$.

Proof. The series is well-defined since $\mathfrak{h}_b(\mathfrak{h}_b, \mathfrak{h}_b)$ consists of finitely many discrete points in the complex fiber and Kudla's Green currents have only logarithmic singularities at these points. Moreover, the generating series converges because $\mathfrak{S}_{\mathfrak{h}_b}$ is defined as the product of the Kudla-Millson series with a function that has exponential decay in $\mathfrak{h}_b$ (see (4.2.1)). The modularity of the generating series follows directly from the modularity of the Kudla-Millson series. $\square$

7. EISENSTEIN SERIES OF UNITARY GROUPS

In this section, we review the Siegel Eisenstein series and their Fourier-Jacobi expansion. Let $\mathbb{H}_{\mathbb{A}_f}$ be the *incoherent* $\mathbb{H}_{\mathbb{A}_f}$ -Hermitian space associated with $\mathbb{H}$, as fixed in §5.1. This means that $\mathbb{H}_{\mathbb{A}_f}$ is totally positive definite and satisfies $\mathbb{H}_{\mathbb{A}_f} \simeq \mathbb{H}_{\mathbb{A}_f}$ for all finite places $\mathfrak{p} \in \mathbb{A}_f$.

7.1. Siegel Eisenstein series. Let $\mathbb{H}$ be the standard split $\mathbb{H}/\mathbb{H}$ -skew-hermitian space of dimension $2n$. Let $\mathbb{H}^* = \mathbb{H}^*$ and let $\mathbb{H}^*(\mathbb{A}_f) = \mathbb{H}^*(\mathbb{A}_f) \mathbb{H}^*(\mathbb{A}_f)$ be the standard Siegel parabolic subgroup of $\mathbb{H}^*(\mathbb{A}_f)$, where

$$\begin{aligned} \mathbb{H}^*(\mathbb{A}_f) &= \left\{ \mathbb{H}^*(\mathbb{A}_f) = \begin{pmatrix} \mathbb{A}_f & \mathbb{A}_f \\ \mathbb{A}_f & \mathbb{A}_f \end{pmatrix} : \mathbb{A}_f \in \mathbb{H}^*(\mathbb{A}_f) \right\}, \\ \mathbb{H}^*(\mathbb{A}_f) &= \left\{ \begin{pmatrix} \mathbb{A}_f & \mathbb{A}_f \\ \mathbb{A}_f & \mathbb{A}_f \end{pmatrix} : \mathbb{A}_f \in \mathbb{H}^*(\mathbb{A}_f) \right\}. \end{aligned}$$

We define

$$\mathbb{H}^*(\mathbb{A}_f) = \begin{pmatrix} \mathbb{A}_f & \mathbb{A}_f \\ -\mathbb{A}_f & \mathbb{A}_f \end{pmatrix} \in \mathbb{H}^*(\mathbb{A}_f).$$

Let $\chi : \mathbb{H}^{\times} / \mathbb{H}^{\times} \rightarrow \mathbb{H}^{\times}$ be the quadratic character associated with the extension $\mathbb{H}/\mathbb{H}$. Fix a character $\chi : \mathbb{H}^{\times} \rightarrow \mathbb{H}^{\times}$ such that $\chi|_{\mathbb{H}^{\times}} = \chi$.

We may regard χ as a character on $\mathbb{H}^*(\mathbb{A}_f)$ via $\chi(\mathbb{H}^*(\mathbb{A}_f)) = \chi(\mathbb{H}^*(\mathbb{A}_f))$ and extend it to $\mathbb{H}^*(\mathbb{A}_f)$ by setting it to be trivial on $\mathbb{H}^*(\mathbb{A}_f)$. The *degenerate principal series* is then defined as the unnormalized smooth induction

$$\mathbb{H}^*(\mathbb{A}_f, \chi) := \mathbb{H}^*(\mathbb{A}_f, \chi)(\mathbb{H}^*(\mathbb{A}_f) | - |^{\chi+1/2}), \quad \chi \in \mathbb{H}^*(\mathbb{A}_f).$$

For a standard section $\mathbb{H}^*(\mathbb{A}_f, \chi) \in \mathbb{H}^*(\mathbb{A}_f, \chi)$ (i.e., a section whose restriction to the standard maximal compact subgroup of $\mathbb{H}^*(\mathbb{A}_f)$ is independent of $\mathbb{H}$), we define the associated *Siegel Eisenstein series* by

$$\mathbb{H}^*(\mathbb{A}_f, \chi, \mathbb{H}^*(\mathbb{A}_f)) := \sum_{\mathbb{H}^*(\mathbb{A}_f) \setminus \mathbb{H}^*(\mathbb{A}_f)} \mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f), \chi), \quad \mathbb{H}^*(\mathbb{A}_f) \in \mathbb{H}^*(\mathbb{A}_f).$$

This series converges for $\chi(\mathbb{H}) \gg \mathbb{H}$, admits a meromorphic continuation to the entire complex plane, and is holomorphic at $\chi = \mathbb{H}$. Note that $\mathbb{H}^*(\mathbb{A}_f, \chi, \mathbb{H}^*(\mathbb{A}_f))$ depends on the choice of χ .

For $\chi \in \mathbb{H}^*(\mathbb{A}_f)$ and $\mathbb{H}^*(\mathbb{A}_f) \in \mathbb{H}^*(\mathbb{A}_f)$, define

$$\mathbb{H}^*(\mathbb{A}_f, \chi, \mathbb{H}^*(\mathbb{A}_f)) = \int_{\mathbb{H}^*(\mathbb{A}_f) \setminus \mathbb{H}^*(\mathbb{A}_f)} \mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f), \chi, \mathbb{H}^*(\mathbb{A}_f)) \mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f)) \mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f)).$$

Then, we have the decomposition

$$\mathbb{H}^*(\mathbb{A}_f, \chi, \mathbb{H}^*(\mathbb{A}_f)) = \sum_{\chi \in \mathbb{H}^*(\mathbb{A}_f)} \mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f), \chi, \mathbb{H}^*(\mathbb{A}_f)). \quad (7.1.1)$$

For $\chi(\mathbb{H}) \gg \mathbb{H}$, the integral may be unfolded in the usual way. If $\chi(\mathbb{H}) \neq \mathbb{H}$ and $\mathbb{H}^*(\mathbb{A}_f) = \mathbb{H}^*(\mathbb{A}_f)$ is factorizable, the Fourier coefficient admits a factorization into a product:

$$\mathbb{H}^*(\mathbb{A}_f, \chi, \mathbb{H}^*(\mathbb{A}_f)) = \prod_{\mathfrak{p} \in \mathbb{A}_f} \mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f), \chi, \mathbb{H}^*(\mathbb{A}_f)),$$

where

$$\mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f), \chi, \mathbb{H}^*(\mathbb{A}_f)) = \int_{\mathbb{H}^*(\mathbb{A}_f)} \mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f), \chi, \mathbb{H}^*(\mathbb{A}_f)) \mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f)) \mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f))$$

is the *local Whittaker function*, as defined in §3.6. Furthermore, we have the equivariance property

$$\mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f), \chi, \mathbb{H}^*(\mathbb{A}_f)) = \mathbb{H}^*(\mathbb{H}^*(\mathbb{A}_f), \chi, \mathbb{H}^*(\mathbb{A}_f)),$$

where $\mathbb{H}^*(\mathbb{A}_f)$ denotes the action of right translation by $\mathbb{H}^*(\mathbb{A}_f) \in \mathbb{H}^*(\mathbb{A}_f)$ in the local induced representation $\mathbb{H}^*(\mathbb{A}_f)$.

Recall that, for χ nonsingular, these local integrals have an entire analytic continuation. In other words, we have the following result:

Proposition 7.1.1 ([Kud97]). For $\star \in \mathcal{O} \setminus \mathcal{O}^\times$, the function $\mathbb{E}_\star(\vartheta, \mathbb{I}, \varphi)$ admits a meromorphic analytic continuation and is holomorphic at $\mathbb{I} = \mathcal{O}$. In particular, the following expression holds:

$$\mathbb{E}_\star(\vartheta, \mathbb{I}, \varphi) = \prod_{\mathfrak{p} \in \mathcal{O}} \mathbb{E}_{\star, \mathfrak{p}}(\vartheta_{\mathfrak{p}}, \mathbb{I}_{\mathfrak{p}}, \varphi_{\mathfrak{p}})$$

for $\mathbb{I}$ near $\mathcal{O}$. The vanishing of the nonsingular coefficients at $\mathbb{I} = \mathcal{O}$ is determined by the vanishing of the local factors $\mathbb{E}_{\star, \mathfrak{p}}(\vartheta_{\mathfrak{p}}, \mathbb{I}_{\mathfrak{p}}, \varphi_{\mathfrak{p}})$. $\square$

Next, we study the central derivative of the Siegel Eisenstein series. The main argument follows from [Kud97, §6]. The Fourier expansion in (7.1.1) is well-defined near $\mathbb{I} = \mathcal{O}$, and from now on, we will restrict our attention to this region.

Theorem 7.1.2. The central derivative $\partial \mathbb{E}_\star(\vartheta, \mathcal{O}, \varphi)$ is well-defined and is an automorphic form on $\mathcal{O}^\times \backslash \mathcal{O}^\times$. Moreover, we have

$$\partial \mathbb{E}_\star(\vartheta, \mathcal{O}, \varphi) = \sum_{\star} \frac{\star}{\star} \Big|_{\mathbb{I}=\mathcal{O}} \mathbb{E}_\star(\vartheta, \mathbb{I}, \varphi),$$

which is absolutely convergent. Furthermore, for nonsingular $\star$, we have

$$\frac{\star}{\star} \Big|_{\mathbb{I}=\mathcal{O}} \mathbb{E}_\star(\vartheta, \mathbb{I}, \varphi) = \partial \mathbb{E}_\star(\vartheta, \mathcal{O}, \varphi),$$

i.e., the Fourier coefficients are preserved under differentiation.

Proof. Since the function $\star \mapsto \mathbb{E}_\star(\vartheta, \mathbb{I}, \varphi)$ is smooth on $\mathcal{O} \setminus \mathcal{O}^\times / \mathcal{O}^\times$ and holomorphic in a neighborhood of $\mathbb{I} = \mathcal{O}$, the Fourier expansion is absolutely convergent and uniform for $\mathbb{I}$ in a closed disk around $\mathbb{I} = \mathcal{O}$ (see [Kud97, §6]). Hence, it can be differentiated term by term with respect to $\mathbb{I}$, and the resulting series for the derivative remains absolutely convergent. By differentiating term by term with respect to $\mathbb{I}$, evaluating at $\mathbb{I} = \mathcal{O}$, and collecting the terms for nonsingular $\star$, we obtain the first assertion. The second assertion follows from Fubini's theorem. $\square$

7.2. Weil representation. The Eisenstein series of interest to us arise from the Weil representations. By class field theory, we may, and will, choose an additive character $\mathfrak{f} : \mathcal{O}^\times / \mathcal{O}^\times \rightarrow \mathbb{C}^\times$ such that $\mathfrak{f}$ is unramified outside of $\mathcal{O}^\times$, the set of finite places of $\mathcal{O}^\times$ that split in $\mathcal{O}$.

Let $\mathcal{H}$ be an $\mathcal{O}^\times / \mathcal{O}^\times$ -hermitian space of rank r , we say that the space $\mathcal{H}$ (resp., the Schwartz function $\varphi \in \mathcal{S}(\mathcal{H})$) and the Eisenstein series $\mathbb{E}_\star(\vartheta, \mathbb{I}, \varphi)$ is *coherent* if $\mathcal{H} = \mathcal{O} \otimes_{\mathcal{O}^\times} \mathcal{H}_0$ for some $\mathcal{O}^\times / \mathcal{O}^\times$ -Hermitian space $\mathcal{H}_0$, and *incoherent* otherwise.

Let $\mathcal{S}(\mathcal{H})$ be the space of Schwartz functions on $\mathcal{H}$. The fixed choice of $\mathfrak{f}$ and $\mathfrak{f}$ gives a *Weil representation* $\otimes = \otimes_{\mathfrak{f}, \mathfrak{f}}$ of $\mathcal{O}^\times \times \mathcal{H}$ on $\mathcal{S}(\mathcal{H})$. Explicitly, for $\varphi \in \mathcal{S}(\mathcal{H})$ and $\mathfrak{h} \in \mathcal{H}$,

$$\begin{aligned} \otimes(\varphi)(\mathfrak{h}) &= \mathfrak{f}(\varphi(\mathfrak{h})) \prod_{\mathfrak{p} \in \mathcal{O}^\times} \mathfrak{f}(\mathfrak{h}_{\mathfrak{p}})^{r/2} \varphi(\mathfrak{h} \cdot \mathfrak{h}_{\mathfrak{p}}), & \otimes(\varphi) &= \left(\varphi, \frac{\cdot}{\cdot} \right) \in \mathcal{S}(\mathcal{H}), \\ \otimes(\varphi)(\mathfrak{h}) &= \mathfrak{f}(\varphi(\mathfrak{h})) \varphi(\mathfrak{h}), & \otimes(\varphi) &= \left(\varphi, \frac{\cdot}{\cdot} \right) \in \mathcal{S}(\mathcal{H}), \\ \otimes(\varphi)(\mathfrak{h}) &= \varphi(\mathfrak{h}) \cdot \widehat{\varphi}(\mathfrak{h}), & \otimes &= \left(\varphi, \frac{\cdot}{\cdot} \right), \\ \otimes(\varphi)(\mathfrak{h}) &= \varphi(\mathfrak{h}^{-1} \cdot \mathfrak{h}), & \otimes &\in \mathcal{H}(\mathcal{H}). \end{aligned}$$

Here $\mathfrak{f}$ is the Weil constant (see [KR14, (10.3)]), and $\widehat{\varphi}$ is the Fourier transform of φ using the self-dual Haar measure on $\mathcal{H}$ with respect to $\mathfrak{f} \circ \mathfrak{f}$.

For $\varphi \in \mathcal{S}(\mathcal{H})$, define the function

$$\mathbb{E}_\star(\vartheta, \mathbb{I}) := \otimes(\vartheta)(\mathbb{I}), \quad \vartheta \in \mathcal{O}^\times \backslash \mathcal{O}^\times.$$

Then, $\mathbb{E}_\star \in \mathcal{S}(\mathcal{O}^\times \backslash \mathcal{O}^\times)$. Let $\mathbb{E}_\star(-, \mathbb{I}) \in \mathcal{S}(\mathbb{I}, \mathbb{I})$ be the associated standard section, known as the *standard Siegel-Weil section* associated to $\mathbb{E}_\star$.

For $\varphi \in \mathcal{S}(\mathcal{H})$, we define the Eisenstein series and its Fourier coefficients as follows:

$$\mathbb{E}_\star(\vartheta, \mathbb{I}, \varphi) := \mathbb{E}_\star(\vartheta, \mathbb{I}, \mathbb{E}_\star), \quad \mathbb{E}_\star(\vartheta, \mathbb{I}, \varphi) := \mathbb{E}_\star(\vartheta, \mathbb{I}, \mathbb{E}_\star).$$

Similarly, we write

$$\mathbb{E}_{\star, \mathfrak{p}}(\vartheta_{\mathfrak{p}}, \mathbb{I}_{\mathfrak{p}}, \varphi_{\mathfrak{p}})$$

for the local Whittaker function.

For a finite place $\mathfrak{p}$, we define the space of *regular test functions* as

$$\mathcal{W}(\mathfrak{p}) = \{ \psi \in \mathcal{W}(\mathfrak{p}) \mid \psi(\mathfrak{p}) = 0 \text{ if } \mathfrak{p}(\mathfrak{p}) = 0 \}.$$

Fix a finite subset $S \subset \mathfrak{p}$ with $|S| = n > 0$ and define

$$\mathcal{W}(\mathfrak{p}) = \bigotimes_{\mathfrak{p} \in S} \mathcal{W}(\mathfrak{p}).$$

A Schwartz function $\psi \in \mathcal{W}(\mathfrak{p})$ is *regular* at $\mathfrak{p}$ place if there exists a finite subset $S \subset \mathfrak{p}$ such that $|S| = n$ and that $\psi \in \mathcal{W}(\mathfrak{p})$. The following result is due to [YZZ12, §6] and [Liu11b, 2.B]:

Theorem 7.2.1. *Suppose $\psi \in \mathcal{W}(\mathfrak{p})$ is regular at $\mathfrak{p}$ finite places, and that $\mathfrak{p}$ is positive definite, and that $\psi \in \mathcal{W}(\mathfrak{p})$ is the product of standard Gaussian functions as in (4.3.1).*

Then, for $\mathfrak{p}$ singular and $\mathfrak{p} \in \mathfrak{p}(\mathfrak{p})$, we have

$$\mathfrak{p}(\mathfrak{p}) = \mathfrak{p}(\mathfrak{p}) \geq 0.$$

Proof. The proof in [Liu11b, 2.B] applies directly, although it is originally stated for the case when $\mathfrak{p}$ is even-dimensional. See also [LZ22a, (12.3.0.2)]. $\square$

Remark 7.2.2. We can construct a regular test function $\psi_{\mathfrak{p}}$ of the form

$$\psi_{\mathfrak{p}} = \bigotimes_{\mathfrak{p} \in S} \psi_{\mathfrak{p}}, \quad \psi_{\mathfrak{p}} = \mathfrak{p} \in \mathcal{W}(\mathfrak{p}),$$

where $\mathfrak{p} \subset \mathfrak{p}$ are open subsets. This follows from the fact that any open subset in a non-Archimedean metric space can be expressed as a union of boxes. Furthermore, we can choose $\psi_{\mathfrak{p}}$ to be a $\mathfrak{p}$ -valued function.

7.3. Classical Eisenstein series associated to the Shimura datum. Recall from §4.1 that we defined the Hermitian upper half-space for $\mathfrak{p} = \mathfrak{p}(\mathfrak{p})$, with $\mathfrak{p} = \mathfrak{p} \otimes \mathfrak{p} \simeq \mathfrak{p}$.

Notation 7.3.1. For the remaining part of the paper, we will always assume the $\mathfrak{p}/\mathfrak{p}$ -hermitian space $\mathfrak{p}$ is positive definite and fix $\psi \in \mathcal{W}(\mathfrak{p})$ as the product of standard Gaussian functions in (4.3.1).

For any $\psi \in \mathcal{W}(\mathfrak{p})$, the *classical Eisenstein series* for $\mathfrak{p} = \mathfrak{p} + \mathfrak{p} \in \mathfrak{p}(\mathfrak{p})$ is defined as

$\mathfrak{p}(\mathfrak{p}, \mathfrak{p}) := \mathfrak{p}(\mathfrak{p})^{-1} \mathfrak{p}(\mathfrak{p})^{-1/2}$. $\mathfrak{p}(\mathfrak{p}, \mathfrak{p}) := \mathfrak{p}(\mathfrak{p}) \in \mathfrak{p}(\mathfrak{p})$, where $\mathfrak{p} \in \mathfrak{p}(\mathfrak{p})$ is chosen such that $\mathfrak{p} = \mathfrak{p} \cdot \mathfrak{p}$. See §7.2 for notations. Notice that $\mathfrak{p}(\mathfrak{p}, \mathfrak{p})$ does not depend on the choice of $\mathfrak{p}$, and that $\mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \mathfrak{p} \in \mathfrak{p}(\mathfrak{p})$.

The Fourier expansion (7.1.1) can then be rewritten as

$$\mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \sum_{\mathfrak{p} \in \mathfrak{p}} \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \sum_{\mathfrak{p} \in \mathfrak{p}} \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) \mathfrak{p}(\mathfrak{p}), \quad \mathfrak{p} = \mathfrak{p}(\mathfrak{p}).$$

We write the central derivative as

$$\partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) := \partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}), \quad \partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) := \partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}).$$

A similar notation will be used for Whittaker functions.

Proposition 7.3.2. *Assume that $\psi \in \mathcal{W}(\mathfrak{p})$ is regular at least at two finite places. Then:*

(1) *The Eisenstein series admits a Fourier expansion:*

$$\mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \sum_{\mathfrak{p} \in \mathfrak{p}} \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \sum_{\mathfrak{p} \in \mathfrak{p}} \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) \mathfrak{p}(\mathfrak{p}),$$

where $\mathfrak{p}(\mathfrak{p}, \mathfrak{p}) := \mathfrak{p}(\mathfrak{p}, \mathfrak{p})/\mathfrak{p}$ is the $\mathfrak{p}$ -th Fourier coefficient in the classical sense.

(2) *The central derivative of the Eisenstein series also has a Fourier expansion:*

$$\partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \sum_{\mathfrak{p} \in \mathfrak{p}} \partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \sum_{\mathfrak{p} \in \mathfrak{p}} \partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) \mathfrak{p}(\mathfrak{p}),$$

where $\partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) := \partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p})/\mathfrak{p}$. For each term, we have the expression

$$\partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \sum_{\mathfrak{p} \in \mathfrak{p}} \partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}), \quad \partial \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \mathfrak{p}'(\mathfrak{p}, \mathfrak{p}) \cdot \mathfrak{p}(\mathfrak{p}, \mathfrak{p}),$$

where $\mathfrak{f}_{\mathfrak{g}, \mathfrak{h}}(\mathfrak{a}, \mathfrak{b})$ depends on $\mathfrak{a}$ only when $\mathfrak{h} \mid \infty$.

Proof. Part (1) follows from Theorem 7.2.1 and the fact that $\mathfrak{g}_{\mathfrak{h}, \mathfrak{h}} = \mathfrak{a} \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h}, \mathfrak{h})$. Part (2) follows from Theorem 7.1.2 and Proposition (7.1.1). $\square$

Let $\mathfrak{g} \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})$ be either totally positive definite ($\mathfrak{g} \in \mathfrak{g}_{\mathfrak{h}}^+(\mathfrak{h})$) or have signature $(\mathfrak{h} - \mathfrak{a}, \mathfrak{a})$ at one place while being positive definite at all other places ($\mathfrak{g} \in \mathfrak{g}_{\mathfrak{h}}^{\dagger}(\mathfrak{h})$). By Theorem 4.3.1(1), only these two cases yield non-vanishing Fourier coefficients for $\mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}, \mathfrak{b})$ and $\partial \mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}, \mathfrak{b})$.

7.4. Fourier-Jacobi series. In this subsection, we recall the Fourier-Jacobi type expansion for Hermitian Siegel modular forms. For a more detailed definitions and references, we refer the reader to [Bra49].

Let $\mathfrak{f}(\mathfrak{a}) : \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h}) \rightarrow \mathfrak{h}$ be a classical Siegel modular form (not necessarily holomorphic) with a Fourier expansion that is absolutely convergent:

$$\mathfrak{f}(\mathfrak{a}) = \sum_{\mathfrak{g} \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})} \mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}) \mathfrak{g}, \quad \mathfrak{f}_{\mathfrak{g}} = \mathfrak{f}_{\mathfrak{g}}(\mathfrak{g}(\mathfrak{a})).$$

We write

$$\mathfrak{a} = \begin{pmatrix} \mathfrak{a} & \mathfrak{b} \\ \mathfrak{c} & \mathfrak{d} \end{pmatrix}, \quad \text{and} \quad \mathfrak{g} = \begin{pmatrix} \mathfrak{g} & \mathfrak{h} \\ \mathfrak{i} & \mathfrak{j} \end{pmatrix} = \mathfrak{g}(\mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{b}).$$

Then the modular form $\mathfrak{f}(\mathfrak{a})$ has a Fourier-Jacobi type expansion:

$$\mathfrak{f}(\mathfrak{a}) = \sum_{\mathfrak{g}_b \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})} \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b}) \mathfrak{g}_b, \quad \mathfrak{f}_{\mathfrak{g}_b} = \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{g}_b(\mathfrak{a}, \mathfrak{b})),$$

where

$$\mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b}) = \sum_{\mathfrak{g} = \begin{pmatrix} \mathfrak{g} & \mathfrak{h} \\ \mathfrak{i} & \mathfrak{j} \end{pmatrix}} \mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}) \cdot \mathfrak{f}_{\mathfrak{g}}(\mathfrak{h} + \mathfrak{g}_b \mathfrak{b}) \mathfrak{f}_{\mathfrak{g}}(\mathfrak{g}_b(\mathfrak{a}, \mathfrak{b})).$$

If we set $\mathfrak{a} = \mathfrak{a} = \mathfrak{a}$, we obtain the function with the Fourier expansion:

$$\mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}) := \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{a}) = \sum_{\mathfrak{h} \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})} \sum_{\mathfrak{g} = \begin{pmatrix} \mathfrak{g} & \mathfrak{h} \\ \mathfrak{i} & \mathfrak{j} \end{pmatrix}} \mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}) \mathfrak{g}, \quad \mathfrak{f}_{\mathfrak{g}} = \mathfrak{f}_{\mathfrak{g}}(\mathfrak{g}(\mathfrak{a})).$$

Note that this function also depends on $\mathfrak{b}$, the imaginary part of $\mathfrak{a}$. We will fix $\mathfrak{a} = \mathfrak{a}$ from now on. Since $\mathfrak{f}_{\mathfrak{g}_b}$ is the pure tensor of standard Gaussian functions, the Fourier-Jacobi expansion shows that $\mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b})$ is a weight- $\mathfrak{h}$ modular form, where $\mathfrak{g} \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})$ acts by

$$\mathfrak{g} \cdot \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b}) = \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{g}(\mathfrak{a}), \mathfrak{g}(\mathfrak{b})).$$

In particular, the associated automorphic function of $\mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a})$ is $\mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})$ -invariant.

Recall from Proposition 7.3.2 that the Eisenstein series and its central derivative admit a Fourier expansion, which is absolutely convergent by Theorem 7.1.2:

$$\mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}, \mathfrak{b}) = \sum_{\mathfrak{g} \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})} \mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{g}, \quad \text{and} \quad \partial \mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}, \mathfrak{b}) = \sum_{\mathfrak{g} \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})} \partial \mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{g}.$$

We consider the functions

$$\mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b}) = \sum_{\mathfrak{h} \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})} \sum_{\mathfrak{g} = \begin{pmatrix} \mathfrak{g} & \mathfrak{h} \\ \mathfrak{i} & \mathfrak{j} \end{pmatrix}} \mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{g}, \quad \text{and} \quad \partial \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b}) = \sum_{\mathfrak{h} \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})} \sum_{\mathfrak{g} = \begin{pmatrix} \mathfrak{g} & \mathfrak{h} \\ \mathfrak{i} & \mathfrak{j} \end{pmatrix}} \partial \mathfrak{f}_{\mathfrak{g}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{g}.$$

Consider the Weil representation $\mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})$ acting on $\mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b})$ in the first factor. By [Kud97, A.2], this action is compatible with the Weil representation $\mathfrak{g}$. In particular, for any $\mathfrak{h} \in \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})$, we have

$$\mathfrak{g}_{\mathfrak{h}}(\mathfrak{h}) \cdot \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b}) = \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{h}(\mathfrak{a}), \mathfrak{h}(\mathfrak{b})), \quad \text{and} \quad \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h}) \cdot \partial \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b}) = \partial \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{h}(\mathfrak{a}), \mathfrak{h}(\mathfrak{b})).$$

We conclude the following:

Proposition 7.4.1. *Suppose that $\mathfrak{f}_{\mathfrak{g}_b} \in \mathfrak{f}_{\mathfrak{h}}(\mathfrak{h})$ is invariant under $\mathfrak{g}_{\mathfrak{h}}(\mathfrak{h}) \subset \mathfrak{g}_{\mathfrak{h}}(\mathfrak{h})$ under the Weil representation. Then for any $\mathfrak{h} = \mathfrak{h} \otimes \mathfrak{h} \in \mathfrak{f}_{\mathfrak{h}}(\mathfrak{h})$, the generating series $\mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b})$ and $\partial \mathfrak{f}_{\mathfrak{g}_b}(\mathfrak{a}, \mathfrak{b})$ lie in $\mathfrak{f}_{\mathfrak{h}}(\mathfrak{h})$.* $\square$

Part 3. Proof of Kudla-Rapoport conjecture

8. ARITHMETIC INTERSECTION OF SPECIAL CYCLES

In this section, we review the concept of almost intersection of admissible cycles, as established in [MZ24] and further generalized in [Zha25, LMZ25]. Due to the non-regularity of the integral model $\widehat{\mathcal{M}}$, we must instead work with the arithmetic Picard group.

8.1. Arithmetic Picard group and admissible line bundles. Let Ψ be a number field and let $\mathfrak{s} \in \mathbb{R}_{>0}$. Let $\mathcal{M}$ be a flat, proper scheme of finite type over $\mathbb{A}_{\Psi}[\mathfrak{s}^{-1}]$ with smooth generic fiber $\mathcal{M}_{\mathfrak{s}}$.

Following [BGS94, §2.1.2], a *hermitian vector bundle* on $\mathcal{M}$ is a pair $\widehat{\mathcal{E}} = (\mathcal{E}, (\|\cdot\|_{\mathfrak{s}})_{\mathfrak{s} \in \mathbb{R}_{>0}}(\Psi, \Psi))$, where $\mathcal{E}$ is a locally free coherent $\mathcal{O}_{\mathcal{M}}$ -module, and for each complex embedding $\mathfrak{s} \in \mathbb{R}_{>0}(\Psi, \Psi)$, the holomorphic vector bundle $\mathcal{E}_{\mathfrak{s}}$ over the complex fiber $\mathcal{M}_{\mathfrak{s}}$ is equipped with a hermitian metric $\|\cdot\|_{\mathfrak{s}}$.

Let $\widehat{\mathcal{L}}(\mathcal{M}) = \widehat{\mathcal{L}}(\mathcal{M}) \otimes \mathbb{C}^*$ be the set of isomorphism classes of hermitian line bundles on $\mathcal{M}$. The tensor product endows $\widehat{\mathcal{L}}(\mathcal{M})$ with the structure of an abelian group. The identity element in $\widehat{\mathcal{L}}(\mathcal{M})$ is the structure sheaf $\mathcal{O}_{\mathcal{M}}$ equipped with the trivial metric, and the inverse of the class of $\widehat{\mathcal{E}}$ is given by its dual bundle. For a detailed treatment, see [GS90, §1 and 2].

Fix a Kähler metric $\mathfrak{b}_{\mathfrak{s}}$ on the complex fiber $\mathcal{M}_{\mathfrak{s}}$ for every $\mathfrak{s} \in \mathbb{R}_{>0}(\Psi, \Psi)$ that is compatible with the complex conjugation. This metric induces a Laplace operator on differential forms on $\mathcal{M}_{\mathfrak{s}}$. Let $\mathcal{Z}(\mathcal{M}_{\mathfrak{s}}, \mathbb{R})$ denote the $\mathbb{R}$ -vector space of real differential forms of type $(\bullet, \bullet)$ on $\mathcal{M}_{\mathfrak{s}}$. A hermitian line bundle $\widehat{\mathcal{E}} = (\mathcal{E}, (\|\cdot\|_{\mathfrak{s}})_{\mathfrak{s} \in \mathbb{R}_{>0}}(\Psi, \Psi))$ is called *admissible* if its first Chern form $\mathfrak{c}_1(\mathcal{E}_{\mathfrak{s}}, \|\cdot\|_{\mathfrak{s}}) \in \mathcal{Z}(\mathcal{M}_{\mathfrak{s}}, \mathbb{R})$, as defined in [GS90, §2.3], is harmonic for each $\mathfrak{s} \in \mathbb{R}_{>0}(\Psi, \Psi)$.

Let $\widehat{\mathcal{L}}^{\text{ad}}(\mathcal{M})$ be the subgroup consisting of classes of admissible hermitian line bundles on $\mathcal{M}$. According to [GS90, §2.6], the admissible metric $\|\cdot\|_{\mathfrak{s}}$ is unique up to a constant scalar on each factor.⁸

Note that while [GS90] assumes that the scheme $\mathcal{M}$ is regular with $\mathfrak{s} = \mathfrak{s}$, the existence and uniqueness of the admissible metric depend only on the complex fiber. Therefore, these assumptions are not relevant in this context, see [GS92, §2] for the most general treatment.

Definition 8.1.1. (1) For any finite place $\mathfrak{p}$ of Ψ , let $\widehat{\mathcal{L}}_{\mathfrak{p}}(\mathcal{M}) \subset \widehat{\mathcal{L}}(\mathcal{M})$ be the subgroup generated by hermitian line bundles $(\mathcal{E}, (\|\cdot\|_{\mathfrak{s}})_{\mathfrak{s} \in \mathbb{R}_{>0}}(\Psi, \Psi))$, where $\mathcal{E}_{\mathfrak{p}, \mathfrak{s}}$ are effective Cartier divisors of $\mathcal{M}$ whose underlying topological space is supported on the special fiber $\mathcal{M} \otimes \mathbb{F}_{\mathfrak{p}}$ and where $\mathfrak{b}_{\mathfrak{s}}$ is the fixed Kähler metric on the complex fiber $\mathcal{E}_{\mathfrak{p}, \mathfrak{s}} \cong \mathcal{E}_{\mathfrak{p}, \mathfrak{s}}$, for each $\mathfrak{s} \in \mathbb{R}_{>0}(\Psi, \Psi)$.

By definition, we have an isomorphism

$$\widehat{\mathcal{L}}_{\mathfrak{p}}(\mathcal{M}) \cong \widehat{\mathcal{L}}_{\mathfrak{p}}(\mathcal{M}),$$

where the RHS is the subgroup of $\widehat{\mathcal{L}}(\mathcal{M})$ generated by the same classes but without the hermitian metric.

(2) Let $\widehat{\mathcal{L}}^{\text{ad}}_{\infty}(\mathcal{M}) \subset \widehat{\mathcal{L}}^{\text{ad}}(\mathcal{M})$ be the subgroup generated by $(\mathcal{E}, (\|\cdot\|_{\mathfrak{s}})_{\mathfrak{s} \in \mathbb{R}_{>0}}(\Psi, \Psi))$ where $\mathfrak{c}_{\mathfrak{s}}$ is a constant and $\mathfrak{b}_{\mathfrak{s}}$ is the Kähler metric we fixed on $\mathcal{M}$ for each $\mathfrak{s} \in \mathbb{R}_{>0}(\Psi, \Psi)$.

We define $\widehat{\mathcal{L}}^{\text{ad}}_{\text{fin}}(\mathcal{M})$ as the subgroup of $\widehat{\mathcal{L}}^{\text{ad}}(\mathcal{M})$ generated by $\widehat{\mathcal{L}}_{\mathfrak{p}}(\mathcal{M})$ for all finite places $\mathfrak{p}$ of Ψ together with the subgroup $\widehat{\mathcal{L}}^{\text{ad}}_{\infty}(\mathcal{M})$.

Lemma 8.1.2. (1) The projection $\widehat{\mathcal{L}}^{\text{ad}}(\mathcal{M}) \rightarrow \widehat{\mathcal{L}}_{\text{fin}}(\mathcal{M})$ given by $(\mathcal{E}, (\|\cdot\|_{\mathfrak{s}})_{\mathfrak{s} \in \mathbb{R}_{>0}}(\Psi, \Psi)) \mapsto \mathcal{E}_{\text{fin}}$ induces a left exact sequence

$$0 \longrightarrow \widehat{\mathcal{L}}^{\text{ad}}_{\infty}(\mathcal{M}) \longrightarrow \widehat{\mathcal{L}}^{\text{ad}}(\mathcal{M}) \longrightarrow \widehat{\mathcal{L}}_{\text{fin}}(\mathcal{M}),$$

where $\mathcal{M}_{\text{fin}}$ is the generic fiber of the integral model and $\mathcal{E}_{\text{fin}}$ is the pull-back of the line bundle $\mathcal{E}$ to the generic fiber.

(2) For any finite place $\mathfrak{p}$ such that $\mathcal{M} \otimes \mathbb{A}_{\Psi, \mathfrak{p}}$ is smooth over $\mathbb{A}_{\Psi, \mathfrak{p}}$, we have $\widehat{\mathcal{L}}_{\mathfrak{p}}(\mathcal{M}) \subset \widehat{\mathcal{L}}^{\text{ad}}_{\text{fin}}(\mathcal{M})$.

Proof. See [LMZ25, Lemma 8.2]. $\square$

⁸In [GS90, §2.6] and its references, this constant is fixed by the choice of Kähler form; see [Ara74, §1, equation (*)].

Let $\widetilde{\text{pink}}_{\bullet}(\text{apple})$ be the (blue -valued) group of apple -cycles on apple and let $\text{pink}_{\bullet}(\text{apple})$ be the quotient of $\widetilde{\text{pink}}_{\bullet}(\text{apple})$ by the subgroup generated by apple -cycles that are rationally trivial on a closed fiber of apple . Recall from [BGS94, Proposition 2.3.1] that we have a well-defined intersection pairing:

$$\widehat{\text{green}}_{\bullet}(\text{apple}) \times \text{pink}_{\bullet}(\text{apple}) \longrightarrow \text{rainbow} := \text{rainbow} / \text{mushroom} \cdot \text{blue} \{ \text{brown} : \text{orange} : \text{red} \mid \text{pink} \}, \quad (8.1.1)$$

see also remarks in the p.941 of [BGS94].

Let $\widetilde{\text{pink}}_{\bullet}(\text{apple})_{\text{yellow}=\text{orange}}$ be the subgroup of apple -cycles that have zero degree on each connected component of the generic fiber of apple . This subgroup is the orthogonal complement of $\widehat{\text{green}}_{\bullet}(\text{apple})$. Let $\widetilde{\text{pink}}_{\bullet}(\text{apple})^{\perp} \subseteq \widetilde{\text{pink}}_{\bullet}(\text{apple})$ be the orthogonal complement of $\widehat{\text{green}}_{\text{vert}}^{\text{red}}(\text{apple})$. Let $\text{green}^{\text{red}}(\text{apple}_{\text{pink}})$ be the image of the map $\widehat{\text{green}}^{\text{red}}(\text{apple}) \rightarrow \text{green}(\text{apple}_{\text{pink}})$ in Lemma 8.1.2. By Lemma 8.1.2, the truncated arithmetic intersection pairing (8.1.2) induces a bilinear pairing:

$$(-, -) : \text{green}(\text{apple}_{\text{pink}})^{\text{adm}} \times \widetilde{\text{pink}}_{\bullet}(\text{apple})^{\perp} \longrightarrow \text{rainbow}. \quad (8.1.2)$$

8.2. Arithmetic Kudla-Rapoport special divisors. Let $\text{pink} \subset \text{apple}$ be an effective Cartier divisor with associated line bundle $\text{red}(-\text{pink})$. On a complex fiber, let $\text{green}_{\text{pink}}$ be a Green current associated to the complex subvariety $\text{pink} \subset \text{apple}_{\text{pink}}$. Then the pair $(\text{pink}_{\text{pink}}, \text{green}_{\text{pink}})$ determines a hermitian line bundle $(\text{red}(-\text{pink})_{\text{pink}}, \|\cdot\|_{\text{pink}})$ over the complex fiber (see [GS90, §2]). By abuse of notation, let $(\text{pink}, (\text{green}_{\text{pink}})_{\text{pink}} \in \mathbb{C})$ be the corresponding hermitian line bundle in the arithmetic Picard group.

We go back to our global notation in §5 and §6. Let $\text{brown} \in \text{brown}(\text{blue}(\text{apple}_{\text{pink}}))_{\text{brown}}$ be an admissible Schwartz function and let $\text{brown} \subset \text{brown}(\text{apple}_{\text{pink}})$ be any compact open subgroup under which brown is invariant via the Weil representation. For $\text{brown} \in \text{green}_{\text{pink}}$, we have the Kudla-Rapoport divisor $\widetilde{\text{pink}}(\text{brown}, \text{brown}) \rightarrow \widehat{\text{blue}}$.

For $\text{pink} \in \text{red}(\text{apple}_{\text{pink}})$ and $\text{brown} \in \text{green}_{\text{pink}}^+$, we have the *automorphic Green function* [Bru12] $\text{green}_{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown}, \text{brown}) = \text{green}_{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown})$ introduced in [Bru12]. By the work of Bruinier [Bru12, Corollary 5.16], this automorphic Green function is admissible. While its definition is similar to that of the Kudla's Green function, the precise details are not essential for this paper.

For $\text{brown} \in \text{green}_{\text{pink}}^+$, define *arithmetic Kudla-Rapoport divisors*

$$\widetilde{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown}) = (\widetilde{\text{pink}}(\text{brown}, \text{brown}), (\text{green}_{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown}))_{\text{pink}}) \in \widehat{\text{green}}(\widehat{\text{blue}}).$$

Definition 8.2.1. (1) We take $\text{blue}_{\text{pink}}$ and $\text{blue}_{\text{brown}}$ as in [LMZ25, Definition 6.7] as extensions of the automorphic line bundle blue to $\widehat{\text{blue}}$. Define the arithmetic line bundle

$$\widetilde{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown}, \text{brown}) := -\text{brown}(\text{brown})(\text{blue}_{\text{pink}}, \|\cdot\|_{\text{green}_{\text{pink}}}), \quad \text{and} \quad \widetilde{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown}, \text{brown}) := -\text{brown}(\text{brown})(\text{blue}_{\text{brown}}, \|\cdot\|_{\text{green}_{\text{pink}}})$$

where $\|\cdot\|_{\text{green}_{\text{pink}}}$ denotes the natural Peterson metric.

(2) Let $\text{brown} \in \text{brown}(\text{blue}(\text{apple}_{\text{pink}}))$ be any admissible Schwartz function, we have generating series

$$\widetilde{\text{pink}}^{\text{brown}}(\text{brown}) = \widetilde{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown}) + \sum_{\text{brown} \in \text{green}_{\text{pink}}^+} \widetilde{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown}, \text{brown}), \quad \text{brown} = \text{brown}(\text{brown}).$$

It is defined as a formal series in the arithmetic Picard group $\widehat{\text{green}}(\widehat{\text{blue}})[[\text{brown}]]$.

Proposition 8.2.2. Let $\text{brown} \in \text{brown}(\text{blue}(\text{apple}_{\text{pink}}))_{\text{brown}}$ be brown -invariant under the Weil representation and let $\text{pink} \in \widetilde{\text{pink}}_{\bullet}(\text{apple})^{\perp}$ be any apple -cycle. The arithmetic intersection product

$$(\widetilde{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown}), \text{pink}) \in \text{rainbow}[[\text{brown}]]$$

is well-defined and is a weight- pink holomorphic Hilbert modular form up to a constant term (for which we use the same notation). More precisely, the generating function $\text{brown} \in \text{brown}(\text{apple}_{\text{pink}}) \mapsto (\widetilde{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown}), \text{pink})$ lies in $\text{brown}(\text{apple}_{\text{pink}}) \otimes \text{brown}(\text{apple}_{\text{pink}}) \otimes \text{rainbow}[[\text{brown}]]$.

Proof. Since $\widetilde{\text{pink}}^{\text{brown}}(\text{brown}, \text{brown}) \in \widehat{\text{green}}^{\text{red}}(\widehat{\text{blue}})[[\text{brown}]]$, the result follows from (8.1.2) and [MZ24, Theorem 2.4]. $\square$

Let $\text{brown} \rightarrow \widehat{\text{blue}}$ be any finite map with one-dimensional image and zero dimensional complex fiber. The composition $\text{brown}_{\text{pink}}(\text{brown}) \rightarrow \text{brown}_{\text{pink}}(\widehat{\text{blue}}) \rightarrow \text{brown}_{\text{pink}}(\widehat{\text{blue}}) \cong \text{brown}_{\text{pink}}(\widehat{\text{blue}})$ induces a natural map:

$$\text{brown}_{\text{pink}}(\text{brown}) \longrightarrow \text{pink}(\widehat{\text{blue}}), \quad (8.2.1)$$

where the middle isomorphism is due to [Sta18, 0FEW]. In particular, let $\mathfrak{w}_b \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_b, \mathfrak{w}_b))^{*-}$ be an admissible Schwartz function. We define a $\mathfrak{w}$ -cycle $\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f))$ via Definition 6.2.1. Hence we obtain the arithmetic intersection pairing

$$(\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b), \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b)) \in \mathbb{R}[\mathfrak{w}]. \quad (8.2.2)$$

9. LOCAL INTERSECTION NUMBERS

In this section, we calculate the non-archimedean local contribution of the generating series in Definition 8.2.1 to the arithmetic intersection product given in equation (8.2.2). We also compute the archimedean local contribution for its variant “Kudla generating series”.

9.1. Intersection of Kudla-Rapoport cycles. Recall the group data from §5.1. In particular, we have a global CM field $\mathbb{K}/\mathbb{Q}$ with a fixed CM type Φ , along with an $\mathbb{K}/\mathbb{Q}$ -hermitian space of dimension n and an integral lattice $\mathfrak{L} \subset \mathbb{K}$.

For the remainder of the paper, we impose the following assumptions on Schwartz functions $\mathfrak{w} = \mathfrak{w}_b \otimes \mathfrak{w}_b \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_b, \mathfrak{w}_b))^{*-}$:

- $\mathfrak{w}_b = \mathfrak{w}_{b, \mathfrak{p}} \otimes \mathfrak{w}_{b, \mathfrak{p}'} \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_{b, \mathfrak{p}}, \mathfrak{w}_{b, \mathfrak{p}'}))^{*-}$ such that $\mathfrak{w}_{b, \mathfrak{p}}$ is admissible.
 - $\mathfrak{w}_b = \mathfrak{w}_{b, \mathfrak{p}} \otimes \mathfrak{w}_{b, \mathfrak{p}'} \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_{b, \mathfrak{p}}, \mathfrak{w}_{b, \mathfrak{p}'}))^{*-}$ such that $\mathfrak{w}_{b, \mathfrak{p}'}$ is admissible.
 - The function $\mathfrak{w} = \mathfrak{w}_b \otimes \mathfrak{w}_b$ is regular in at least two places over $\mathfrak{p}$.
- (★)

Under this assumption, for any singular $\mathfrak{w} \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_b, \mathfrak{w}_b))^{*-}$, the derived averaged weighted special cycle $\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b)$ vanishes by Definition 6.2.1(2). Consequently, for any positive element $\mathfrak{w} \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_b, \mathfrak{w}_b))^{*+}$ and any positive definite Hermitian matrix $\mathfrak{w}_b \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_b, \mathfrak{w}_b))^{*+}$, the derived intersection can be expressed as

$$\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) \cap \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) = \sum_{\substack{\mathfrak{w} = \begin{pmatrix} \mathfrak{w} & * \\ * & \mathfrak{w}_b \end{pmatrix} \\ \mathfrak{w} \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_b, \mathfrak{w}_b))^{*+}}} \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b). \quad (9.1.1)$$

By [KR14, Lemma 2.7], any nonsingular matrix $\mathfrak{w}$ appearing in the summation is positive semi-definite and therefore must be positive definite.

For each $\mathfrak{w} \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_b, \mathfrak{w}_b))^{*+}$, equation (8.1.1) defines an intersection number of $\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b)$ and $\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b)$ valued in $\mathbb{R}$. For $\mathfrak{w} \neq \mathfrak{w}$, the intersection is proper over the complex fiber by [KR14, Lemma 2.20], and we can define a more specific intersection number

$$\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) := \frac{\mathfrak{w}}{\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b)} \left(\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b), \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) \right) \in \mathbb{R}, \quad (9.1.2)$$

which lifts the $\mathbb{R}$ -valued intersection number (this depends on the choice of the section $\mathfrak{w}$ in the line bundle). Here

$$\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) = \#(\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) \setminus \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) / \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b)). \quad (9.1.3)$$

Moreover, the intersection number (9.1.2) is defined as the sum of an archimedean and a non-archimedean part, such that:

- The *archimedean contribution* is defined in the usual way by evaluating the Green function at the complex point of the KR cycle:

$$\sum_{\mathfrak{w} \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_b, \mathfrak{w}_b))^{*+}} \langle \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b), \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) \rangle_{\mathfrak{w}} \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b),$$

where by definition $\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) = \mathfrak{w}$ for complex places $\mathfrak{w}$ (and $\mathfrak{w}$ is $\mathfrak{w}$ were a real place).

- The *non-archimedean contribution* is given by the arithmetic Euler-Poincaré characteristic:

$$\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) := \sum_{\mathfrak{w} \in \mathfrak{w}(\mathbb{G}(\mathbb{A}_f)(\mathfrak{w}_b, \mathfrak{w}_b))^{*+}} \langle \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b), \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) \rangle_{\mathfrak{w}} \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b),$$

where the local intersection number $\langle -, - \rangle_{\mathfrak{w}}$ is defined for a non-archimedean place $\mathfrak{w}$ through the Euler-Poincaré characteristic of a derived tensor product on $\mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b) \otimes_{\mathfrak{w}} \mathfrak{w}(\mathfrak{w}_b, \mathfrak{w}_b)$.

With this definition of $\mathbb{I}_{\mathfrak{p}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$, we obtain a well-defined place-by-place decomposition:

$$\mathbb{I}_{\mathfrak{p}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) = \sum_{\mathfrak{a} \in \mathfrak{p}, \mathfrak{b} \in \mathfrak{p}} \mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) + \sum_{\mathfrak{a} \in \mathfrak{p}, \mathfrak{b} \in \mathfrak{p}} \mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}),$$

where $\mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ is the sum of the local contributions from all $\mathfrak{p} \in \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}$ lying above $\mathfrak{a}$ and $\mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ is the sum of the local contributions from all $\mathfrak{p} \in \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}$ lying above $\mathfrak{a}$.

For a finite place $\mathfrak{a} \in \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}$, by the same proof as in [KR14, Lemma 2.21], the local contribution $\mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ is nonzero only if the quadratic extension $\mathfrak{p}/\mathfrak{a}$ remains non-split over $\mathfrak{a}$, hence

$$\mathbb{I}_{\mathfrak{p}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) = \sum_{\mathfrak{a} \in \mathfrak{p}, \mathfrak{b} \in \mathfrak{p}} \mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) + \sum_{\mathfrak{a} \in \mathfrak{p}, \mathfrak{b} \in \mathfrak{p}} \mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}),$$

where $\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}$ denote the set of all finite places $\mathfrak{p} \nmid \mathfrak{p}$ over $\mathfrak{p}$ such that $\mathfrak{p}/\mathfrak{a}$ is unramified.

Recall from §6.5 that we have Kudla's Green functions $\mathfrak{g}_{\mathfrak{p}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ associated to the special divisor $\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ on $\mathfrak{p}_{\mathfrak{p}}$, where $\mathfrak{p}_{\mathfrak{p}} = \mathfrak{p}_{\mathfrak{p}} + \mathfrak{p}_{\mathfrak{p}} \in \mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}_{\mathfrak{p}})$ is a variable. For our application, we also consider

$$\mathbb{I}_{\mathfrak{p}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) = \sum_{\mathfrak{a} \in \mathfrak{p}, \mathfrak{b} \in \mathfrak{p}} \mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) + \sum_{\mathfrak{a} \in \mathfrak{p}, \mathfrak{b} \in \mathfrak{p}} \mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}),$$

where the non-archimedean contribution is still given by the arithmetic Euler-Poincaré characteristic, and the archimedean contribution is defined by evaluating the Kudla's Green function $\mathfrak{g}_{\mathfrak{p}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ on $\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ rather than the automorphic Green function, which is not holomorphic in general.

Next, consider the central value of the Eisenstein series at a nonsingular $\mathfrak{p}$ and its first derivative. By the archimedean Siegel-Weil formula (Theorem 4.3.1(1)), only two classes of $\mathfrak{p}$ contribute nonzero terms: either $\mathfrak{p}$ is totally positive definite ($\mathfrak{p} \in \mathfrak{p}_{\mathfrak{p}}^+(\mathfrak{p})$), or $\mathfrak{p}$ has signature $(\mathfrak{p} - \mathfrak{p}, \mathfrak{p})$ at some place and is positive definite at all other places ($\mathfrak{p} \in \mathfrak{p}_{\mathfrak{p}}^+(\mathfrak{p})$).

Based on this, we define $\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) = \mathfrak{p}_{\mathfrak{p}}(\mathfrak{p})$ as follows (recall from §7 that $\mathfrak{p}_{\mathfrak{p}}$ is the incoherent $\mathfrak{p}/\mathfrak{a}$ -hermitian space associated to $\mathfrak{p}$):

- When $\mathfrak{p} \in \mathfrak{p}_{\mathfrak{p}}^+(\mathfrak{p})$, we define $\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ to be the set of finite places $\mathfrak{p} \in \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}$ such that $\mathfrak{p}_{\mathfrak{p}}$ does not represent $\mathfrak{p}$. This set is nonempty since $\mathfrak{p}_{\mathfrak{p}}$ is not totally positive definite.
- When $\mathfrak{p} \in \mathfrak{p}_{\mathfrak{p}}^+(\mathfrak{p})$, we define $\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ to be the set of finite places $\mathfrak{p} \in \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}$ such that $\mathfrak{p}_{\mathfrak{p}}$ does not represent $\mathfrak{p}$, together with the set of archimedean places $\mathfrak{p} \in \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}$ where the base change of $\mathfrak{p}$ to the nearby hermitian space $\mathfrak{p}_{\mathfrak{p}}$ is positive definite.

For a more general definition using representation theory, see [Kud97, §5]. By the same proof as in [KR14, Lemma 9.1], the derivative of the local Whittaker function at a finite place $\mathfrak{p}$ is nonzero only if the quadratic extension $\mathfrak{p}/\mathfrak{a}$ remains non-split over $\mathfrak{a}$.

9.2. Local intersection: inert non-archimedean places. Now let $\mathfrak{p} \in \mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}$ be a finite place of $\mathfrak{p}$ that is inert in $\mathfrak{p}$, and let $\mathfrak{p}$ be the unique place of $\mathfrak{p}$ lying above $\mathfrak{p}$.

Proposition 9.2.1. *For the Schwartz function $\mathfrak{p} = \mathfrak{p} \otimes \mathfrak{p} \in \mathfrak{p}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ satisfying $(\star)$, we have*

$$\begin{aligned} \mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) &= \frac{\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})}{\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p})} \sum_{\mathfrak{p} \in \mathfrak{p}(\mathfrak{p}) \setminus (\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}) \times \mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}))} \mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) \cdot \int_{\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p})} \mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) \cdot \mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}), \\ &= \frac{\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})}{\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p})} \sum_{\substack{\mathfrak{p} = (\mathfrak{p} \otimes \mathfrak{p}) \in \mathfrak{p}_{\mathfrak{p}}^+(\mathfrak{p}) \\ \mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}) = \{\mathfrak{p}\}}} \mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) \cdot \mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}). \end{aligned}$$

Here, $\mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ is the weighted arithmetic intersection number defined in Definition 2.3.3. It is independent of the choice of representative elements $\mathfrak{p}$ in $\mathfrak{p}$, and we set $\mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s}) := \mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}(\mathfrak{p}, \mathfrak{q}, \mathfrak{r}, \mathfrak{s})$ for some $\mathfrak{p} \in \mathfrak{p}$.

Proof. Following the proof of [KR14, Proposition 2.22], the terms in (9.1.1) contribute to the intersection number $\mathbb{I}_{\mathfrak{p}, \mathfrak{a}}^{\text{loc}}$ only when $\mathfrak{p}$ is positive definite, $\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}) = \{\mathfrak{p}\}$, and $\mathfrak{p} \nmid \mathfrak{p}$. Moreover,

by the argument in [KR14, Lemma 2.21], the intersection in this case is supported in the basic locus. Consequently, we may reduce our computation to the formal completion along the basic locus using §6.3. For any two $\mathfrak{a}, \mathfrak{b} \in \mathfrak{A}$ where $\mathfrak{A}$ is nonsingular, we can find $\mathfrak{c} \in \mathfrak{A}(\mathfrak{a}, \mathfrak{b})$ such that $\mathfrak{a} \mathfrak{b} = \mathfrak{c}^2$, hence the induced automorphism on the RZ space induces an identification between $\mathfrak{A}(\mathfrak{a}, \mathfrak{b})$ and $\mathfrak{A}(\mathfrak{a}, \mathfrak{c})$. Hence the notation $\mathfrak{A}(\mathfrak{a}, \mathfrak{b})$ does not depend on the choice of $\mathfrak{c} \in \mathfrak{A}(\mathfrak{a}, \mathfrak{b})$.⁹

Fix a place $\mathfrak{p}$ of $\mathfrak{F}$ above $\mathfrak{p}_0$. Then, by Proposition 6.3.1, we can compute the intersection number fiberwise using the projection (5.4.2) and then multiply the result by the factor $\mathfrak{A}(\mathfrak{a}, \mathfrak{b})$ defined in (9.1.3). Therefore, we focus on the intersection over the fiber $\mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}}$ defined in (5.4.3).

By Proposition 6.3.1, the restrictions of the special cycles $\tilde{\mathfrak{a}}(\mathfrak{a}, \mathfrak{b})$ and $\tilde{\mathfrak{b}}(\mathfrak{a}, \mathfrak{b})$ to $\mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}}$ take the forms

$$\sum_{(\mathfrak{a}, \mathfrak{b}) \in \mathfrak{A}(\mathfrak{a}) \setminus [\mathfrak{A}(\mathfrak{a}) \times \mathfrak{A}(\mathfrak{a})]} \mathfrak{a}(\mathfrak{a}, \mathfrak{b}) \cdot [\mathfrak{a}(\mathfrak{a}, \mathfrak{b})]_{\mathfrak{A}(\mathfrak{a})},$$

and

$$\sum_{(\mathfrak{a}, \mathfrak{b}) \in \mathfrak{A}(\mathfrak{a}) \setminus [\mathfrak{A}(\mathfrak{a}) \times \mathfrak{A}(\mathfrak{a})]} \mathfrak{b}(\mathfrak{a}, \mathfrak{b}) \cdot [\mathfrak{b}(\mathfrak{a}, \mathfrak{b})]_{\mathfrak{A}(\mathfrak{a})},$$

respectively. We compute the intersection number by pulling back to the covering formal scheme $\mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}} \times \mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}} / \mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}}$ in the $\mathfrak{A}$ -uniformization setting, as described in §6.3.

The intersection number of $\tilde{\mathfrak{a}}(\mathfrak{a}, \mathfrak{b}) \cap \tilde{\mathfrak{b}}(\mathfrak{a}, \mathfrak{b})$ (restricted to $\mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}}$) is given by the sum

$$\mathfrak{a}(\mathfrak{a}, \mathfrak{b}) \cdot \mathfrak{b}(\mathfrak{a}, \mathfrak{b}) \cdot \mathfrak{A}(\mathfrak{a}, \mathfrak{b}) \cap \mathfrak{A}(\mathfrak{a}, \mathfrak{b}),$$

taken over the $\mathfrak{A}(\mathfrak{a})$ -orbits (under the diagonal action) of tuples $(\mathfrak{a}, \mathfrak{b})$ satisfying

$$(\mathfrak{a}, \mathfrak{b}) \in \mathfrak{A}(\mathfrak{a}) \times \mathfrak{A}(\mathfrak{a}) / \mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}} \quad \text{and} \quad (\mathfrak{a}, \mathfrak{b}) \in \mathfrak{A}(\mathfrak{a}) \times \mathfrak{A}(\mathfrak{a}) / \mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}}.$$

Here, $\mathfrak{A}$ denotes the Euler–Poincaré characteristic of the derived tensor product of the corresponding complexes in the $\mathfrak{A}$ -group. Its well-definedness follows from the same reasoning as in Proposition 6.2.2.

Let $\mathfrak{a} = (\mathfrak{a}, \mathfrak{b})$. By (6.3.1), we obtain

$$\mathfrak{A}(\mathfrak{a}, \mathfrak{b}) \cap \mathfrak{A}(\mathfrak{a}, \mathfrak{b}) = \mathfrak{A}(\mathfrak{a}, \mathfrak{b}) \cdot \mathfrak{A}(\mathfrak{a}, \mathfrak{b}). \quad (9.2.1)$$

The first term of (9.2.1) is equal to

$$[\mathfrak{A} : \mathfrak{A}] \cdot \mathfrak{A}(\mathfrak{a}, \mathfrak{b}).$$

In particular, it is invariant under the (diagonal) action of $\mathfrak{A}(\mathfrak{a})$ on the product $(\mathfrak{A}(\mathfrak{a}) \times \mathfrak{A}(\mathfrak{a}))$.

The second term of (9.2.1) as a function

$$(\mathfrak{a}, \mathfrak{b}) \in (\mathfrak{A}(\mathfrak{a}) \times \mathfrak{A}(\mathfrak{a})) / \mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}} \mapsto \mathfrak{A}(\mathfrak{a}, \mathfrak{b})$$

is also invariant under the (diagonal) $\mathfrak{A}(\mathfrak{a})$ -action. For a fixed pair $\mathfrak{a} = (\mathfrak{a}, \mathfrak{b})$, we obtain

$$\sum_{(\mathfrak{a}, \mathfrak{b}) \in (\mathfrak{A}(\mathfrak{a}) \times \mathfrak{A}(\mathfrak{a})) / \mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}}} \mathfrak{A}(\mathfrak{a}, \mathfrak{b}) \cdot \mathfrak{A}(\mathfrak{a}, \mathfrak{b}) = \frac{\mathfrak{A}(\mathfrak{a}, \mathfrak{b})}{\mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}}} \int_{\mathfrak{A}^\wedge_{\mathfrak{p}, \mathfrak{F}}} \mathfrak{A}(\mathfrak{a}, \mathfrak{b}) \cdot \mathfrak{A}(\mathfrak{a}, \mathfrak{b}).$$

This concludes the first line of the assertion. To conclude the second line of the assertion, we decompose

$$\mathfrak{A}(\mathfrak{a}) \times \mathfrak{A}(\mathfrak{a}) = \coprod_{\mathfrak{c} = \begin{pmatrix} \mathfrak{a} & * \\ * & \mathfrak{b} \end{pmatrix}} \mathfrak{A}(\mathfrak{a}, \mathfrak{c}),$$

⁹This does not relate to the linear invariance of the special cycle.

where the index ranges over all $\star \in \mathcal{O}^+_{\mathbb{F}}(\mathbb{F})$, we may rewrite the summation as:

$$\frac{\text{box}(\text{box})}{\text{box}(\text{box})} \sum_{\star = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix}} \sum_{\mathfrak{o} \in \mathcal{O}^+(\mathbb{F}) / \mathcal{O}^+(\mathbb{F})} \text{box}_{\mathfrak{o}}(\star, \star) \cdot \int_{\mathcal{O}^+(\mathbb{F})} \star(\star^{-1}\star) \star.$$

By the regularity assumption, if $\star$ is singular, then for any $\mathfrak{o} \in \mathcal{O}^+(\mathbb{F})$, the local orbital integral $\int_{\mathcal{O}^+(\mathbb{F})} \star(\star^{-1}\star) \star$ vanishes at a place $\mathfrak{o} \mid \star$. Hence the global orbital integral $\int_{\mathcal{O}^+(\mathbb{F})} \star(\star^{-1}\star) \star$ vanishes for all $\mathfrak{o} \in \mathcal{O}^+(\mathbb{F})$ with singular $\star$. By [KR14, Lemma 2.7], we further conclude that $\star \in \mathcal{O}^+_{\mathbb{F}}(\mathbb{F})$, hence:

$$\text{box}_{\mathfrak{o}}(\star, \star) = \frac{\text{box}(\text{box})}{\text{box}(\text{box})} \sum_{\substack{\star = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix} \in \mathcal{O}^+_{\mathbb{F}}(\mathbb{F}) \\ \star(\star, \star) = \{\star\}}} \text{box}_{\mathfrak{o}}(\star, \star) \cdot \star(\star),$$

where we use the fact that $\text{box}_{\mathfrak{o}}(\star, \star) = \text{box}_{\mathfrak{o}}(\star, \star)$ whose well-definedness is discussed in the beginning of the proof. $\square$

9.3. Local intersection: archimedean places. In this part, we compute the archimedean contribution of the local intersections. Let $\mathfrak{o} \in \mathcal{O}_{\mathbb{F}}$ be a archimedean place of $\mathbb{F}$ and let $\mathfrak{o}$ be the unique place in $\mathcal{O}$ above $\mathfrak{o}$.

Theorem 9.3.1. *For the Schwartz function $\star = \star \otimes \star_b \in \mathcal{O}(\mathcal{O}_{\mathbb{F}}) \otimes \mathcal{O}(\mathcal{O}_{\mathbb{F}})$ satisfying $(\star)$, and $\star \in \mathbb{F}^{\times}$, we have*

$$\begin{aligned} \text{box}_{\mathfrak{o}}(\star, \star, \star) &= \frac{\text{box}(\text{box})}{\text{box}(\text{box})} \sum_{\substack{\star = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix} \in \mathcal{O}^+_{\mathbb{F}}(\mathbb{F}) \\ \star(\star, \star) = \{\star\}}} \text{box}_{\mathfrak{o}}(\star, \star) \star(\star) \\ &\quad - \frac{\text{box}(\text{box})}{\text{box}(\text{box})} \int_{\mathcal{O}_{\mathbb{F}}} \star(\star, \star, \star) \wedge \star_{\mathfrak{o}}(\star, \star, \star). \end{aligned}$$

Here $\star = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix} = \star + \star \in \mathcal{O}_{\mathbb{F}}(\mathbb{F})$, and functions in the second line of equations are defined in Proposition 6.5.2. Also $\text{box}_{\mathfrak{o}}(\star, \star) = \text{box}_{\mathfrak{o}}(\star, \star)$ is the local height function on $\mathcal{O}_{\mathbb{F}}(\mathbb{F})$ (Theorem 4.3.1), which is independent of the choice of $\mathfrak{o}$ with $(\mathfrak{o}, \mathfrak{o}) = \star$.

Note that the archimedean uniformization a priori is taken for some $\mathfrak{o} \mid \mathfrak{o}$ over $\mathbb{F}$, but the complex uniformization for the RSZ Shimura variety (§5.5) and Kudla-Rapoport divisors and their Green functions (§6) only depend on their restriction $\mathfrak{o}$ to $\mathbb{F}$ (and the fixed CM type $\mathcal{O}$). Hence we index the functions in the main statement by $\mathfrak{o}$ for simplicity.

Proof. Similar to the computation in Proposition 9.2.1, we have

$$\text{box}_{\mathfrak{o}}(\star, \star, \star) = \frac{\text{box}(\text{box})}{\text{box}(\text{box})} \sum_{\substack{\star = \begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix} \\ \in \mathcal{O}^+_{\mathbb{F}}(\mathbb{F})}} \sum_{\mathfrak{o} \in \mathcal{O}^+(\mathbb{F}) / \mathcal{O}^+(\mathbb{F})} \star(\star, \star, \star)(\star(\mathfrak{o})(\star_b)) \cdot \int_{\mathcal{O}^+(\mathbb{F})} \star(\star^{-1}\star) \star.$$

While the summation initially ranges over all Hermitian matrices $\star$ of the form $\begin{pmatrix} \star & \star \\ \star & \star \end{pmatrix}$, the regularity assumption on $\star$ ensures that only nonsingular terms contribute. Moreover, the set $\mathcal{O}^+(\mathbb{F})$ is non-empty if and only if $\star$ represents the space $\mathcal{O}^+(\mathbb{F})$, hence we may restrict to $\star \in \mathcal{O}^+_{\mathbb{F}}(\mathbb{F})$ satisfying $\star(\star, \star) = \{\star\}$.

By Proposition 6.4.1 and our regularity assumption $(\star)$ on $\star$, the complex fibers $\star_{\mathfrak{o}}(\star, \star)$ and $\star_{\mathfrak{o}}(\star, \star_b)$ intersect properly (including the case where one of them are empty). We have

$$\star_{\mathfrak{o}}(\star, \star, \star) = \int_{\mathcal{O}_{\mathbb{F}}} \star_{\mathfrak{o}}(\star, \star) \wedge \star_{\mathfrak{o}}(\star, \star_b).$$

Using the definition of the star product (4.2.2) and Green currents for special cycles (4.2.1), we have

$$\int_{\mathcal{Y}(\mathfrak{a}_b)} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) \wedge \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) = \int_{\mathcal{Y}(\mathfrak{a}_b)} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) \star \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) - \int_{\mathcal{Y}(\mathfrak{a}_b)} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) \wedge \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b). \quad (9.3.1)$$

The summation of the first term in (9.3.1) gives

$$\frac{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)}{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)} \sum_{\substack{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) \\ \in \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)}} \sum_{\substack{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) \\ \in \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)}} \int_{\mathcal{Y}(\mathfrak{a}_b)} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) \left(\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) \right) \cdot \int_{\mathcal{Y}(\mathfrak{a}_b)} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b).$$

Recall from the definition of the local height function (Theorem 4.3.1) that $\int_{\mathcal{Y}(\mathfrak{a}_b)} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) = \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$. Taking the sum, this gives

$$\frac{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)}{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)} \sum_{\substack{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) \\ \in \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)}} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b).$$

Taking the summation over the second terms of (9.3.1) and applying the definitions of $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$ from (6.5.1) and of $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$ after (6.5.1), we obtain the desired result. $\square$

10. GENERATING SERIES AND MODULARITY

In this section, we combine the results in §7, §8 and §9 to construct the generating series $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$ introduced in (1.4.1). We will also prove that it defines a holomorphic modular form valued in $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) \otimes \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$, where $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) := \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$. The proof is based on the modification method introduced in [MZ24] for the hyperspecial level and generalized by [Zha25] for general maximal parahoric level.

10.1. Modified Fourier-Jacobi series. In this subsection, we modify the Fourier-Jacobi series introduced in §7.4 to incorporate error terms. We then compare the local factors with the generating series on the geometric side.

Fix basis elements $\mathfrak{g}_{\mathfrak{a}_b} \in \wedge^{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$ and $\mathfrak{g}_{\mathfrak{a}_b} \in \wedge^{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$. Recall from (H5) that there exists a form $\mathfrak{g}_{\mathfrak{a}_b}$ of degree $\mathfrak{g}_{\mathfrak{a}_b}$ on $\mathfrak{g}_{\mathfrak{a}_b}$ such that its pullback, $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) = \mathfrak{g}_{\mathfrak{a}_b}^* \mathfrak{g}_{\mathfrak{a}_b}$, defines a gauge form on $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$, which is independent of the choice of $\mathfrak{g}_{\mathfrak{a}_b}$. This form determines a Tamagawa measure on $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$ via a choice of convergence factors $\mathfrak{g}_{\mathfrak{a}_b}$. Furthermore, recall from (H6) and (H7) that we have the constants

$$\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) = \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b), \quad \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b) = \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b),$$

which compare with self-dual Haar measures.

Let $\mathfrak{g}_{\mathfrak{a}_b} = \mathfrak{g}_{\mathfrak{a}_b} \otimes \mathfrak{g}_{\mathfrak{a}_b} \in \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b, \mathfrak{a}_b)$ be an admissible Schwartz function of type $\mathfrak{g}_{\mathfrak{a}_b}$ satisfying $(\star)$, and let $\mathfrak{g}_{\mathfrak{a}_b}$ be the finite set of finite places $\mathfrak{a}_b \nmid \mathfrak{p}$ of $\mathfrak{g}_{\mathfrak{a}_b}$ where $\mathfrak{g}_{\mathfrak{a}_b}$ is not self-dual.

Recall from (G2) that $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b)$ denotes the $\mathfrak{a}_b$ -nearby hermitian space of $\mathfrak{g}_{\mathfrak{a}_b}$. Define

$$\mathfrak{g}_{\mathfrak{a}_b} := \mathfrak{g}_{\mathfrak{a}_b} \sum_{\mathfrak{a}_b \in \mathfrak{g}_{\mathfrak{a}_b}} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b) \in \bigoplus_{\mathfrak{a}_b \in \mathfrak{g}_{\mathfrak{a}_b}} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b)(\mathfrak{g}_{\mathfrak{a}_b, \mathfrak{a}_b}) \cdot \mathfrak{g}_{\mathfrak{a}_b},$$

where $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b) = \mathfrak{g}_{\mathfrak{a}_b} \otimes \mathfrak{g}_{\mathfrak{a}_b}$, $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b)$ is a $\mathfrak{g}_{\mathfrak{a}_b}$ -valued Schwartz function in $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b)(\mathfrak{g}_{\mathfrak{a}_b, \mathfrak{a}_b})$ such that

$$\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b) = -\frac{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b)}{\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b)} \sum_{\substack{\mathfrak{a}_b \leq \mathfrak{a}_b \text{ and} \\ \mathfrak{a}_b - \mathfrak{a}_b \in \mathfrak{g}_{\mathfrak{a}_b}}} \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b) \in \mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b)(\mathfrak{g}_{\mathfrak{a}_b, \mathfrak{a}_b}), \quad (10.1.1)$$

where $\mathfrak{g}_{\mathfrak{a}_b}$ is the type of the vertex lattice $\mathfrak{g}_{\mathfrak{a}_b}$ at the place $\mathfrak{a}_b$, and where $\mathfrak{g}_{\mathfrak{a}_b}(\mathfrak{a}_b)$ are admissible Schwartz functions at $\mathfrak{a}_b$ that match with $\mathfrak{g}_{\mathfrak{a}_b}$ and are of type $\mathfrak{g}_{\mathfrak{a}_b}$.

For each $\mathfrak{a}_\bullet$, the tensor product $\mathfrak{w}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet) \otimes \mathfrak{w}_{\mathfrak{a}_\bullet}$ defines a Schwartz function on the coherent hermitian space $\mathfrak{e}^{(\mathfrak{a}_\bullet)}(\mathfrak{w}_{\mathfrak{a}_\bullet})$, and gives rise to the (coherent) Eisenstein series (recall Notation 7.3.1 for our convention):

$$\mathfrak{E}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) = \sum_{\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)}} \mathfrak{E}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})(\mathfrak{a}_\bullet).$$

Let $\mathfrak{F}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})$ denote the $\mathfrak{a}_\bullet$ -th Fourier coefficient of $\mathfrak{E}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})$.

Lemma 10.1.1. $\mathfrak{F}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) = \mathfrak{a}_\bullet$ unless $\mathfrak{a}_\bullet = \{\mathfrak{a}_\bullet\}$.

Proof. By construction, $\mathfrak{w}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet)$ defines a Schwartz function in $\mathfrak{e}^{(\mathfrak{a}_\bullet)}$. It is positive definite at the archimedean places and is isomorphic to $\mathfrak{e}$ at all finite places $\mathfrak{p}$ except at $\mathfrak{a}_\bullet$. In particular, a matrix $\mathfrak{a}_\bullet$ represents this space if and only if it represents $\mathfrak{a}_\bullet$ but does not represent $\mathfrak{a}_\bullet$. $\square$

Define the *modified Eisenstein series* as

$$\partial \mathfrak{E}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) := \partial \mathfrak{E}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) + \mathfrak{E}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}). \quad (10.1.2)$$

We denote its $\mathfrak{a}_\bullet$ -th Fourier coefficient by

$$\partial \mathfrak{F}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) = \partial \mathfrak{F}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) + \mathfrak{F}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}).$$

Theorem 10.1.2. *There exists a constant $\mathfrak{c}_{\mathfrak{a}_\bullet}$, depending only on $\mathfrak{e}$ (see (10.1.3)), such that for any nonsingular $\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)}$ with $\mathfrak{a}_\bullet = \{\mathfrak{a}_\bullet\}$, we have*

$$\frac{\partial \mathfrak{F}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}{\mathfrak{E}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})} = \begin{cases} \mathfrak{c}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \partial \mathfrak{E}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \cdot \mathfrak{a}_\bullet(\mathfrak{w}_{\mathfrak{a}_\bullet}), & \mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)}, \\ \mathfrak{c}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \mathfrak{a}_\bullet(\mathfrak{w}_{\mathfrak{a}_\bullet}), & \mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)}. \end{cases}$$

Here, $\mathfrak{c}_{\mathfrak{a}_\bullet}$ denotes the cardinality of the residue field of $\mathfrak{e}$ at $\mathfrak{a}_\bullet$ when $\mathfrak{a}_\bullet$ is non-archimedean and equals $\mathfrak{a}_\bullet$ when $\mathfrak{a}_\bullet$ is an real embedding. Furthermore, $\partial \mathfrak{F}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) = \mathfrak{a}_\bullet$ for all remaining values of $\mathfrak{a}_\bullet$.

Proof. The last assertion follows from assumption $(\star)$. For any nonsingular $\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)}$ satisfying $\mathfrak{a}_\bullet = \{\mathfrak{a}_\bullet\}$ with $\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)}$, by the non-archimedean Siegel–Weil formula (Theorem 3.7.1) and the archimedean (arithmetic) Siegel–Weil formula (Theorem 4.3.1), we have

$$\begin{aligned} \partial \mathfrak{F}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) &= \mathfrak{a}'_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \prod_{\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)} \setminus \{\mathfrak{a}_\bullet\}} \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \prod_{\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)}} \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}), \\ &= \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \frac{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})} \prod_{\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)} \setminus \{\mathfrak{a}_\bullet\}} \frac{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})} \prod_{\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)}} \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}). \end{aligned}$$

Define the constant

$$\mathfrak{c}_{\mathfrak{a}_\bullet} = \frac{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}. \quad (10.1.3)$$

Then we have

$$\mathfrak{c}_{\mathfrak{a}_\bullet} \partial \mathfrak{F}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})^{-\mathfrak{a}_\bullet} = \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}).$$

Now suppose $\mathfrak{a}_\bullet = \{\mathfrak{a}_\bullet\}$ for $\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)}$. If $\mathfrak{a}_\bullet \notin \mathfrak{e}^{(\mathfrak{a}_\bullet)}$, by the local Siegel–Weil formula for both non-archimedean and archimedean places, we obtain

$$\begin{aligned} \partial \mathfrak{F}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) &= \prod_{\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)} \setminus \{\mathfrak{a}_\bullet\}} \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \mathfrak{a}'_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \prod_{\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)} \setminus \{\mathfrak{a}_\bullet\}} \mathfrak{a}'_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}), \\ &= \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \left(\frac{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})} \right)^{\mathfrak{a}_\bullet} \mathfrak{a}'_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \prod_{\mathfrak{a}_\bullet \in \mathfrak{e}^{(\mathfrak{a}_\bullet)} \setminus \{\mathfrak{a}_\bullet\}} \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}), \\ &= \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \frac{\mathfrak{a}'_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})} \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}). \end{aligned}$$

By Theorem 3.7.1, Proposition 3.7.2, and the definition of $\partial \mathfrak{E}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})$ in Theorem 3.4.4, we obtain

$$\frac{\mathfrak{a}'_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})} = \frac{\mathfrak{a}'_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})} \cdot \frac{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})} = \partial \mathfrak{E}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}) \cdot \frac{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})}{\mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet})} \mathfrak{a}_{\mathfrak{a}_\bullet}(\mathfrak{a}_\bullet, \mathfrak{w}_{\mathfrak{a}_\bullet}),$$

where $\mathfrak{d}_{\mathfrak{p}}$ is the Haar measure induced by self-dual Haar measures at the place $\mathfrak{p}$ (see §3.1). Changing to the Tamagawa measure, we conclude that

$$\partial \mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1} = \mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \partial \mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1} \mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}).$$

On the other hand, if $\mathfrak{p} \in \mathfrak{p}$, we have

$$\mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1} = \mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1} \mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1} \mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}).$$

Furthermore,

$$\frac{\mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1}}{\mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1}} = \frac{\mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1}}{\mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1}} \cdot \frac{\mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1}}{\mathfrak{d}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})^{-1}}.$$

This will contribute the error term of $\partial \mathfrak{d}_{\mathfrak{p}}$ in Theorem 3.4.4, by the way we define the correction term in equation (10.1.1). $\square$

10.2. Geometric modularity of the difference Green function. We recall the geometric modularity of the difference Green function, proved in [ES18] for $\mathfrak{p} = \mathfrak{p}$ and generalized in [MZ24] for totally real $\mathfrak{p}$.

Let $\mathfrak{p} \in \mathfrak{p}$ be any place of $\mathfrak{p}$. Consider the difference of two Green functions:

$$\mathfrak{d}_{\mathfrak{p}}^{-1}(\mathfrak{a}, \mathfrak{b}) = \begin{cases} \mathfrak{d}_{\mathfrak{p}}^{-1}(\mathfrak{a}, \mathfrak{b}) - \mathfrak{d}_{\mathfrak{p}}^{-1}(\mathfrak{a}, \mathfrak{b}), & \text{for } \mathfrak{p} \in \mathfrak{p}^+, \\ \mathfrak{d}_{\mathfrak{p}}^{-1}(\mathfrak{a}, \mathfrak{b}), & \text{otherwise.} \end{cases}$$

We define the generating series of Green functions as the archimedean part of the generating series of arithmetic special divisors ($\mathfrak{p} \in \mathfrak{p}$):

$$\mathfrak{d}_{\mathfrak{p}}^{-1}(\mathfrak{a}, \mathfrak{b}) := \sum_{\mathfrak{p} \in \mathfrak{p}} \mathfrak{d}_{\mathfrak{p}}^{-1}(\mathfrak{a}, \mathfrak{b}) \mathfrak{d}_{\mathfrak{p}}^{-1}(\mathfrak{a}, \mathfrak{b}), \quad \mathfrak{p} = \mathfrak{p}, \mathfrak{p}, \mathfrak{p} - \mathfrak{p}.$$

Theorem 10.2.1 ([MZ24, Theorem 3.13]). *Suppose that $\mathfrak{d}_{\mathfrak{p}} \in \mathfrak{p}(\mathfrak{p}, \mathfrak{p})$ is invariant under $\mathfrak{p} \subset \mathfrak{p}(\mathfrak{p}, \mathfrak{p})$ via the Weil representation. Then the generating series $\mathfrak{d}_{\mathfrak{p}}^{-1}(\mathfrak{a}, \mathfrak{b})$, when paired with a degree-zero $\mathfrak{p}$ -cycle on $\mathfrak{p}$, lies in $\mathfrak{p}(\mathfrak{p}, \mathfrak{p})$.*

10.3. Auxiliary cycles and modularity. We now construct the auxiliary cycles and prove the modularity of the generating series in our setup. Following the approach in [Zha25], the main idea is to modify the given $\mathfrak{p}$ -cycle $\mathfrak{p} \in \mathfrak{p}(\mathfrak{p})$ by some auxiliary $\mathfrak{p}$ -cycles $\mathfrak{p} \in \mathfrak{p}(\mathfrak{p})$ whose modularity is already known. This modification ensures that the resulting cycle $\mathfrak{p} + \mathfrak{p}$ lies in $\mathfrak{p}(\mathfrak{p})^{\perp}$, allowing us to apply the admissible intersection pairing defined in (8.1.2).

We choose certain Kudla–Rapoport special cycles as auxiliary cycles at the archimedean places. Let $\mathfrak{p} = (\mathfrak{p}, \dots, \mathfrak{p})$ and let $\mathfrak{p} = \mathfrak{p} - \mathfrak{p}$ be the identity matrix of rank $\mathfrak{p} - \mathfrak{p}$. Let $\mathfrak{p} = \mathfrak{p} \otimes \mathfrak{p} \in \mathfrak{p}(\mathfrak{p}, \mathfrak{p})$ such that:

- The complex fiber $\mathfrak{p}(\mathfrak{p}, \mathfrak{p})$ has nonzero degree for one place $\mathfrak{p} \in \mathfrak{p}$ over $\mathfrak{p}$.
- The prime-to- $\mathfrak{p}$ part $\mathfrak{p}$ is admissible of weight $\mathfrak{p}$.
- It is regular at two split places $\mathfrak{p}, \mathfrak{p}$.

For such $\mathfrak{p}$ and $\mathfrak{p}$, by Proposition 3.4.9, we see that:

- For any $\mathfrak{p} \in \mathfrak{p}$, the local intersection number vanishes, $\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{p}) = \mathfrak{p}$. Similarly, for the analytic side $\partial \mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{p}) = \mathfrak{p}$. Recall here from §10.1 that $\mathfrak{p}$ is the set of inert places $\mathfrak{p}$ over $\mathfrak{p}$ where the vertex lattice $\mathfrak{p}$ is not self-dual.
- For any $\mathfrak{p} \in \mathfrak{p} \setminus \mathfrak{p}$, the local intersection number $\mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{p})$ equals the analytic side $\partial \mathfrak{p}_{\mathfrak{p}}(\mathfrak{p}, \mathfrak{p})$.

Remark 10.3.1. In [Zha25, §13] and [LMZ25, §8], the auxiliary cycle at the archimedean place is chosen as a Hecke CM cycle $\mathfrak{p}(\mathfrak{p}, \mathfrak{p})$, where $\mathfrak{p}$ is a maximal order at any $\mathfrak{p} \in \mathfrak{p}$. To ensure the existence of such a cycle, one must enlarge the truncation $\mathfrak{p}$, as discussed in [Zha25, Theorem 13.5]. On the other hand, the auxiliary cycles we construct here do not require enlarging $\mathfrak{p}$. This will help improve the almost modularity theorem in [Zha25], see Remark 10.4.1.

Next, we recall the non-archimedean auxiliary cycles from [Zha25].

Theorem 10.3.2. *For any $\bullet$ -cycle $\mathfrak{c} \in \widetilde{\mathcal{C}}(\widetilde{\mathcal{H}})$ and any place $\mathfrak{p} \mid \mathfrak{p}_0 \in \mathcal{O}_{\mathfrak{p}}^{\times}$ of $\mathfrak{F}$, there exists a $\bullet$ -cycle $\mathfrak{c}_{\mathfrak{p}}$, supported in the basic locus of $\widetilde{\mathcal{H}} \otimes_{\mathfrak{F}} \mathfrak{F}_{\mathfrak{p}}$, and a Schwartz function $\mathfrak{f}_{\mathfrak{p}}(\mathfrak{c}) \in \mathcal{S}(\mathcal{O}_{\mathfrak{p}}^{\times})(\mathfrak{c}_{\mathfrak{p}})$ such that:*

- (1) *The pairing of $\mathfrak{c} - \mathfrak{c}_{\mathfrak{p}}$ with each vertical component of every geometric mod- $\mathfrak{p}$ fiber $\widetilde{\mathcal{H}} \otimes_{\mathfrak{F}} \mathfrak{F}_{\mathfrak{p}}$ of $\widetilde{\mathcal{H}}$ is zero.*
- (2) *The arithmetic intersection pairing*

$$(\widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}), \mathfrak{f}_{\mathfrak{p}}(\mathfrak{c})) \in \mathcal{S}(\mathcal{O}_{\mathfrak{p}}^{\times})$$

is well-defined and is a weight- $\mathfrak{w}$ holomorphic Hilbert modular form given by $\mathfrak{f}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}_{\mathfrak{p}}(\mathfrak{c}))$ up to a constant term (for which we use the same notation).

Proof. The modification follows the same argument as in [Zha25, §12.4], using the *very special* $\bullet$ -cycles defined in [Zha25, Definition 12.14]. A detailed and corrected exposition is provided in [LMZ25, §8.3, Step 3], we omit the proof here. $\square$

10.4. Modular forms. In this subsection, we construct the auxiliary generating series

$$\mathfrak{f}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}_{\mathfrak{p}}) \in \mathcal{S}(\mathcal{O}_{\mathfrak{p}}^{\times})(\mathfrak{c}_{\mathfrak{p}}), \mathfrak{f}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}_{\mathfrak{p}}) \in \mathcal{S}(\mathcal{O}_{\mathfrak{p}}^{\times}).$$

Recall our global notation in §5.1. Let $\mathfrak{F}/\mathfrak{F}_0$ be a global CM extension with a fixed CM type Φ and a distinguished element $\mathfrak{d} \in \mathfrak{F}_0^{\times}$. Let $(\mathfrak{V}, \mathfrak{h})$ be a hermitian space of dimension $\mathfrak{w}$ over $\mathfrak{F}$ such that the signature of $\mathfrak{V}$ is $(\mathfrak{w} - \mathfrak{r}, \mathfrak{r})$ at $\mathfrak{p}_0$ and $(\mathfrak{r}, \mathfrak{w} - \mathfrak{r})$ otherwise. In this case, the associated Shimura datum $(\mathfrak{G}, \mathfrak{X}, \{\mathfrak{h}_{\mathfrak{p}}\})$ is of *strict fake Drinfeld type* (see [RSZ21, Example 2.3]). By [Zha25, Corollary 12.6], the associated canonical model $\mathfrak{H}(\mathfrak{G}, \mathfrak{X}, \{\mathfrak{h}_{\mathfrak{p}}\})$ is connected. For each $\mathfrak{p} \in \mathcal{O}_{\mathfrak{p}}^{\times}$, the complex fiber $\mathfrak{H}_{\mathfrak{p}}$ defined in §5.5 is identical to the complex fiber of $\mathfrak{H}(\mathfrak{G}, \mathfrak{X}, \{\mathfrak{h}_{\mathfrak{p}}\})$. Fix a distinguished prime $\mathfrak{p} \in \mathcal{O}_{\mathfrak{p}}^{\times}$.

Fix an ideal $\mathfrak{a} \subset \mathcal{O}_{\mathfrak{p}}^{\times}$ containing all places above $\mathfrak{p}$ (to be enlarged later). The data specified so far can be completed to define a Shimura datum. We find a lattice $\mathfrak{M}$ and enlarge $\mathfrak{a}$ so that $\mathfrak{M}_{\mathfrak{p}}$ are vertex lattices for any $\mathfrak{p} \in \mathcal{O}_{\mathfrak{p}}^{\times} \setminus \mathfrak{a}$. We require that the level structure satisfies the assumptions in §5.3.

Let $\mathfrak{g}_{\mathfrak{p}} \in \mathcal{O}_{\mathfrak{p}}^{\times}(\mathfrak{F})$ be a totally positive definite hermitian matrix of rank $\mathfrak{w} - \mathfrak{r}$. Let $\mathfrak{c} : \mathfrak{F} \rightarrow \mathfrak{F}$ be any complex embedding with restriction $\mathfrak{c} \in \mathcal{O}_{\mathfrak{p}}^{\times}$ over $\mathfrak{F}$ and $\mathfrak{c}_{\mathfrak{p}}$ over $\mathfrak{F}_{\mathfrak{p}}$. By the complex uniformization of the special cycles (§6.4), we see that the degree (over $\mathfrak{F}$) of $\widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})$ (resp. $\widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})$) is identical on each geometric connected component $\mathfrak{H}_{\mathfrak{p}}$ of the projection (5.5.1) since $\mathfrak{H}_{\mathfrak{p}}$ is connected.

Since $\widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})$ has nonzero degree on $\mathfrak{H}_{\mathfrak{p}}$, we may arrange for a suitable constant $\mathfrak{c}_{\mathfrak{p}}$ such that the degree function of

$$\widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}} + \mathfrak{c}_{\mathfrak{p}} \widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}}$$

is identically zero along each fiber of the projection $\widetilde{\mathcal{H}} \rightarrow \mathfrak{H}(\mathfrak{G}, \mathfrak{X}, \{\mathfrak{h}_{\mathfrak{p}}\})/\mathfrak{H}_{\mathfrak{p}}$. By the uniformization, it follows that $\widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}} + \mathfrak{c}_{\mathfrak{p}} \widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}}$ has zero degree for all $\mathfrak{p} \in \mathcal{O}_{\mathfrak{p}}^{\times}$.

For each finite $\mathfrak{p} \mid \mathfrak{p}_0 \in \mathcal{O}_{\mathfrak{p}}^{\times}$, by Theorem 10.3.2, we can find modifications $\mathfrak{f}_{\mathfrak{p}}$ for each $\mathfrak{p} \in \mathcal{O}_{\mathfrak{p}}^{\times}$.

To summarize, we define a suitable modification:

$$\widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}} = \widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}} + \mathfrak{c}_{\mathfrak{p}} \widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}} + \sum_{\mathfrak{p} \mid \mathfrak{p}_0 \in \mathcal{O}_{\mathfrak{p}}^{\times}} \mathfrak{c}_{\mathfrak{p}} \widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}}, \quad (10.4.1)$$

which satisfies the following properties:

- (1) (Over $\mathfrak{F}$) It has degree $\mathfrak{w}$ on each connected component of every complex fiber $\mathfrak{H}_{\mathfrak{p}}$.
- (2) (Over $\mathfrak{F}_{\mathfrak{p}}$) Its pairing with each vertical component of every geometric mod- $\mathfrak{p}$ fiber $\widetilde{\mathcal{H}} \otimes_{\mathfrak{F}} \mathfrak{F}_{\mathfrak{p}}$ of $\widetilde{\mathcal{H}}$ is zero for all finite places $\mathfrak{p} \nmid \mathfrak{p}_0$ over $\mathfrak{F}$.

Therefore, $\widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}} \in \widetilde{\mathcal{C}}(\widetilde{\mathcal{H}})^{\perp}$, and we may apply (8.1.2). By Proposition 8.2.2, the arithmetic intersection pairing

$$(\widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}), \widetilde{\mathfrak{c}}_{\mathfrak{p}}(\mathfrak{g}_{\mathfrak{p}}, \mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}}) \in \mathcal{S}(\mathcal{O}_{\mathfrak{p}}^{\times})$$

is well-defined and is a weight- $\frac{1}{2}$ holomorphic Hilbert modular form up to a constant term (for which we use the same notation).

Remark 10.4.1. The modification procedure applies to any $\mathfrak{p}$ -cycle $\mathfrak{c} \in \mathfrak{C}(\widetilde{\mathcal{H}})$ satisfying the following condition: for each $\mathfrak{a} \in \mathfrak{A}$, the degree of the complex fiber $\mathfrak{c}_{\mathfrak{a}}$ is constant for all place $\mathfrak{p} \mid \mathfrak{a}$ over $\mathfrak{P}$.

This condition holds for Kudla–Rapoport special cycles and CM cycles. Let $\mathcal{H}(\mathfrak{p})(\mathfrak{a})$ be the integral model of the abelian-type unitary Shimura variety, we have a natural projection $\mathcal{H}(\mathfrak{p})(\mathfrak{a}) \rightarrow \mathcal{H}(\mathfrak{p})(\mathfrak{a})$. Let $\mathfrak{c}$ be any $\mathfrak{p}$ -cycle in $\mathcal{H}(\mathfrak{p})(\mathfrak{a})$, then the pull-back $\mathfrak{c}^*$ also satisfies this condition.

Let $\mathfrak{c} = \left(\begin{smallmatrix} \mathfrak{a} \\ \mathfrak{a} \end{smallmatrix} \right) \in \mathfrak{C}(\mathfrak{p})$, recall from §7.4 the Fourier–Jacobi coefficients of the Eisenstein series and its derivative:

$$\partial \mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) = \sum_{\mathfrak{c} \in \mathfrak{C}(\mathfrak{p})} \sum_{\mathfrak{d} \in \mathfrak{D}(\mathfrak{p})} \partial \mathfrak{c}_{\mathfrak{d}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{d}}^*, \quad \mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) = \sum_{\mathfrak{c} \in \mathfrak{C}(\mathfrak{p})} \sum_{\mathfrak{d} \in \mathfrak{D}(\mathfrak{p})} \mathfrak{c}_{\mathfrak{d}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{d}}^*.$$

Define the *modified Fourier–Jacobi series* as

$$\partial \mathfrak{E}_{\mathfrak{p}}^{\bullet}(\mathfrak{a}, \mathfrak{b}) := \partial \mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) + \mathfrak{E}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}).$$

This is the Fourier–Jacobi series of the modified Eisenstein series introduced in (10.1.2).

Consider also (where $\mathfrak{c}_{\mathfrak{p}}$ is the constant defined in (10.1.3))

$$\mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}) := \mathfrak{c}_{\mathfrak{p}} \partial \mathfrak{E}_{\mathfrak{p}}^{\bullet}(\mathfrak{a}, \mathfrak{b}) - \sum_{\mathfrak{c} \in \mathfrak{C}(\mathfrak{p})} \mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{p}}^* + \sum_{\mathfrak{p} \mid \mathfrak{a} \in \mathfrak{A}} \mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{p}}^*,$$

where $\mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b})$ is defined in Proposition 6.5.2. This expression relates to the intersection number of auxiliary cycles defined on the geometric side in §10.3.

Now, consider the generating series (recall the index convention in Theorem 9.3.1, and note that modifications over special fibers $\mathfrak{c}_{\mathfrak{p}}$ has empty complex fiber):

$$\begin{aligned} \mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}) = \mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) := & \sum_{\mathfrak{c} \in \mathfrak{C}(\mathfrak{p})} \left[\mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{p}}^* + \mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{p}}^* \right] \\ & + \mathfrak{c}_{\mathfrak{p}} \partial \mathfrak{E}_{\mathfrak{p}}^{\bullet}(\mathfrak{a}, \mathfrak{b}) + \mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}). \end{aligned} \quad (10.4.2)$$

By Theorem 10.2.1, Proposition 7.4.1, and Proposition 6.5.2, we conclude that

$$\mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}) \in \mathfrak{C}(\mathfrak{p}) \mathfrak{C}(\mathfrak{p})^*.$$

Proposition 10.4.2. $\mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}) \in \mathfrak{C}(\mathfrak{p}) \mathfrak{C}(\mathfrak{p})^*.$

Proof. For an automorphic function $\mathfrak{f}$, the holomorphicity condition is characterized by the vanishing of the archimedean lowering operator at each point $\mathfrak{a} \in \mathfrak{A}$. Since the differential operator commutes with the right action of $\mathfrak{C}(\mathfrak{p})$, by the strong approximation theorem for the special linear group, it suffices to verify holomorphicity when $\mathfrak{a} = \mathfrak{a}$. This reduces our analysis to classical Hilbert modular forms.

We now show that $\mathfrak{c}_{\mathfrak{p}}(\mathfrak{a})$ admits a Fourier expansion of the form (noting that $\mathfrak{c}_{\mathfrak{p}} \neq \mathfrak{c}_{\mathfrak{p}}^*$):

$$\mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}) = \sum_{\mathfrak{c} \in \mathfrak{C}(\mathfrak{p}), \mathfrak{c} \geq \mathfrak{a}} \mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{c}), \quad \mathfrak{c}_{\mathfrak{p}} \in \mathfrak{C}(\mathfrak{p}). \quad (10.4.3)$$

Since $\mathfrak{c}_{\mathfrak{p}}$ in (10.4.1) are supported on the special fiber, we have

$$\mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{p}}^* = \mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{p}}^* + \mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{p}}^*.$$

Since $\mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{p}}^*$ is holomorphic for each $\mathfrak{a} \in \mathfrak{A}$, it remains to show the holomorphicity of the function:

$$\sum_{\mathfrak{c} \in \mathfrak{C}(\mathfrak{p})} \left[\mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{p}}^* - \mathfrak{c}_{\mathfrak{p}}(\mathfrak{a}, \mathfrak{b}) \mathfrak{c}_{\mathfrak{p}}^* \right] - \mathfrak{c}_{\mathfrak{p}} \partial \mathfrak{E}_{\mathfrak{p}}^{\bullet}(\mathfrak{a}, \mathfrak{b}), \quad \mathfrak{c}_{\mathfrak{p}} = \mathfrak{c}_{\mathfrak{p}} \text{ or } \mathfrak{c}_{\mathfrak{p}}^*.$$

Since $\heartsuit$ is regular at two places, for any $\heartsuit \in \heartsuit_{\heartsuit}^{\times}$, the intersection $\heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \cap \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b)$ is proper (and is empty when $\heartsuit$ is not positive). The analogous statement holds for $\heartsuit_{\heartsuit}$. Consequently, we can write

$$\begin{aligned} & \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \left(\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) + \heartsuit_{\heartsuit}(\heartsuit_{\heartsuit}, \heartsuit_b) \right) \\ &= \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \left(\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) + \heartsuit_{\heartsuit}(\heartsuit_{\heartsuit}, \heartsuit_{\heartsuit}) \right) \\ & \quad - \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \left(\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) + \heartsuit_{\heartsuit}(\heartsuit_{\heartsuit}, \heartsuit_{\heartsuit}) \right), \end{aligned} \quad (10.4.4)$$

where all terms are well-defined. Since the automorphic Green function is independent of $\heartsuit$, it suffices to show that the following function (and its $\heartsuit_{\heartsuit}$ variant, which follows analogously) has a Fourier expansion of the form given in (10.4.3):

$$\sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \left[\heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \left(\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) \right) - \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) \right] - \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b). \quad (10.4.5)$$

By Theorem 10.1.2, modulo constant terms, we have:

$$\begin{aligned} \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) &= \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \cdot \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) \\ & \quad + \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \partial \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \cdot \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b). \end{aligned}$$

By the proof of Theorem 9.3.1 (noting that the intersection is proper), we obtain:

$$\begin{aligned} & \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \left(\heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \left(\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) \right) - \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) \right) \\ &= \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \cdot \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b). \end{aligned}$$

Therefore, all non-positive terms in the expansion in (10.4.5) cancel out, and the remaining expression takes the form given in (10.4.3). $\square$

Moreover, the proof of Proposition 10.4.2 yields the following conclusion:

Corollary 10.4.3. *For any $\heartsuit > 0$, we have*

$$\begin{aligned} & \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) - \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) = \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \left(\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) + \heartsuit_{\heartsuit}(\heartsuit_{\heartsuit}, \heartsuit_b) \right) \\ & \quad + \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \partial \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \cdot \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) \\ & \quad + \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \sum_{\heartsuit \in \heartsuit_{\heartsuit}^{\times}} \partial \heartsuit_{\heartsuit}(\heartsuit, \heartsuit_b) \cdot \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b). \quad \square \end{aligned}$$

We may project $\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b)$ from $\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b)$ to

$$\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) \otimes \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b),$$

which we will still denote by $\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b)$. We define the function

$$\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) := \frac{\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b)}{\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b)} \left(\heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) \right) + \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) \in \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b) \otimes \heartsuit_{\heartsuit}(\heartsuit_b, \heartsuit_b). \quad (10.4.6)$$

Proposition 10.4.4. *Let $\mathfrak{v} \in \mathfrak{o}_\bullet^\times$. The Fourier coefficients of $\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b})$ are given by:*

$$\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) := \sum_{\substack{\mathfrak{c} \in \mathfrak{o}_\bullet^\times \\ \mathfrak{c} \equiv \mathfrak{v} \pmod{\mathfrak{p}_\bullet}}} \mathfrak{f}_{\mathfrak{c},\mathfrak{b}} := \sum_{\substack{\mathfrak{c} \in \mathfrak{o}_\bullet^\times \\ \mathfrak{c} \equiv \mathfrak{v} \pmod{\mathfrak{p}_\bullet}}} \mathfrak{f}_{\mathfrak{c},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) = \sum_{\substack{\mathfrak{c} \in \mathfrak{o}_\bullet^\times \\ \mathfrak{c} \equiv \mathfrak{v} \pmod{\mathfrak{p}_\bullet}}} \left(\mathfrak{f}_{\mathfrak{c},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) - \partial \mathfrak{f}_{\mathfrak{c},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) \right) \cdot \mathfrak{f}_{\mathfrak{c},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) \in \mathfrak{o}_\bullet^\times.$$

Proof. As discussed in §9.1, for each $\mathfrak{v} > \mathfrak{o}_\bullet$, there exists an element

$$\frac{\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b})}{\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b})} \left(\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) \right) - \mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) \in \mathfrak{o}_\bullet^\times$$

which lifts the Fourier coefficient $\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) \in \mathfrak{o}_\bullet^\times$. The intersection pairing can then be decomposed into local contributions as described in §9.1.

The archimedean contribution to the geometric side consists of special values of automorphic Green functions

$$\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) \left(\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) + \mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}, \mathfrak{v}_\mathfrak{b}) \right),$$

which agree with the analytic side as shown in Corollary 10.4.3.

The non-archimedean contributions to the geometric side are addressed in Proposition 9.2.1. By our construction of $\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}$, all terms involving $\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}$ vanish by Lemma 11.0.1, leaving only the terms stated in Proposition 10.4.4. $\square$

11. GLOBALIZATION AND THE PROOF OF KUDLA-RAPOPORT CONJECTURE

In this section, we prove Theorem 3.4.4 by induction. The case $\mathfrak{v} = \mathfrak{o}_\bullet$ follows from Corollary 3.5.2. Assuming the theorem holds for all local hermitian spaces of dimension $\leq \mathfrak{v} - \mathfrak{o}_\bullet$ ($\mathfrak{v} \neq \mathfrak{o}_\bullet$), we now establish the Kudla-Rapoport conjecture for $\mathfrak{v}$ and for an integer $\mathfrak{b} \leq \mathfrak{v}$, for a prime $\mathfrak{p} > \mathfrak{v}$, and for an admissible Schwartz function of weight $\mathfrak{w} = (\mathfrak{w}_\mathfrak{b}, \mathfrak{w}_\mathfrak{v})$, where $\mathfrak{w}_\mathfrak{b} \in \{\mathfrak{o}_\bullet, \mathfrak{p}_\bullet\}^{\mathfrak{v}-\mathfrak{b}}$. We write this case as $\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}/\mathfrak{p}_\bullet, \mathfrak{w}, \mathfrak{b}, \mathfrak{w}_\mathfrak{b}, \mathfrak{v}_\mathfrak{b})$.

Recall that $\mathfrak{p}_\bullet$ is the $\mathfrak{o}_\bullet/\mathfrak{p}_\bullet$ -hermitian space that contains a type $\mathfrak{b}$ vertex lattice, and $\mathfrak{p}_\bullet$ is the $\mathfrak{o}_\bullet/\mathfrak{p}_\bullet$ -hermitian space that does not. By Proposition 3.4.9, we have the base case:

Lemma 11.0.1. *Assume $\mathfrak{v} \neq \mathfrak{o}_\bullet$, and let $\mathfrak{w} \in \mathfrak{w}(\mathfrak{p}_\bullet)$ be an admissible Schwartz function of type $\mathfrak{b}$. Let $\mathfrak{a} \in \mathfrak{p}_\bullet$ be a set of linearly independent vectors such that for one of the entries $\mathfrak{a}_\mathfrak{v}$, the following holds:*

- (1) If $\mathfrak{b} \neq \mathfrak{v}$, then $\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}_\mathfrak{v}, \mathfrak{a}_\mathfrak{v}) = \mathfrak{o}_\bullet$; if $\mathfrak{b} = \mathfrak{v}$, then $\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}_\mathfrak{v}, \mathfrak{a}_\mathfrak{v}) = \mathfrak{p}_\bullet$.
- (2) If $\mathfrak{b} \neq \mathfrak{o}_\bullet$, then $\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}_\mathfrak{v}, \mathfrak{a}_\mathfrak{v}) = -\mathfrak{p}_\bullet$; if $\mathfrak{b} = \mathfrak{o}_\bullet$, then $\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}_\mathfrak{v}, \mathfrak{a}_\mathfrak{v}) = \mathfrak{o}_\bullet$.

Then, we have

$$\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}) = \partial \mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}, \mathfrak{w}).$$

In other words, Theorem 3.4.4 holds for $\mathfrak{f}_{\mathfrak{v},\mathfrak{b}}(\mathfrak{a}/\mathfrak{p}_\bullet, \mathfrak{w}, \mathfrak{b}, \mathfrak{w}_\mathfrak{b}, \mathfrak{v}_\mathfrak{b})$. $\square$

Suppose we aim to prove Theorem 3.4.4 for a $\mathfrak{p}$ -adic extension $\mathfrak{o}_\bullet/\mathfrak{p}_\bullet$ with $\mathfrak{p} > \mathfrak{v}$, a vector $\mathfrak{a} \in \mathfrak{p}_\bullet$, and all admissible Schwartz functions $\mathfrak{w} \in \mathfrak{w}(\mathfrak{p}_\bullet)$ of type $\mathfrak{b}$.

We now change to global notation. We start with a global CM extension $\mathfrak{o}_\bullet/\mathfrak{p}_\bullet$ satisfying the following conditions:

- CM 1** $\mathfrak{o}_\bullet$ has a place $\mathfrak{p}_\bullet$ above $\mathfrak{p}$ that is unramified in $\mathfrak{o}_\bullet$, and the induced local extension $\mathfrak{o}_\bullet/\mathfrak{p}_\bullet$ is the one for which we aim to prove the Kudla-Rapoport conjecture.
- CM 2** All places $\mathfrak{p}_\bullet \neq \mathfrak{p}_\bullet$ of $\mathfrak{o}_\bullet$ lying above $\mathfrak{p}$ are split in $\mathfrak{o}_\bullet$.
- CM 3** If we want to verify the local Kudla-Rapoport conjecture over $\mathfrak{p}_\bullet$, we choose $\mathfrak{o}_\bullet$ to be a quadratic extension of $\mathfrak{p}_\bullet$ such that $\mathfrak{o}_\bullet, \mathfrak{p}_\bullet = \mathfrak{p}_\bullet \times \mathfrak{p}_\bullet$ is a split place.

We fix a CM type $\mathfrak{p}_\bullet$ for $\mathfrak{o}_\bullet/\mathfrak{p}_\bullet$. By enlarging $\mathfrak{p}_\bullet \subset \mathfrak{p}_\bullet$ while ensuring that $\mathfrak{p}_\bullet \nmid \mathfrak{p}_\bullet$, we further assume that $\mathfrak{o}_\bullet/\mathfrak{p}_\bullet$ are unramified for any finite places $\mathfrak{p}_\bullet \nmid \mathfrak{p}_\bullet$ over $\mathfrak{p}_\bullet$.

Now, we choose an $\mathfrak{v}$ -dimensional $\mathfrak{o}_\bullet/\mathfrak{p}_\bullet$ -hermitian vector space $\mathfrak{p}_\bullet$ satisfying the following conditions:

- HM 1** The signature of $\mathfrak{p}_\bullet$ is $(\mathfrak{v} - \mathfrak{o}_\bullet, \mathfrak{o}_\bullet)$ at a unique place $\mathfrak{p}_\bullet : \mathfrak{o}_\bullet \rightarrow \mathfrak{p}_\bullet$ and $(\mathfrak{v}, \mathfrak{o}_\bullet)$ at all other places $\mathfrak{p}_\bullet \in \mathfrak{p}_\bullet \setminus \{\mathfrak{p}_\bullet\}$.

HM 2 The localization $\mathbb{A}_{\mathfrak{p}_0}$ is isomorphic to the hermitian space we started with.

HM 3 The space $\mathbb{A}_{\mathfrak{p}_0} = \prod_{\mathfrak{p} \in \mathfrak{p}_0} \mathbb{A}_{\mathfrak{p}} \mathbb{A}_{\mathfrak{p}_0}$ contains a lattice $\mathbb{M}_{\mathfrak{p}_0} = \prod_{\mathfrak{p} \in \mathfrak{p}_0} \mathbb{M}_{\mathfrak{p}} \mathbb{M}_{\mathfrak{p}_0}$ such that $\mathbb{M}_{\mathfrak{p}_0}$ is a vertex lattice of type $\mathfrak{p}_0$. We extend this into a lattice $\mathbb{M}^{[\mathfrak{p}]} \subset \mathbb{A}$ (may extend $\mathfrak{p}$ if necessary). Similarly, we construct $\mathbb{M}^{[\mathfrak{q}]} \subset \mathbb{A}(\mathfrak{q}_0)$ such that it is of type $\mathfrak{q} \equiv \mathfrak{p} - \mathfrak{r}$ ($\mathfrak{r} \in \mathfrak{p}_0$) at $\mathfrak{p}_0$.

By local constancy, we may replace $\mathfrak{p} \in \mathbb{A}(\mathfrak{p}_0)$ with a vector in $\mathbb{A}(\mathfrak{p}_0)$ without changing the intersection number on the geometric side (Theorem 2.3.6) and the corresponding modified local density function on the analytic side (by continuity). We still denote this vector by $\mathfrak{p}$. Let $\mathfrak{p} \in \mathbb{A}(\mathfrak{p}_0)$ be its moment matrix. It is non-degenerate and totally positive definite, with $\mathfrak{p}(\mathfrak{p}) = \{\mathfrak{p}_0\}$. We write

$$\mathfrak{p} = \begin{pmatrix} \mathfrak{p}_a & \mathfrak{p}_b \\ \mathfrak{p}_c & \mathfrak{p}_d \end{pmatrix}$$

where $\mathfrak{p}_b$ is a matrix of size $(\mathfrak{p} - \mathfrak{r}) \times (\mathfrak{p} - \mathfrak{r})$.

We further enlarge $\mathfrak{p}$ so that the following conditions hold:

LR 1 By enlarging $\mathfrak{p}$, we may assume that $\mathbb{M}^{[\mathfrak{p}]}$ and $\mathbb{M}^{[\mathfrak{q}]}$ are self-dual for all inert places $\mathfrak{p}_0 \in \mathbb{A} \setminus \{\mathfrak{p}_0\}$.

LR 2 By enlarging $\mathfrak{p}$, we may assume that for all inert places $\mathfrak{p}_0 \in \mathbb{A} \setminus \{\mathfrak{p}_0\}$, there exists some $\mathfrak{r} \in \{\mathfrak{r}_1, \dots, \mathfrak{r}_k\}$ such that

$$\star_{\mathfrak{p}_0}(\mathfrak{r}_i, \mathfrak{r}_j) = \mathfrak{p}_0,$$

where $\mathfrak{r}_i$ is an entry of the vector $\mathfrak{p}$.

We now complete the given data to a Shimura datum as in §5.1, ensuring only that $\mathfrak{p} \nmid \mathfrak{p}$. Then, we construct a class of Schwartz functions. First, we write

$$\mathfrak{p}_0 := \mathfrak{p} \mathbb{M}^{[\mathfrak{p}]} \mathfrak{p} \quad \text{and} \quad \mathfrak{p}_0 := \mathfrak{p} \mathbb{M}^{[\mathfrak{q}]} \mathfrak{p}.$$

They are admissible Schwartz functions in $\mathbb{A}(\mathbb{A}(\mathfrak{p}_0))$. Similarly, let $\mathfrak{p}_0 \in \mathbb{A}(\mathbb{A}(\mathfrak{p}_0))$ be an admissible Schwartz function of weight $\mathfrak{p}_0$.

We then construct $\mathbb{A}$ -valued Schwartz functions $\mathfrak{p}_0 \in \mathbb{A}(\mathbb{A}(\mathfrak{p}_0))$ and $\mathfrak{p}_0 \in \mathbb{A}(\mathbb{A}(\mathfrak{p}_0))$ such that:

SW 1 The Schwartz function $\mathfrak{p}_0 \otimes \mathfrak{p}_0$ is regular at at least two places (see Theorem 7.2.1). Such a test function always exists, as we can construct a regular test function $\mathfrak{p}_0$ of the form

$$\mathfrak{p}_0 = \bigotimes_{\mathfrak{p}} \mathfrak{p}_0, \quad \mathfrak{p}_0 = \mathfrak{p}_0 \in \mathbb{A}(\mathbb{A}(\mathfrak{p}_0)),$$

where $\mathfrak{p}_0$ is the characteristic function of a small region $\mathfrak{p}_0$ (see Remark 7.2.2). Moreover, we can choose $\mathfrak{p}_0$ to be $\mathbb{A}$ -valued.

SW 2 The orbit integrals $\mathfrak{p}_0(\mathfrak{p}_0)$ and $\mathfrak{p}_0(\mathfrak{p}_0)$ are non-vanishing, where

$$\mathfrak{p}_0 = (\mathfrak{p}_0, \mathfrak{p}_0) \otimes \mathfrak{p}_0 \in \mathbb{A}(\mathbb{A}(\mathfrak{p}_0)), \quad \mathfrak{p}_0 \in \{\mathfrak{p}_0, \mathfrak{p}_0\}.$$

SW 3 For all positive definite Hermitian matrices $\mathfrak{p} \neq \mathfrak{p}$ of the form

$$\mathfrak{p} = \begin{pmatrix} \mathfrak{p}_a & * \\ * & \mathfrak{p}_b \end{pmatrix} \in \mathbb{A}(\mathfrak{p}_0), \quad \mathfrak{p}(\mathfrak{p}, \mathfrak{p}) = \{\mathfrak{p}_0\},$$

the orbital integrals $\mathfrak{p}_0(\mathfrak{p}_0)$ and $\mathfrak{p}_0(\mathfrak{p}_0)$ vanish. Since there are only finitely many such $\mathfrak{p}$ satisfying these conditions, we can achieve this by shrinking the support of the Schwartz function at a single place so that it intersects only with $\mathfrak{p}_0$.

We choose gauge forms $\mathfrak{p} \in \mathbb{A}(\mathbb{A}(\mathfrak{p}_0))^*$ and $\mathfrak{p} \in \mathbb{A}(\mathbb{A}(\mathfrak{p}_0))^*$ such that the volumes $\mathfrak{p}(\mathfrak{p}_0)$ are rational numbers for all $\mathfrak{p}_0 \in \mathbb{A}(\mathfrak{p}_0)$, where the volumes are computed using the Tamagawa measure induced by $\mathfrak{p}$ and $\mathfrak{p}$ (see §10.1) (this is possible, see, e.g. [KR14] or [BY21]).

By Proposition 10.4.4 we have generating series

$$\begin{aligned} \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?, \mathfrak{w}_b) &= \sum_{\mathfrak{w} \in \mathfrak{w}_b} \sum_{\mathfrak{w}_b \in \mathfrak{w}_b} \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) \cdot \mathfrak{w}_b(\mathfrak{w}_b?) \\ &= \sum_{\substack{\mathfrak{w} = \begin{pmatrix} * & * \\ * & \mathfrak{w}_b \end{pmatrix} \in \mathfrak{w}_b \\ \mathfrak{w}_b(\mathfrak{w}) = \{\mathfrak{w}_b\}}} \left(\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) - \partial \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) \right) \cdot \mathfrak{w}_b(\mathfrak{w}_b?), \quad ? \in \{\mathfrak{w}_b, \mathfrak{w}_b\}. \end{aligned}$$

Proposition 11.0.2. *The following dualizing relation holds:*

$$\begin{pmatrix} \mathfrak{w}_b & -\mathfrak{w}_b \\ \mathfrak{w}_b & \mathfrak{w}_b \end{pmatrix} \cdot \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?, \mathfrak{w}_b) = \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) \cdot \mathfrak{w}_b(\mathfrak{w}_b?).$$

Proof. Recall from (10.4.6) that $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?, \mathfrak{w}_b)$ is the summation of terms (all terms with $\mathfrak{w}_b$ vanish):

$$\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?), \quad \partial \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?), \quad \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?), \quad \left(\widehat{\mathfrak{w}_b}(\mathfrak{a}, \mathfrak{w}_b?), \mathfrak{w}_b(\mathfrak{w}_b?) \right).$$

Each of these terms satisfies the claimed modularity via the action of the Weil representation, as established in Theorem 10.2.1, Proposition 6.5.2, Proposition 7.4.1 (combine with Proposition 3.4.10 for error terms) and Proposition 8.2.2, respectively. Therefore, the assertion follows from the relation in $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?, \mathfrak{w}_b)$

$$\begin{pmatrix} \mathfrak{w}_b & -\mathfrak{w}_b \\ \mathfrak{w}_b & \mathfrak{w}_b \end{pmatrix} \cdot \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) = \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) \cdot \mathfrak{w}_b(\mathfrak{w}_b?). \quad \square$$

Theorem 11.0.3. $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?, \mathfrak{w}_b) = \mathfrak{w}_b$ for $? \in \{\mathfrak{w}_b, \mathfrak{w}_b\}$.

Proof. For notational convenience, we write $\mathfrak{w}_b(\mathfrak{a})$ for $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?, \mathfrak{w}_b)$, where $? \in \{\mathfrak{w}_b, \mathfrak{w}_b\}$. Its Fourier expansion is given by

$$\mathfrak{w}_b(\mathfrak{a}) = \sum_{\mathfrak{w} > \mathfrak{w}_b} \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}), \quad ? \in \{\mathfrak{w}_b, \mathfrak{w}_b\}.$$

By Proposition 10.4.4, for $\mathfrak{w} \in \mathfrak{w}_b^\times$, we have:

$$\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}) = \sum_{\mathfrak{w}_b \in \mathfrak{w}_b} \frac{\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) \cdot \mathfrak{w}_b(\mathfrak{w}_b?)}{\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) \cdot \mathfrak{w}_b(\mathfrak{w}_b?)}. \quad \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) = \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) - \partial \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) \cdot \mathfrak{w}_b(\mathfrak{w}_b?), \quad ? \in \{\mathfrak{w}_b, \mathfrak{w}_b\}.$$

Let $\mathfrak{w}_b = \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?)$, we apply the double induction lemma from [Zha25, Lemma 15.1] to the pair $(\mathfrak{w}_b(\mathfrak{a}), \mathfrak{w}_b(\mathfrak{a}))$ with $\mathfrak{w}_b = \{\mathfrak{w}_b\}$. The dual relation [Zha25, Lemma 15.1(1)] follows from Proposition 11.0.2, while the support condition [Zha25, Lemma 15.1(2)] follows from (10.4.6), where we choose $\mathfrak{w}_b(\mathfrak{a})$ as $\mathfrak{w}_b(\mathfrak{a}) := \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?) \rightarrow \mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b?)$. It remains to verify the boundary condition [Zha25, Lemma 15.1(3)], it boils down to the following cases:

(1) When $\mathfrak{w}_b = \mathfrak{w}_b$, we have $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = -\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b)$ and $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = -\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b)$. It is sufficient to prove the following two claims:

- (i) For $\mathfrak{w}_b$ with $\mathfrak{w}_b(\mathfrak{w}_b) = \mathfrak{w}_b$, we have $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = \mathfrak{w}_b$;
- (ii) For $\mathfrak{w}_b$ with $\mathfrak{w}_b(\mathfrak{w}_b) = \mathfrak{w}_b$, we have $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = \mathfrak{w}_b$.

(2) When $\mathfrak{w}_b < \mathfrak{w}_b < \mathfrak{w}_b$, we have $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = -\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b)$ and $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = -\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) + \mathfrak{w}_b$. It is sufficient to prove the following two claims:

- (i) For $\mathfrak{w}_b$ with $\mathfrak{w}_b(\mathfrak{w}_b) = \mathfrak{w}_b$, we have $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = \mathfrak{w}_b$;
- (ii) For $\mathfrak{w}_b$ with $\mathfrak{w}_b(\mathfrak{w}_b) = -\mathfrak{w}_b$, we have $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = \mathfrak{w}_b$.

(3) When $\mathfrak{w}_b = \mathfrak{w}_b$, we have $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = -\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) - \mathfrak{w}_b$ and $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = -\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) + \mathfrak{w}_b$. It is sufficient to prove the following two claims:

- (i) For $\mathfrak{w}_b$ with $\mathfrak{w}_b(\mathfrak{w}_b) = \mathfrak{w}_b$, we have $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = \mathfrak{w}_b$;
- (ii) For $\mathfrak{w}_b$ with $\mathfrak{w}_b(\mathfrak{w}_b) = -\mathfrak{w}_b$, we have $\mathfrak{w}_b(\mathfrak{a}, \mathfrak{w}_b) = \mathfrak{w}_b$.

Let us verify case (2); the other cases follow by analogous arguments. By Proposition 10.4.4, it suffices to show that

$$\sum_{\substack{(\mathfrak{a}, \mathfrak{b}) \in \mathcal{O}_K^\times / \mathcal{O}_K^\times; \\ \mathfrak{a} \neq \mathfrak{b}}} \left(\text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) - \partial \text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) \right) \cdot \text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) = \mathfrak{a}, \quad \mathfrak{a} \in \{\mathfrak{a}, \mathfrak{b}\} \quad (11.0.1)$$

for any inert places $\mathfrak{a} \nmid \mathfrak{p}$ in case (i) and (ii). For case (i):

- If $\mathfrak{a} = \mathfrak{b}$, then (11.0.1) follows from Lemma 11.0.1 and assumption SW 3.
- If $\mathfrak{a} \neq \mathfrak{b}$, then since $\mathfrak{a} \nmid \mathfrak{p}$, assumption LR 2 implies that one of the diagonal entries of $\mathfrak{a}$ has unit length, and we may again apply Lemma 11.0.1.

Case (ii) follows by the same reasoning, and we complete the proof. $\square$

Corollary 11.0.4. *The Kudla-Rapoport conjecture (Theorem 3.4.4) holds for $\mathfrak{a} \in \mathcal{O}_K^\times / \mathcal{O}_K^\times$.*

Proof. By Theorem 11.0.3, we have $\mathfrak{a} = \text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) = \text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) = \mathfrak{a}$. Taking $\mathfrak{a}$'s Fourier coefficient, we have

$$\sum_{\substack{(\mathfrak{a}, \mathfrak{b}) \in \mathcal{O}_K^\times / \mathcal{O}_K^\times; \\ \mathfrak{a} \neq \mathfrak{b}}} \left(\text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) - \partial \text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) \right) \cdot \text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) = \mathfrak{a}, \quad \mathfrak{a} \in \{\mathfrak{a}, \mathfrak{b}\}.$$

By our appropriate choice of gauge form in $\mathfrak{a}$ and $\mathfrak{b}$ in the assumption, the $\text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a})$ takes values in $\mathfrak{a}$. The log terms $\text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a})$ for $\mathfrak{a} \in \mathcal{O}_K^\times / \mathcal{O}_K^\times$ are $\mathfrak{a}$ -linearly independent. By SW 1, $\mathfrak{a}$ takes values in $\mathfrak{a}$, hence

$$\sum_{\substack{(\mathfrak{a}, \mathfrak{b}) \in \mathcal{O}_K^\times / \mathcal{O}_K^\times; \\ \mathfrak{a} \neq \mathfrak{b}}} \left(\text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) - \partial \text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) \right) \cdot \text{Tr}_{\mathfrak{a}, \mathfrak{b}}(\mathfrak{a}) = \mathfrak{a}, \quad \mathfrak{a} \in \{\mathfrak{a}, \mathfrak{b}\}.$$

By SW 3 and Lemma 11.0.1, in the summation only the term with index $\mathfrak{a} = \mathfrak{b}$ is nonzero. Specifically, we have

$$\left(\text{Tr}_{\mathfrak{a}, \mathfrak{a}}(\mathfrak{a}) - \partial \text{Tr}_{\mathfrak{a}, \mathfrak{a}}(\mathfrak{a}) \right) \cdot \text{Tr}_{\mathfrak{a}, \mathfrak{a}}(\mathfrak{a}) = \mathfrak{a}, \quad \mathfrak{a} \in \{\mathfrak{a}, \mathfrak{b}\}.$$

By SW 2, $\text{Tr}_{\mathfrak{a}, \mathfrak{a}}(\mathfrak{a}) \neq \mathfrak{a}$, and we have

$$\text{Tr}_{\mathfrak{a}, \mathfrak{a}}(\mathfrak{a}) = \partial \text{Tr}_{\mathfrak{a}, \mathfrak{a}}(\mathfrak{a}), \quad \mathfrak{a} \in \{\mathfrak{a}, \mathfrak{b}\}.$$

Since $\mathfrak{a}$ can be chosen to be any admissible Schwartz function, this completes the proof. $\square$

Remark 11.0.5. When $\Psi_{\mathfrak{a}}/\Psi_{\mathfrak{b}}$ is unramified for all places $\mathfrak{a} \mid \mathfrak{p} \in \mathcal{O}_K^\times$, the integral model $\widehat{\text{Tr}}_{\mathfrak{a}, \mathfrak{b}} \rightarrow \widehat{\text{Tr}}_{\mathfrak{a}, \mathfrak{b}}[\frac{1}{p}]$ is regular. In the globalization procedure when proving the KR conjecture, one can always choose a global CM extension $\mathfrak{a}/\mathfrak{b}$ with CM type $\mathfrak{a}$ and a hermitian space $\mathfrak{a}$ as described in §5.1, with reflex field $\mathfrak{a}$ such that the associated reflex field $\mathfrak{a}/\mathfrak{b}$ is unramified over $\mathfrak{a}$. This CM type is called *unramified at a prime $\mathfrak{a}$* as defined in [MZ24, §6.2], and its existence is established in [MZ24, after (10.1)].

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Named upright tokens

Original notation is shown in TeX form. Font style is part of the identifier; superscripts, subscripts and decorations are applied as in the paper.

	Aut	2		Spec	7
	CM	1		Spf	10
	Ch	3		Spl	2
	Den	46		Stab	6
	Diff	31		U	41
	Ei	2		adm	6
	Eis	65		aux	71
	End	7		charac	1
	FJ	27		corr	23
	Fil	1		d	12
	GL	4		dd	5
	GU	2		deg	4
	Gal	3		det	16
	Gr	2		diag	6
	Herm	125		dim	1
	Hom	28		hol	11
	Ind	1		im	2
	Int	68		inert	8
	KM	24		ker	5
	KR	3		lim	1
	Lie	7		log	27
	Mat	1		max	16
	Nm	4		mod	35
	Orb	31		ord	2
	Pet	3		reg	42
	Pic	33		rel	2
	Re	2		self	5
	Res	17		sign	6
	Ros	1		tr	25
	SL	5		val	32
	$\mathrm{S0}$	3		vert	2
	Sh	8		vol	41
	Span	3			

Italic symbols

Original notation is shown in TeX form. Font style is part of the identifier; superscripts, subscripts and decorations are applied as in the paper.

	A	55
	B	5
	C	29
	D	7
	E	87
	F	620
	G	260
	I	11
	J	11
	K	213
	L	205
	M	20
	N	8
	O	118
	P	5
	R	3
	S	147
	T	840
	U	2
	V	369
	W	44
	X	75
	Y	3
	Z	74
	a	136
	b	232
	c	64
	d	76
	e	5
	f	180
	g	159
	h	418
	i	185
	j	19
	k	57
	m	116
	n	785

	p	76
	q	119
	r	154
	s	91
	t	268
	u	306
	v	766
	w	114
	x	84
	y	44
	z	34
	\mu	47
	\varpi	87
	\Gamma	7
	\Theta	5
	\Lambda	28
	\Sigma	78
	\Phi	77
	\alpha	194
	\beta	65
	\gamma	23
	\delta	9
	\eta	40
	\iota	47
	\lambda	125
	\nu	226
	\xi	179
	\pi	12
	\rho	9
	\sigma	4
	\tau	159
	\phi	910
	\chi	38
	\psi	60
	\omega	72
	\varphi	48

Styled symbols and constants

Original notation is shown in TeX form. Font style is part of the identifier; superscripts, subscripts and decorations are applied as in the paper.

	$\mathbb{A}$	164		$\mathbf{z}$	2
	$\mathbb{C}$	53		$\boldsymbol{\tau}$	84
	$\mathbb{E}$	23		$\mathcal{A}$	55
	$\mathbb{F}$	17		$\mathcal{C}$	12
	$\mathbb{G}$	8		$\mathcal{D}$	26
	$\mathbb{H}$	19		$\mathcal{E}$	38
	$\mathbb{L}$	100		$\mathcal{F}$	1
	$\mathbb{P}$	3		$\mathcal{G}$	72
	$\mathbb{Q}$	165		$\mathcal{L}$	8
	$\mathbb{R}$	43		$\mathcal{M}$	98
	$\mathbb{V}$	83		$\mathcal{N}$	40
	$\mathbb{W}$	3		$\mathcal{O}$	39
	$\mathbb{X}$	24		$\mathcal{S}$	102
	$\mathbb{Z}$	11		$\mathcal{V}$	34
	$\mathbf{B}$	44		$\mathcal{X}$	63
	$\mathbf{H}$	66		$\mathcal{Y}$	59
	$\mathbf{K}$	69		$\mathcal{Z}$	293
	$\mathbf{T}$	19		$\varnothing$	1
	$\mathbf{a}$	25		∞	186
	$\mathbf{b}$	9		$\mathfrak{a}$	12
	$\mathbf{i}$	2		$\mathfrak{d}$	323
	$\mathbf{k}$	1		$\mathfrak{p}$	3
	$\mathbf{t}$	5		$\emptyset$	1744
	$\mathbf{u}$	260		1	1183
	$\mathbf{v}$	4		2	328
	$\mathbf{w}$	114		3	1
	$\mathbf{x}$	11		4	2
	$\mathbf{y}$	78		$\mathscr{0}$	1